[]

Contents

Chapter 1

Introduction: Mott Transitions and Kondo Breakdown

The rich physics of Mott metal-insulator transitions in strongly correlated systems has been an active subject of study for quite some time [1–4]. Their appearance in various materials such as the nickelates, cuprates, iridates and manganites, among others, is marked by complex phase diagrams that display dramatic changes in properties across changes in filling and/or bandwidth, even at low temperatures. The physics of Mott transitions is driven by the competition between two energy scales in the system: (i) the scale of itineracy, quantified by the bandwidth, and (ii) the scale of electron-electron repulsion. A Mott insulator arises when the electronic correlations are strong enough that low-energy electronic excitations become gapped, and the metallic valence band gets split into lower and upper bands; for an initially half-filled valence band, the lower band is filled at $T = 0$ while the upper band is vacant. Because of strong electronic repulsion, the electrons in the lower band form local moments which can show a variety of behaviour. This includes various symmetry-broken ground states such as an antiferromagnet [5], or symmetry-preserved states such as quantum spin liquids [6]. The physics of Mottness is also responsible for several very interesting phenomena, such as high-temperature superconductivity [5], non-Fermi liquids [7] and the various phases in magic-angle twisted bilayer graphene [8, 9].

1.1 Metal-insulator transition arising from strong correlation

Initial attempts at explaining metal-insulator transitions was based on a non-interacting picture of electrons (through the work of Bethe, Sommerfeld, Bloch and Wilson among others); in such a picture, electronic bands can arise only from the scattering of electrons off the periodic lattice potential. One then distinguishes between metals and insulators at zero temperature by the position of the Fermi level E_F with respect to the bands: if E_F lies within a band gap, an integer number of bands will be completely filled and the system will be an insulator, whereas if E_F lies within one of the bands, the system will be a metal because there are vacant gapless excitations within the band.

It was first pointed out by Boer and Verwey [10] that certain insulators and semiconductors (such as nickel oxides) have a particle filling that, purely from band theory, would lead to partially filled

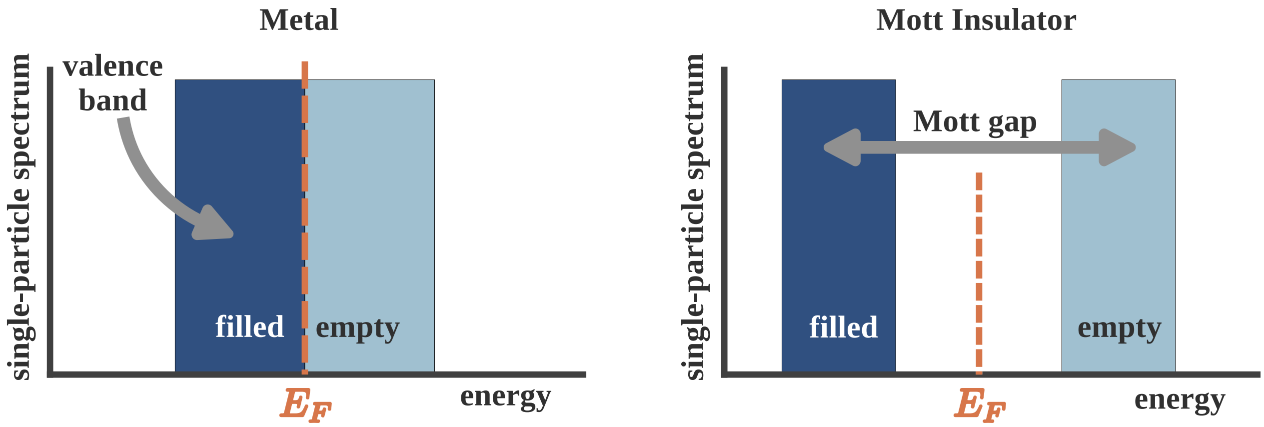

Figure 1.1

bands. This poses a challenge because a partially filled band necessarily leads to a metallic phase (see Fig. 1.2 (left)). The insulating behaviour of these materials therefore signals a fundamental limitation of single-particle band theory and points to a qualitatively different mechanism for electron localization—one driven by strong electron–electron interactions rather than by scattering from the lattice potential alone. This realization led to the concept of a Mott insulator, in which charge localization emerges as a genuine many-body effect despite the absence of a band gap in the non-interacting electronic structure. A very simple mechanism for this was put forth by Mott [11–13]: in the presence of a large local Coulomb repulsion cost U on a lattice site, the quantum states $|\Psi_1\rangle$ in which the site is singly occupied get separated in energy from the states $|\Psi_2\rangle$ where the site is doubly occupied by a gap of size U; the former class of states $|\Psi_1\rangle$ form the lower band (LB) while the latter $|\Psi_2\rangle$ form the upper band (UB). If we now have a half-filled system (one electron per site), this electron will completely fill LB and leave UB vacant, leading to an insulating state (see Fig. 1.2 (right)).

Given the above discussion, it is clear that a Mott insulator must involve a gap to charge excitations (excitations that add or remove an electron). It does not however determine the fate of the spin, orbital or other degrees of freedom. One scenario is when the spin or orbital degree of freedom undergoes spontaneous symmetry breaking into an ordered state; in the case of spin, this usually results in a superexchange interaction being generated and leads to an long-range ordered antiferromagnetic Neel state coexisting with the Mott state. Examples of such systems include vanadium sesquioxide V_2O_3 and its derivatives and the high-T_c cuprates (such as La_2CuO_4) [4]. The other scenario is one where the spin degree of freedom retains its symmetry, and we end up with a quantum mechanically non-trivial entangled state (in order to quench the entropy). Examples of such states include quantum spin liquids [14]. Mott insulators without long-range order are of particular interest because of the possibility of topological order, fractional excitations, and novel metallic parent states.

A prototypical model for gaining theoretical understanding of Mott metal-insulator transitions on a lattice is the single-band Hubbard model [1, 15–17]

$$H = -t \sum_{\langle i,j \rangle,\sigma} (c^\dagger_{i,\sigma} c_{j,\sigma} + \text{h.c.}) - \frac{U}{2} \sum_i (n_{i,\uparrow} - n_{i,\downarrow})^2 - \mu N \ . \tag{1.1}$$

The first term incorporates nearest-neighbour hopping on the lattice, while the second term introduces inter-electron repulsion when two electrons site on the same lattice site; in this way, the model incorporates the two competing tendencies of delocalisation and localisation that are responsible for the Mott MIT. The third term fixes the chemical potential and determines the filling of the lattice. Many of the materials that are studied for Mott MIT involve $d-$electrons that are five-fold degenerate, and these orbitals can hybridisation with ligand atoms. The Hubbard model avoids these complexities by focusing on only a single band that lies closest to the Fermi energy, and it is the states of this band that are described by the Hubbard model.

This leaves us with two channels for observing Mott MIT [4]: (i) tuning the bandwidth, and (ii) tuning the filling. The former can be done, in experiments, by applying pressure. Filling is tuned by doping specific elements of a compound with elements of a different valence. The former is referred to as bandwidth-control metal-insulator transition, while the latter is called filling-control metal-insulator transition.

1.2 Features of Mott metal-insulator transition

Two complementary pictures of the Mott transition: Brinkman-Rice vs Mott-Hubbard. Initial insights into the Mott MIT were obtained by Hubbard [1], thinking along lines similar to Mott. Within this picture, one starts from the insulating large U limit of the Hubbard model (eq. 1.1) ($U \gg t$. We stay at half-filling for simplicity ($\mu = 0$). The large on-site repulsion leads to the formation of two bands at each site, the lower Hubbard band (LHB) associated with singly-occupied states and the upper Hubbard band (UHB) associated with doubles and holes (states that incur a cost U). In the large U limit, these bands are well-separated, and the dynamics within the bands is a result of the motion of tightly bound holons and doubles (more on this later) that are generated because of the small but non-zero hopping probablity t. As U/t is lowered, the bands come closer and a metal-insulator transition occurs at the critical value of U at which the gap vanishes. For lower U/t, the bands overlap and we get a metal. This picture fails to provide an accurate description of the metal resulting from the melting of such a Mott insulator [18].

An alternative approach to the Mott MIT is due to Brinkman and Rice [19], where the authors applied Gutzwiller's variational method [20] to the half-filled Hubbard model. Gutzwiller's approach involves modifying, through variational factors, the hopping matrix elements that start from doubly occupied states, in order to reflect the electronic correlation at each site. By requiring that the number of doubly-occupied states vanish in the Mott insulating ground state, they obtained a critical value of the interaction strength. In the Brinkman-Rice scenario, therefore, the transition from the metal to the insulating phase occurs when the hopping matrix elements from the doubly-occupied to the singly-occupied states strictly vanishes. Note that in contrast to the Mott-Hubbard picture, this approach starts from a correlated Fermi liquid (renormalised due to the Gutzwiller factors). While it gives a better low-energy description of the correlated metal compared to the Mott-Hubbard picture,

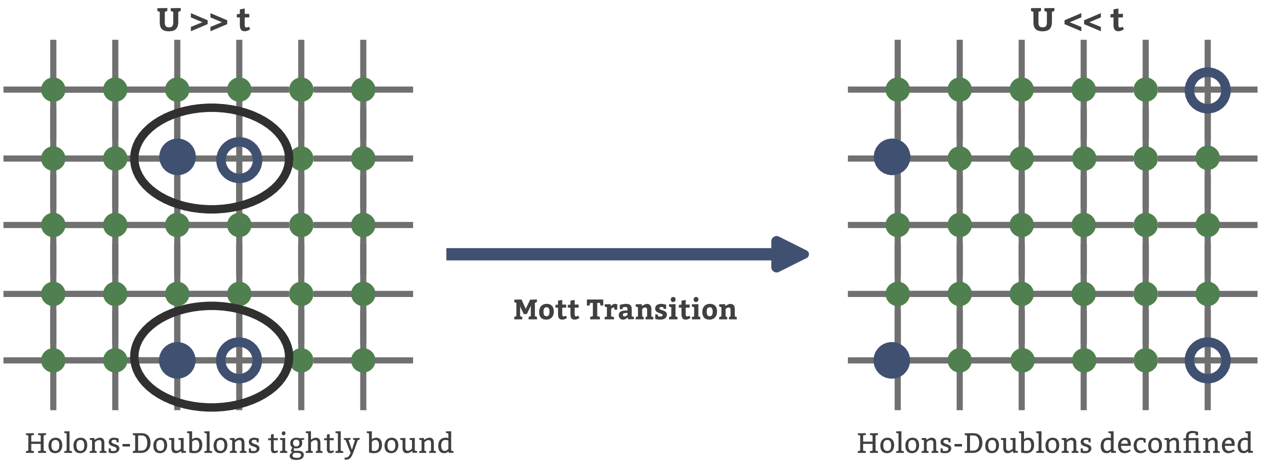

Figure 1.2

it fails to account for the high-energy physics, and treats the Mott insulator as a direct product state of local moments at each state (on account of the strictly zero hopping). This of course means that this method cannot explain antiferromagnetic superexchange processes in the Mott insulator [18].

Holon-doublon unbinding picture of the Mott MIT. In 1949, Mott proposed a zero temperature explanation for a correlation-driven insulator and a mechanism for its transition into a metal in terms of the motion of holes and doubles in the background of a half-filled lattice. For $U \ll t$ (strong correlation), the ground state predominantly involves lattice sites with a single electron. Hopping processes between two nearest-neighbour sites transfer an electron between them and create a pair of hole and double, but the pair remains tightly bound to each other. This is because moving the hole independently involves scattering processes that cross the Mott gap, and these processes are not allowed at $T = 0$. But without the independent motion of the hole (or double), there is no current, and the system is an insulator.

As U/t is lowered, the gap closes and processes that transport the hole without transporting the double become gapless. This essentially means that the pair becomes deconfined, and the hole can be moved far away from the double. Current can then be passed by having the hole move through the lattice, and we get a metallic state. This leads to a picture of the Mott transition into a metal in terms of an unbinding of the hole-double bound states (excitons) that were present in the insulator.

There have been more quantitative studies of this mechanism. I list some of the relatively recent ones here. In [21], Prelovšek et al. studied the motion of holon-doublon pairs in a spin polarised background by rewriting the repulsive Hubbard model in terms of electrons as an attractive Hubbard model of holes and doubles. The model was treated using a BCS-like treatment, and the Mott transition was obtained when the binding energy for the hole-double condensate vanishes. To mark the deconfinement, they show that the approach to the transition from the insulating side is concomitant with an increase in the size of the pairs. Following a slave-boson treatment of holon-doublon unbinding, Zhou et al. [22] show that the Mott transition from a paramagnetic metal to a Mott insulator on the half-field honeycomb and 2D square lattices occur through an intervening Slater insulator with coherent quasiparticles. In particular, their approach sheds light on the nature of the incoherent excitations in the Hubbard sidebands of the local spectral function. Finally, an older work by Watanabe et al. [23] studies the half-filled Hubbard model on an anisotropic triangular lattice using a Monte carlo method with a variational binding parameter between holons and doublons. Their approach reveals a first order Mott transition, along with robust $d-$wave superconductivity and short-range magnetic order.

Spectral features of the Mott transition. One of the important diagnostics of the Mott metal-insulator transition is the spectrum for one-particle electron-like excitations, as expressed through the one-particle retarded Greens function $G(\boldsymbol{k}, \sigma; t) = -i\theta(t)\langle\{c_{\boldsymbol{k},\sigma}(t), c^\dagger_{\boldsymbol{k},\sigma}\}\rangle$. In general, the presence of poles in the Green function at the Fermi energy and in its neighbourhood imply the existence of gapless excitations and hence a metallic phase. It turns out that this statement is equivalent to Luttinger's theorem,

$$\frac{N}{V} = 2\int_{G(0,\boldsymbol{k})>0} \frac{d\boldsymbol{k}}{(2\pi)^3} , \tag{2.1}$$

which states that the particle density in a metal is equal to the number of states in the Fermi volume. As clarified in the equation, the Fermi volume is defined as the region of $k-$space where the Greens function is positive at $\omega = 0$.

In a Mott insulator, strong correlations lead to the destruction of electronic excitations around $\omega = 0$. This results in the Greens function poles transmuting into zeros. This is also tracked by analogous changes in the electronic self-energy $\Sigma(\boldsymbol{k}; \omega)$; Greens function zeros leads to a Mott pole at $\omega = 0$ in $\Sigma(\boldsymbol{k}; \omega)$. The points in $k-$space where $\Sigma(\boldsymbol{k}; 0)$ vanishes is called a Luttinger surface. The Mott metal-insulator transition is therefore about the topological change of a Fermi surface (surface of Greens function poles) into a Luttinger surface (surface of self-energy poles).

As discussed earlier, the appearance of Greens function zeros at $\omega = 0$ leads to the formation of a charge gap that freezes one-particle low-energy excitations. The spectrum gets redistributed into two bands on either side of the gap, the upper and lower Hubbard bands. In the ground state, the lower band is obviously filled while the upper band is empty. Bands of a Mott insulator formed from electronic correlation differ from those formed due to one-particle scattering in a particular way: dynamical spectral weight transfer. Unlike a band insulator, the bands of a Mott insulator are not rigid; tuning the lattice filling results in a continuous transfer of states from one band to another.

The non-rigidity of the bands can be seen most easily from the atomic limit ($t = 0$) of the Hubbard model (eq. 1.1). The local retarded Greens function at an arbitrary site i can be obtained using the equation of motion approach. The final result is [24]

$$G_R(i; \omega) = \frac{1 + x}{\omega + \mu + U/2} + \frac{1 - x}{\omega + \mu - U/2} , \tag{2.2}$$

where $x = 1 - \sum_\sigma \langle n_{i\sigma}\rangle$ is the average number of holes at any site. The spectral weight for the two poles makes it clear that the lower band has $1 + x$ number of states while the upper band has $1 - x$ states. Changing the number of states in the lower band (by tuning μ and hence x) also changes those in the upper band. For example, by adding 0.1 holes per site to the lower band, the spectral weight in the upper band decreases by the same amount. This is the non-rigidity of the bands mentioned previously.

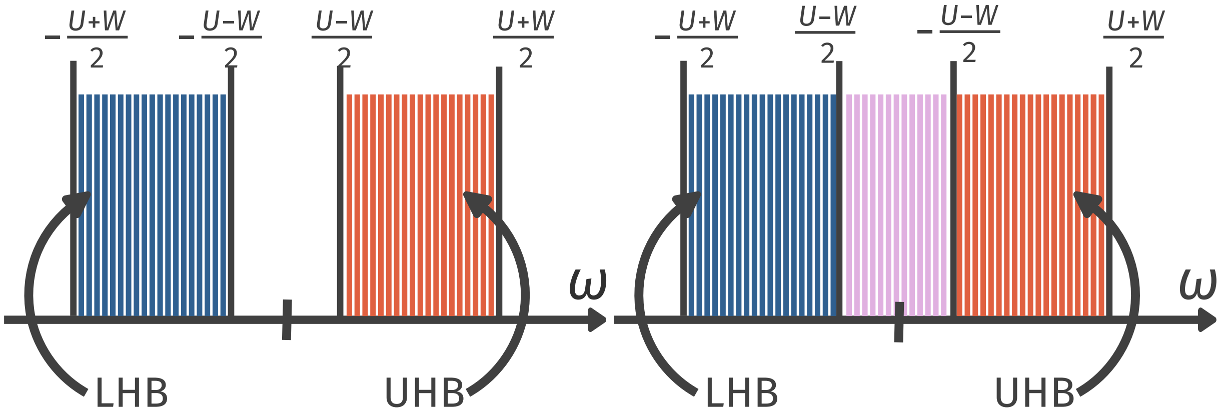

Figure 1.3

1.3 Preliminary Insights Into Mottness: The Baskaran-Hatsugai-Kohmoto model

The Baskaran-Hatsugai-Kohmoto model [**?**, **?**] introduced independently in 1991 and 1992 is an exactly solvable model of electronic correlation that displays a transition from a metallic phase to a Mott insulator. While it's solvability offers an instructive method for studying Mott transition, the infinitely long-ranged nature of its interaction makes it tricky to apply these conclusions to realistic materials.

At half-filling, the model is defined by the following Hamiltonian:

$$H_{\text{BHK}} = \sum_{\mathbf{k},\sigma} \epsilon(\mathbf{k}) n_{\mathbf{k},\sigma} - \frac{U}{2} \sum_{\mathbf{k}} (n_{\mathbf{k},\uparrow} - n_{\mathbf{k},\downarrow})^2 \ . \tag{3.1}$$

Clearly the model is diagonal in momentum-space; the Hamiltonian splits up into commuting 4×4 blocks for every point $\mathbf{k}$:

$$H_{\text{BHK}} = \sum_{\mathbf{k}} H(\mathbf{k}), \quad H(\mathbf{k}) = \epsilon(\mathbf{k}) \sum_{\sigma} n_{\mathbf{k},\sigma} - \frac{U}{2}(n_{\mathbf{k},\uparrow} - n_{\mathbf{k},\downarrow})^2 \ . \tag{3.2}$$

The vacant eigenstate $|0\rangle$ of $H(\mathbf{k})$ has zero energy ($E_0 = 0$), the two singly-occupied states $|\uparrow\rangle$ and $|\downarrow\rangle$ have energy $E_1 = \epsilon(\mathbf{k}) - U/2$ and the doubly-occupied state $|\uparrow, \downarrow\rangle$ has energy $E_2 = 2\epsilon(\mathbf{k})$. This leads to two possible single-particle excitations – one for adding an electron on the vacant state: $(\Delta_1 = E_1 - E_0 = \epsilon(\mathbf{k}) - U/2$, the other for adding an additional electron on a singly-occupied state: $\Delta_2 = E_2 - E_1 = \epsilon(\mathbf{k}) + U/2$.

Upon combining states from other $k-$modes, these isolated excitations broaden into bands around $\pm U/2$ because of the continuum of values available to $\epsilon(\mathbf{k})$. For large enough U (compared to the non-interacting bandwidth W), these bands remain well-separated in energy (Fig. 1.3 left), forming lower and upper Hubbard bands. The half-filled system results in the lower band being completely filled and the upper band empty, leading to a Mott insulator. The states within this band are all singly-occupied (per $k-$state). Upon lowering the correlation strength U, the bands touch each other at the critical value of $U_c = W$ (obtained by equating the top of the lower band to the bottom of the upper band). For lower values of U, the system is a metal due to the presence of gapless excitations within the combined band (Fig. 1.3 right). This therefore provides a simple demonstration of a correlation-driven Mott metal-insulator transition.

1.4 Warming up: Mott transition in infinite dimensions

1.4.1 Introduction to the auxiliary model approach

One of the main results obtained within this thesis is the development of a formalism to study the more realistic case of finite-dimensional interacting models by using appropriately chosen impurity models as auxiliary models. The key to an *auxiliary model method* is to construct an appropriate auxiliary model that while being tractable captures some essential property of the full lattice model [25]. This involves separating the complete degrees of freedom into a system part (H_S) which one treats exactly and the rest of the system (H_R) that one might simplify in order to be able to solve the problem. The non-trivial nature of the problem arises of course from the hybridisation H_{SR} between the system and the bath:

$$H = \begin{bmatrix} H_R & H_{RS} \\ H_{RS}^* & H_S \end{bmatrix} . \tag{4.1}$$

This separation can always be done exactly, if one does not care about tractability. For example, one can take the Hubbard model that describes electrons hopping on a lattice and interacting when two electrons reside on the same lattice site:

$$H_{\text{hub}} = -t \sum_{\langle i,j \rangle,\sigma} (c^\dagger_{i,\sigma} c_{j,\sigma} + \text{h.c.}) + U \sum_i n_{i\uparrow} n_{i\downarrow} , \tag{4.2}$$

and separate this into parts, the system being the local orbitals of a specific site l and the rest of the system being all the other sites:

$$H_S = U n_{l\uparrow} n_{l\downarrow}, \quad H_R = -t \sideset{}{'}\sum_{\langle i,j \rangle,\sigma} (c^\dagger_{i,\sigma} c_{j,\sigma} + \text{h.c.}) + U \sideset{}{'}\sum_i n_{i\uparrow} n_{i\downarrow}, \quad H_{SR} + H_{RS} = -t \sum_{i,\sigma} (c^\dagger_{l,\sigma} c_{i,\sigma} + \text{h.c.}) , \tag{4.3}$$

where the primed summation is carried out over all nearest-neighbour pairs that do not involve l and the summation in $H_{SR} + H_{RS}$ is over all sites adjacent to l.

The smaller system H_S (often called the *cluster*) is typically chosen such that its eigenstates are known exactly. Progress is then made by choosing a simpler form for H_R and its coupling H_{RS} with the smaller system. This combination of the cluster and the simpler bath is then called the *auxiliary system*. A tractable auxiliary system for the Hubbard model is the single-impurity Anderson model (SIAM); the correlated impurity site represents the set of local orbitals of any given site, while the conduction bath represents the rest of the lattice sites in an approximate fashion. Such a construction is shown in fig. 2.1.

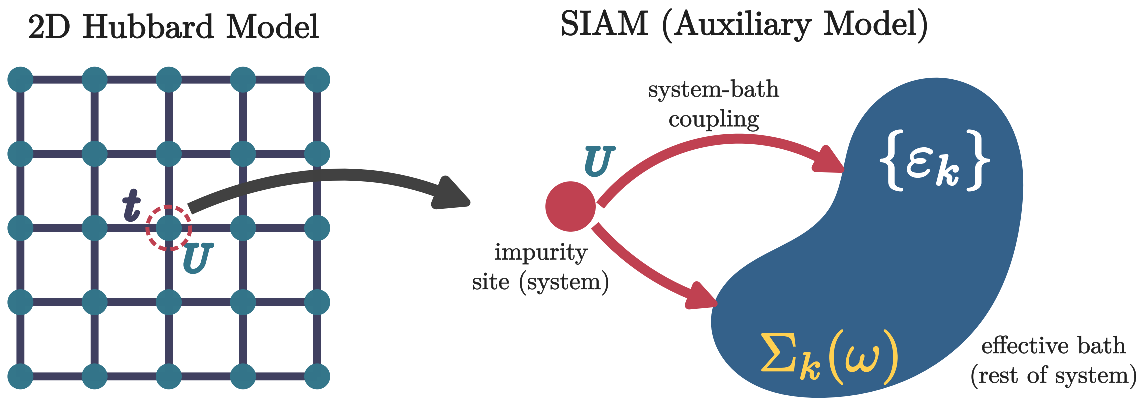

Figure 1.4: *Left: Full Hubbard model lattice with onsite repulsion U^H on all sites and hopping between nearest neighbour sites with strength t^H. Right: Extraction of the auxiliary (cluster+bath) system from the full lattice. The central site on left becomes the impurity site (red) on the right (with an onsite repulsion ϵ_d), while the rest of the $N-1$ sites on the left form a conduction bath (green circles) (with dispersion ϵ_k and correlation modelled by the self-energy $\Sigma_k(\omega)$) that hybridizes with the impurity through the coupling V.*

It should be noted that any reasonable choice of the cluster and bath would break the translational symmetry of the full model: To allow computing quantities, one would need to make the bath (which is a much larger system) simpler than the cluster (which is a single site). This distinction breaks the translational symmetry of the Hubbard model. As a result, calculations that make use of the translation symmetry of the full model (such as $k-$space objects) will require a periodisation operation of an auxiliary model quantity.

1.4.2 Correspondence between impurity and lattice models in $d = \infty$

Dynamical Mean-Field Theory (DMFT) provides a nonperturbative framework for treating strong local electronic correlations in lattice models by mapping them onto a self-consistent quantum impurity problem [**?**, **?**]. The central idea of DMFT is that in the limit of infinite spatial dimension, or equivalently infinite coordination number, the electronic self-energy becomes purely local while retaining its full dynamical (frequency-dependent) structure. As a result, the many-body lattice problem reduces to an effective single-site problem embedded in a self-consistently determined bath, capturing local quantum fluctuations exactly while neglecting nonlocal correlations. DMFT has become a cornerstone of modern correlated-electron theory, successfully describing Mott metal–insulator transitions, quasiparticle formation, and spectral weight transfer in Hubbard-like models [**?**, **?**, **?**].

In order to motivate the impurity model-lattice model mapping inherited by DMFT, we first recall the textbook example of classical mean-field theory. Mean-field theory provides one of the earliest and most instructive examples of controlled approximations in many-body physics. In its classical incarnation, exemplified by the Ising model,

$$H_{\text{Ising}} = -\sum_{i\neq j} J_{ij} S_i S_j - h\sum_i S_i = -\sum_i S_i(h + \sum_{j\neq i} J_{ij} S_j), \tag{4.4}$$

mean-field theory proceeds by replacing the interaction between a given spin and its neighbors by an effective static field proportional to the average magnetization. Writing $S_j = \langle S_j\rangle + \delta S_j$ and neglecting quadratic fluctuations and constant parts, one obtains a mean-field Hamiltonian

$$H^{(\text{MF})}_{\text{Ising}} = -\sum_i (h + h_{\text{MF}}(i)) S_i\ , \tag{4.5}$$

where the mean-field h_{MF} comes out to be

$$h_{\text{MF}}(i) = \sum_{j\neq i} 2\langle S_j\rangle J_{ij}\ . \tag{4.6}$$

The above approach assumes two things: (i) the presence of a non-zero order parameter $m_j = \langle S_j\rangle$ (in this case, a non-zero magnetisation), and (ii) the smallness of the fluctuations $S_j - \langle S_j\rangle$. The mean-field Hamiltonian can be used to compute the partition function (at inverse temperature β) and hence the magnetisation at a site i of the lattice:

$$m_i = \tanh(\beta(h_{\text{MF}}(i) + h)) = \tanh(\beta\sum_{j\neq i} 2m_j J_{ij} + \beta h)\ . \tag{4.7}$$

The problem hasn't yet been solved of course; one needs to determine the appropriate value of m_j. We therefore need an additional equation; this is provided by the so-called self-consistency condition. The translation invariance of the lattice ensures that the magnetisation must be uniform across sites i and j. This allows us to replace m_j with m_i in the above equation:

$$m_i = \tanh(\beta m_i \sum_{j\neq i} 2J_{ij} + \beta h)\ . \tag{4.8}$$

This equation can be solved numerically to obtain the uniform magnetisation $m_i = m$. The mean-field approximation becomes exact in the limit of large coordination, where fluctuations are suppressed and spatial correlations beyond a single site vanish.

The essential feature of classical mean-field theory is therefore the replacement of a many-body problem by an effective *single-site* problem embedded in a self-consistently determined environment. Quantum lattice models admit a closely analogous construction, but with an important conceptual difference: while the classical mean field is static, quantum fluctuations necessarily introduce a tower of excited states that allow for temporal dynamics. This observation underlies the formulation of dynamical mean-field theory (DMFT), which may be viewed as the natural quantum generalization of classical mean-field theory [18, 26].

Similar to the previous case of the Ising model, we will now derive the self-consistency equation for a quantum model. Consider an interacting fermionic lattice Hamiltonian, such as the Hubbard model:

$$H = -\sum_{ij\sigma} t_{ij} c^\dagger_{i\sigma} c_{j\sigma} + U\sum_i n_{i\uparrow} n_{i\downarrow}, \tag{4.9}$$

defined on a lattice with coordination number z.

The point of this exercise is to calculate the interacting self-energy $\Sigma_k(\omega)$ for the fermions. This therefore acts as an "order parameter" and plays the same role as the local magnetisation in the previous exercise. The self-energy defines the interacting k-space Green's function:

$$G_{\mathbf{k}}(\omega) = \frac{1}{\omega - \varepsilon_{\mathbf{k}} - \Sigma_{\mathbf{k}}(\omega)}, \tag{4.10}$$

where $\varepsilon_{\mathbf{k}}$ denotes the noninteracting dispersion, and a real-space local Green's function, obtained by summing over momentum:

$$G_{\text{loc}}(\omega) = \sum_{\mathbf{k}} \frac{1}{\omega - \varepsilon_{\mathbf{k}} - \Sigma_{\mathbf{k}}(\omega)}. \tag{4.11}$$

In line with the overall theme of working with auxiliary models, we consider an auxiliary impurity problem describing the dynamics of a single correlated site with the other environment sites integrated out and described by the "bare" Greens function G_0:

$$S_{\text{imp}} = -\int_0^\beta d\tau d\tau'\, c^\dagger(\tau)\, G_0^{-1}(\tau - \tau')\, c(\tau') + U \int_0^\beta d\tau\, n_\uparrow(\tau) n_\downarrow(\tau). \tag{4.12}$$

Such an impurity problem is solvable, and the impurity Green's function is given by

$$G_{\text{imp}}(\omega) = \frac{1}{G_0^{-1}(\omega) - \Sigma(\omega)}, \tag{4.13}$$

with the same local self-energy $\Sigma(\omega)$ as the lattice problem.

The DMFT approximation becomes exact in infinite dimensions by enforcing the condition that the impurity Green's function reproduces the local lattice Green's function,

$$G_{\text{imp}}(\omega) = G_{\text{loc}}(\omega). \tag{4.14}$$

Combining Eqs. (4.11) and (4.13), one obtains a self-consistency equation that determines the Weiss field $\mathcal{G}_0(\omega)$,

$$\mathcal{G}_0^{-1}(\omega) = \Sigma(\omega) + G_{\text{loc}}^{-1}(\omega). \tag{4.15}$$

Together with the definition of $G_{\text{loc}}(\omega)$ as a functional of $\Sigma(\omega)$, this equation closes the DMFT loop. The interacting lattice problem is thus reduced to the self-consistent solution of a quantum impurity model, capturing all local dynamical correlations exactly while neglecting nonlocal ones [?, ?].

In direct analogy with classical mean-field theory, DMFT replaces a lattice model by an effective single-site problem embedded in a self-consistent environment. The crucial distinction is that the mean field is now dynamical rather than static, encoding quantum fluctuations exactly while neglecting spatial correlations. This controlled limit provides both a conceptual foundation and a practical starting point for extensions beyond DMFT, as well as for alternative auxiliary-model constructions such as the one developed in this thesis.

1.4.3 Examples of strongly correlated materials

Evidently, the route to studying lattice problems via quantum impurity models makes most sense for models with strong local correlations. In practise, the

Transition Metal Oxides (TMOs) Transition Metal Oxides (TMOs) are defined chemically by the presence of partially filled d-shells in transition metals (e.g., Cu^{2+}, Mn^{3+}, V^{4+}) coordinated by oxygen ligands. The physics of TMOs is governed by the competition between the kinetic energy hopping amplitude t and the on-site Coulomb repulsion U; due to the small overlap between the localised $d-$orbitals, the $d-$electrons can delocalise only by hopping onto the ligand atoms. According to the Zaanen-Sawatzky-Allen (ZSA) classification scheme [27], these materials are categorized as either Mott-Hubbard insulators, where the charge gap is determined by the d-d Coulomb interaction ($E_g \approx U$), or charge-transfer insulators, where the gap is dictated by the energy difference between the oxygen p-band and the metal d-band. These derive from considerations of several factors, such as crystal field splitting, arrangement of atoms and lattice symmetries.

For example, in the case of octahedral symmetry, repulsion from the surrounding atoms leads to the raising of energy of the $d_{x^2-y^2}$ and $d_{3z^2-r^2}$ orbitals forming the e_g doublet, while the other three orbitals (d_{xy}, d_{yz} and d_{zx}) form the t_{2g} multiplet. For the heavier TMOs (Ti, V, Cr...), the Fermi level lies within e_g. The increased positive nuclear charge (compared to the lighter elements) lowers the energy of the $d-$orbitals and brings them closer to the $p-$orbitals, reducing the charge transfer energy $\Delta = \varepsilon_d - \varepsilon_p$ of transporting an electron from a $d-$state to a $p-$state, compared to the $d-$level repulsion U ($|\Delta| < U$). Geometric reasons (the e_g orbitals point towards the $p-$orbitals) also enhance the hybridisation between $d-$ and $p-$orbitals in such materials. If Δ becomes larger than the bandwidth, we obtain a *charge-transfer insulator*, where the charge gap is decided by Δ. In the lighter TMOs, the Fermi level passes through t_{2g}, and because they point away from the $p-$orbitals, the hybridisation is weaker. Moreover, the reduced positive nuclear charge leads to an increased $\Delta > U$; a U greater than the bandwidth leads to what's called a *Mott-Hubbard insulator*.

Experimental studies of transition metal oxides (TMOs) have established metal–insulator transitions (MITs) as a defining consequence of strong electronic correlations in systems with partially filled (d)-electron shells [28, 29]. Early experiments on vanadates such as V_2O_3 characterised pressure-driven MITs through the pressure- and temperature-dependent conductivity as well as an anomalous enhancement of the optical conductivity [30–32]. Copper oxides constitute the most extensively studied class of correlated TMOs and provide a canonical example of a doping-driven MIT. Transport measurements on parent compounds such as La_2CuO_4 establish a robust insulating state, while carrier doping leads to anomalous metallic behavior characterized by linear-in-temperature resistivity and unconventional superconductivity [33–35]. Such metal-insulator transition is also reported from angle-resolved photoemission spectroscopy (ARPES) measurements in doped La_2CuO_4 and $Bi_2Sr_2CaCu_2O_8$, along with the emergence of pseudogap near the antinodes at high temperatures and a hole-to-electronlike Fermi surface reconstruction [36–38]. Nickelates ($LaNiO_2$, $La_3Ni_2O_7$, $La_4Ni_3O_{10}$, etc) exhibit related but distinct experimental signatures. With Ni^+ sharing the same $3d^9$ valence configuration as Cu^{2+} in the cuprate superconductors, these materials also display superconductivity, although the mechanism in the nicklelates is different due to the strong coupling between the layers [39, 40]. The optimal doped regime in thin films shows strange metal behaviour, while infinite-layer nickelates show the absence of long-range antiferromagnetic order [41].

Transition Metal Dichalcogenides Transition Metal Dichalcogenides (TMDs) are layered van der Waals materials with the stoichiometry MX_2, where M is a transition metal (e.g., Mo, W) and X is a chalcogen (e.g. S, Se, Te) [42,43]. Recent interest has focused on the interplay between strong spin-orbit coupling (SOC), electronic correlations and direct bandgap – properties that make them particularly useful for electronic applications, spintronics and DNA sequencing. Layered TMDs have the structure $X - M - X$, with the transition metal sandwiched between two hexagonal planes of chalcogenides. Different combinations of the metals with the chalcogens lead to widely varying structures and electronic properties in TMDs. Experimental studies show that metal–insulator transitions (MITs) in TMDs are commonly intertwined with charge density wave (CDW) formation rather than arising from purely on-site Coulomb localization [44, 45]. In prototypical compounds such as 1T-TaS_2, ARPES and transport measurements reveal fluctuating magnetic order parameters in the Mott state and a transition to superconductivity driven by pressure [46–48]. Scanning tunnelling microscope (STM) pulses have been successfully used to manipulate the metal–insulator transition of 1T-TaS2, demonstrating the macroscopic controllability of the Mott phase [49].

ARPES measurements in 2H-$NbSe_2$ and 2H-$TaSe_2$ partial gapping of the Fermi surface restricted to nested momentum regions and a temperature-dependent pseudogap similar to the cuprates [50, 51]. This is further supported by optical conductivity experiments that show spectral weight in the midinfrared region along with a low-frequency Drude contribution [52]. Transport measurements on single crystals of $TaSe_2$ reveal that disorder enhances superconductivity by suppressing other competing orders such as CDW [53]. Collectively, these experimental results establish TMDs as a distinct class of correlated materials in which reduced dimensionality, band narrowing, and lattice reconstruction cooperate to produce metal–insulator behavior beyond conventional band theory.

Iron-Based Superconductors The iron-based superconductors (FeSCs), discovered in 2008, constitute a broad family of layered materials that share a common structure: a layer of iron atoms joined by tetrahedrally coordinated pnictogen (P, As) or chalcogen (S, Se, Te) anions arranged in a stacked sequence; the layers are separated by alkali, alkaline-earth or rare-earth and oxygen/fluorine 'blocking layers' [54–57]. Doping or applying external pressure reveals coupled antiferromagnetic and structural transitions along with superconducting and nematic orders [58–60]. Electrical resistivity measurements in the neighbourhood of the quantum critical point show a crossover from a bad-metal behavior ($T-$linear resistivity) to a Fermi liquid behaviour (T^2-resistivity) [61]. Optical conductivity measurements indicate point to intermediate-to-strong electronic correlations [62].

ARPES measurements reveal a $k-$dependent superconducting gap and de Haas-van Alphen measurements indicate substantial quasiparticle mass enhancements, shedding more light on the correlated nature of the material [63,64]. Strain-tuning measurements of the nematic susceptibility, and differential elastoresistance measurements have provided compelling evidence for the nematic transition and its surrounding fluctuations being the cause of the structural transition as well as the superconductivity [65, 66]. Neutron scattering and NMR measurements further reveal persistent low-energy spin fluctuations across wide regions of the phase diagram, supporting scenarios in which unconventional superconductivity emerges from the interplay of magnetic, orbital, and nematic degrees of freedom [67, 68].

Heavy-Fermion / Kondo-Lattice Systems Heavy-fermion materials are a class of strongly correlated electron systems typically realized in intermetallic compounds of rare earth metals (such as Ce, Sm and Yb) and actinides (such as U, Np and Pu), where localized f electrons hybridize with itinerant conduction bands [69, 70]. At high temperatures, these systems exhibit local-moment behavior governed by Curie–Weiss susceptibilities and incoherent transport, while at low temperatures the Kondo effect leads to the formation of a coherent many-body state with extremely large quasiparticle effective masses [71]. Experimentally, this crossover is most clearly manifested in transport and thermodynamic measurements: the electrical resistivity shows a characteristic maximum followed by a rapid drop signaling the onset of coherence, while the linear specific heat coefficient $\gamma = C/T$ reaches values orders of magnitude larger than in conventional metals, directly reflecting the heavy quasiparticle mass [72].

A defining feature of heavy-fermion materials is their proximity to magnetic quantum phase transitions, which can be accessed experimentally by pressure, chemical substitution, or magnetic field [73]. Near such quantum critical points (QCPs), transport measurements reveal pronounced non-Fermi-liquid behavior ($T-$linear resistivity) and a vanishing Hall response, along with a divergent effective quasiparticle mass as the transition is approached [74, 75]; these are all indications of the breakdown of electronic excitations near the transition. Neutron scattering experiments provide direct evidence for spin fluctuations down to very low temperatures, interplaying with the itinerant electrons at the QCP and potentially driving the superconducting transition [76, 77]. Quantum oscillation measurements through de Haas–van Alphen experiments further demonstrate Fermi surface reconstruction across the quantum critical point, highlighting the drastic change in Fermi volume [78]. Together, these experimental observations establish heavy-fermion compounds as paradigmatic systems in which strong local correlations, hybridization-driven coherence, and quantum critical fluctuations conspire to produce emergent low-energy quasiparticles and unconventional metallic behavior [79].

1.4.4 DMFT results on the Bethe lattice

, and has been studied using an equally diverse array of methods like mean-field theory, renormalisation group approaches, numerical techniques like exact diagonalisation, quantum Monte Carlo and dynamical mean-field theory, and many others. In particular, dynamical mean-field theory (DMFT) [18, 26, 80–86] obtains an exact solution of the Mott metal-insulator transition (MIT) for the 1/2-filled Hubbard model [1, 11, 16, 17, 19] on the Bethe lattice with infinite coordination number, in terms of an Anderson impurity model with a self-consistently determined bath obtained by requiring, in an iterative manner, that its local Greens function be equal to that of the impurity site. The above-mentioned transition can be captured by the local spectral function through (i) the continuous appearance of a Mott gap, followed by (ii) the sharpening and vanishing of the central Kondo resonance. These two features are often referred to, respectively, as the Mott-Hubbard [1] and Brinkman-Rice [19] scenarios of the Mott MIT. The exact nature of the solution arises from the fact that all non-local contributions to the lattice self-energy are observed to vanish upon taking the limit of an infinite coordination number for the lattice model.

1.4.5 Open Questions

1.5 Effect of low dimensionality: Mott MIT in two dimensions

1.5.1 DCA, CDMFT results (among other methods)

1.5.2 Experimental results

1.5.3 Open Questions

1.6 Effect of multiple bands: Orbital selective Mott transition

1.6.1 results on heavy fermions

1.6.2 Experimental results

1.6.3 Open Questions

1.7 Brief introduction to the tiling auxiliary model approach

1.7.1 Why do we need another auxiliary model method? Limitations of existing approaches

1.7.2 Overview

1.7.3 Features

1.8 Entanglement, RG, and Fermionic Criticality

1.8.1 Introduction to entanglement holography

1.8.2 Open questions

1.9 <title about an introduction to our holography work>

The present millennium has seen a rapid surge in the use of quantum entanglement towards the study of quantum and condensed matter systems [87–92]. By tracking quantum correlations among particles across both small and large distances, entanglement measures offer valuable insights into the nature of the ground state and low-energy excitations. This, for example, allows us to distinguish strongly-correlated topologically ordered states characterised by long-ranged entanglement (in the form of a sub-leading topological term in their entanglement entropy [93–96]) from weakly-interacting topologically trivial phases that have short-ranged area-law entanglement [89, 97, 98].

More pertinent to the present work is the entanglement of non-interacting relativistic fermionic systems. Such phases are known to emerge in several condensed matter systems [99], appearing at the surfaces of three-dimensional topological insulators [100–103], quasi-2D organic materials [104–106] and in artificial quantum simulators such as cold atom gases, molecular and microwave crystals [107–113]. Massless critical systems (or conformal field theories (CFTs)) differ from gapped systems due to the fact that their entanglement satisfies a modified area law; in one spatial dimension, the entanglement entropy (S) of a single subsystem of length l scales as $S \sim \frac{c}{3} \ln l$ [114–118], where c is the conformal central charge of the CFT. This violation arises from the non-local nature of the gapless quantum fluctuations that lie proximate to the $d-1$-dimensional Fermi surface of a quantum critical fermionic system in d spatial dimensions. For massive fermions, an entropic $c-$function (spatial derivative of the entanglement entropy) can be expressed in the form of differential equations that need to be solved numerically [119–122].

Unlike topologically ordered phases, there is no geometry-independent topological term in the entanglement entropy of free fermionic systems [123, 124]. It is, however, possible to generate such a term even here by placing the system on a multiply connected manifold such as a cylinder, and inserting a magnetic flux through the cylinder [123, 125–128]. It has been shown that if there are ϕ number of electronic flux quanta piercing the system, an additional term is generated in the entanglement entropy of a subsystem of massless Dirac fermions of sufficiently large length (as defined in Ref. [123]) with the form $\frac{1}{6} \ln \left|2 \sin \frac{\phi}{2}\right|$ [123, 126]. Further, there is evidence from numerical computations on discrete models that such a term is a signature of the specific field theory and encodes universality [128–130].

The notion of entanglement also plays a key role in the context of holographic duality, which posits that there exists a duality relation between a quantum field theory in $d-$spatial dimensions and a gravity theory in $d+1-$spatial dimensions [131–137]. It is widely believed that the additional dimension can be visualised as a renormalisation group flow that starts from the (almost) critical field theory on the boundary and extends into the bulk [138–147]. The most popular realisation of the holographic principle is the AdS/CFT correspondence that links a conformal field theory with a theory of quantum gravity through a strong-weak duality relation [148–155]. This has led to new insights in various areas of high energy and condensed matter physics, such as the viscosity of the quark-gluon plasma [156], the response functions of the Bose-Hubbard model [157, 158], and the phenomenology of non-Fermi liquids [159–162], holographic superconductors [163–168], striped phases [169–171] and holographic Fermi liquids [172–176].

Conversely, the holographic principle also implies that studying the boundary conformal theory can provide insights into the nature of the bulk quantum gravity theory. This requires constructing the emergent dimension from first principles by investigating the RG flow of the (often strongly coupled) quantum field theory, and has proved to be considerably more difficult. Nevertheless, there exist some examples of constructive attempts towards a bulk gravity theory, including [177–186]. These include the momentum-shell renormalisation approach of Refs. [179, 180, 187, 188], the multi-scale entanglement renormalisation ansatz (MERA) approach of Refs. [181–184, 189], the exact holographic mapping (EHM) [190, 191], the Wilsonian RG-based approach of Ki-Seok Kim [192–194] and the unitary renormalisation group (URG) approach [185, 186, 195].

Questions on entanglement and holography become particularly interesting in systems of critical fermions. As mentioned earlier, such systems differ from bosonic and gapped systems through the presence of longer-ranged entanglement that follows a modified area-law [118, 196] as well as topological features arising from the incompressibility of the Fermi volume [197, 198]. Fermi surface gapping phase transitions in such systems are often driven by changes in topological quantum numbers and the nature of the inter-particle entanglement [73, 185, 186]. Constructing a framework for fermionic criticality is therefore an active area of research with consequences for important systems such as unconventional superconductors, strange metals and heavy fermions [74, 199–201]. Still less is known about the holographic implications of such systems. We extend this body of work by providing an explicit demonstration of the holographic principle in the form of a simple and tractable exact holographic mapping (EHM) for a free fermion system with and without a mass gap. Our program yields analytic relations between the RG flow parameters and their dual bulk quantities such as distances and curvature. This also allows us to study the consequences of a critical Fermi surface (lying precisely at the quantum phase transition) on the nature of the holographic space.

Chapter 2

Mapping From Auxiliary Model To Lattice: The Tiling Approach

2.1 The Philosophy of Auxiliary Models

In the previous chapter, we extended the Anderson impurity model to host a correlation-driven delocalisation-localisation transition on the impurity site, from a local Fermi liquid state to a local moment state. We demonstrated that this transition captures very well the phenomenology of the Hubbard dimension in infinite dimensions (as seen from, say, dynamical mean-field theory (DMFT)). In this chapter, we extend this approach by developing a formalism to study the more realistic case of finite-dimensional interacting models by using appropriately chosen impurity models as auxiliary models. The key to an *auxiliary model method* is to construct an appropriate auxiliary model that while being tractable captures some essential property of the full lattice model [25]. This involves separating the complete degrees of freedom into a system part (H_S) which one treats exactly and the rest of the system (H_R) that one might simplify in order to be able to solve the problem. The non-trivial nature of the problem arises of course from the hybridisation H_{SR} between the system and the bath:

$$H = \begin{bmatrix} H_R & H_{RS} \\ H^*_{RS} & H_S \end{bmatrix} . \tag{1.1}$$

This separation can always be done exactly, if one does not care about tractability. For example, one can take the Hubbard model that describes electrons hopping on a lattice and interacting when two electrons reside on the same lattice site:

$$H_{\text{hub}} = -t \sum_{\langle i,j\rangle,\sigma} (c^\dagger_{i,\sigma} c_{j,\sigma} + \text{h.c.}) + U \sum_i n_{i\uparrow} n_{i\downarrow} , \tag{1.2}$$

and separate this into parts, the system being the local orbitals of a specific site l and the rest of the system being all the other sites:

$$H_S = U n_{l\uparrow} n_{l\downarrow}, \quad H_R = -t \sideset{}{'}\sum_{\langle i,j\rangle,\sigma} (c^\dagger_{i,\sigma} c_{j,\sigma} + \text{h.c.}) + U \sideset{}{'}\sum_i n_{i\uparrow} n_{i\downarrow}, \quad H_{SR} + H_{RS} = -t \sum_{i,\sigma} (c^\dagger_{l,\sigma} c_{i,\sigma} + \text{h.c.}) , \tag{1.3}$$

where the primed summation is carried out over all nearest-neighbour pairs that do not involve l and the summation in $H_{SR} + H_{RS}$ is over all sites adjacent to l.

The smaller system H_S (often called the *cluster*) is typically chosen such that its eigenstates are known exactly. Progress is then made by choosing a simpler form for H_R and its coupling H_{RS} with the smaller system. This combination of the cluster and the simpler bath is then called the *auxiliary system*. A tractable auxiliary system for the Hubbard model is the single-impurity Anderson model (SIAM); the correlated impurity site represents the set of local orbitals of any given site, while the conduction bath represents the rest of the lattice sites in an approximate fashion. Such a construction is shown in fig. 2.1.

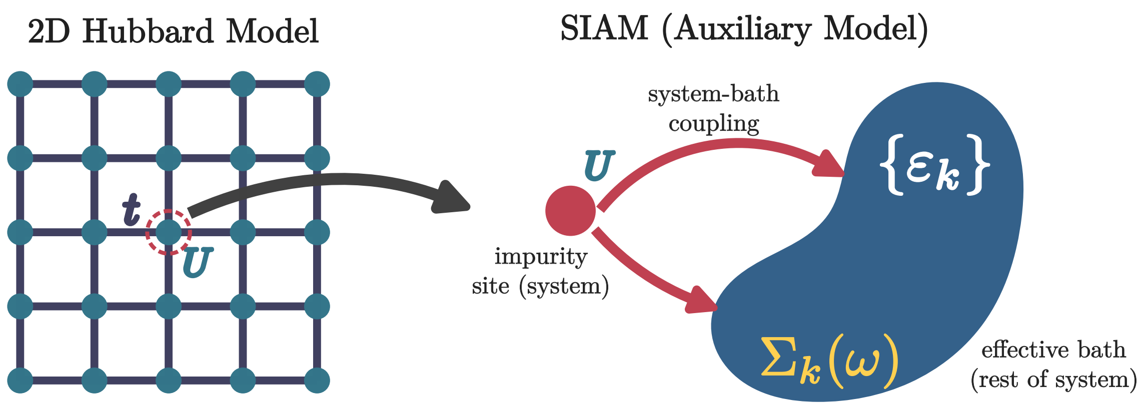

Figure 2.1: *Left: Full Hubbard model lattice with onsite repulsion U^H on all sites and hopping between nearest neighbour sites with strength t^H. Right: Extraction of the auxiliary (cluster+bath) system from the full lattice. The central site on left becomes the impurity site (red) on the right (with an onsite repulsion ϵ_d), while the rest of the $N-1$ sites on the left form a conduction bath (green circles) (with dispersion ϵ_k and correlation modelled by the self-energy $\Sigma_k(\omega)$) that hybridizes with the impurity through the coupling V.*

It should be noted that any reasonable choice of the cluster and bath would break the translational symmetry of the full model: To allow computing quantities, one would need to make the bath (which is a much larger system) simpler than the cluster (which is a single site). This distinction breaks the translational symmetry of the Hubbard model. As a result, calculations that make use of the translation symmetry of the full model (such as $k-$space objects) will require a periodisation operation of an auxiliary model quantity.

The present work is aimed towards developing a new auxiliary model method to study correlated electronic systems. As described above, the first step is to identify the correct impurity model which can faithfully capture the physics of the lattice model we are trying to solve. The local behaviour of this impurity model should reflect the essential local physics of the lattice model. Typically, we will consider impurity model geometries where the real-space bath site connected directly to the impurity hosts some local interaction. We will henceforth refer to this site as the *zeroth site*. This model must be solved using an impurity solver. In the present work, we use the recently-developed unitary renormalisation group method [185,186,199,200,202,203]. The leap to the bulk model is then made by applying **manybody lattice translation operators** on the auxiliary model. This process, referred to as *tiling* here, allows us to reconstruct the lattice model Hamiltonian from that of the auxiliary model and restores the translation invariance of the full lattice model. Once the Hamiltonians can be mapped onto one another, it is possible to relate the eigenstates and correlations functions between the auxiliary model and lattice model, using a manybody version of **Bloch's theorem**.

2.2 What Is The Correct Auxiliary Model For Correlated Electrons on a 2D Square Lattice?

As discussed in more detail in the previous chapter, it was necessary to extend the single-impurity Anderson model (SIAM) in order to obtain a phase transition on the impurity site. Without any modification, the standard Anderson model (at particle-hole symmetry) leads to the emergence of an $\omega = 0$ Kondo resonance in the impurity ($d\sigma$) spectral function at any value of the impurity correlation U_d, at sufficiently low temperatures:

$$H_{\text{SIAM}} = \sum_{k,\sigma} \varepsilon_k n_{k,\sigma} - V \sum_{\sigma} (c^\dagger_{0,\sigma} c_{d,\sigma} + \text{h.c.}) + \varepsilon_d \sum_{\sigma} n_{d,\sigma} + U_d n_{d,\uparrow} n_{d,\downarrow} \ . \tag{2.1}$$

The operator $c_{0,\sigma}$ acts on the local conduction bath density that couples with the impurity site. The low-energy is one of local Fermi liquid for all parameter regimes. Increasing U_d only serves to reduce the width of the central peak at the cost of the appearance of side bands at energy scales of the order of U_d, but the resonance never disappears. The extended Anderson model that we developed augments the SIAM by allowing for local correlation in the local conduction bath charge density that couples with the impurity spin:

$$H_{\text{ESIAM}} = H_{\text{SIAM}} + J_K \boldsymbol{S}_d \cdot \boldsymbol{S}_0 - \frac{U_b}{2}\left(n_{0,\uparrow}^2 - n_{0,\downarrow}^2\right) ; \tag{2.2}$$

tuning this bath correlation to large attractive values (see [204] for some recent findings of non-local effective attractive interactions within the Hubbard model) *does* lead to a localisation transition on the impurity where the local Fermi liquid excitations are replaced by the physics of a gapped local moment on the impurity. We note that the generation of such correlations in the bath is something that occurs in DMFT as well – the conduction bath acquires a *non-trivial self-energy* during the convergence to self-consistency.

While this modified impurity model captures well the Mott metal-insulator transition of the Hubbard model in infinite dimensions, it *lacks any knowledge of the underlying lattice geometry*, and is therefore ill-suited to address the phenomenology of more realistic low-dimensional models. Examples of such phenomenology include the pseudogap phase of the cuprates (with nodal-antinodal dichotomy), strange metal (with $T-$linear resistivity) and high-T_c superconducting phases observed in several families of quantum materials as well as graphene. This leads us to propose the framework of *lattice-embedded quantum impurity models*. Upon embedding the impurity site within the underlying lattice, the coupling between it and its environment must be consistent with the lowered symmetry of the lattice (see Fig. 2.2). Impurity-bath correlations calculated within such an impurity model are sensitive to the lattice geometry, and are much better suited to address the phenomenology mentioned above.

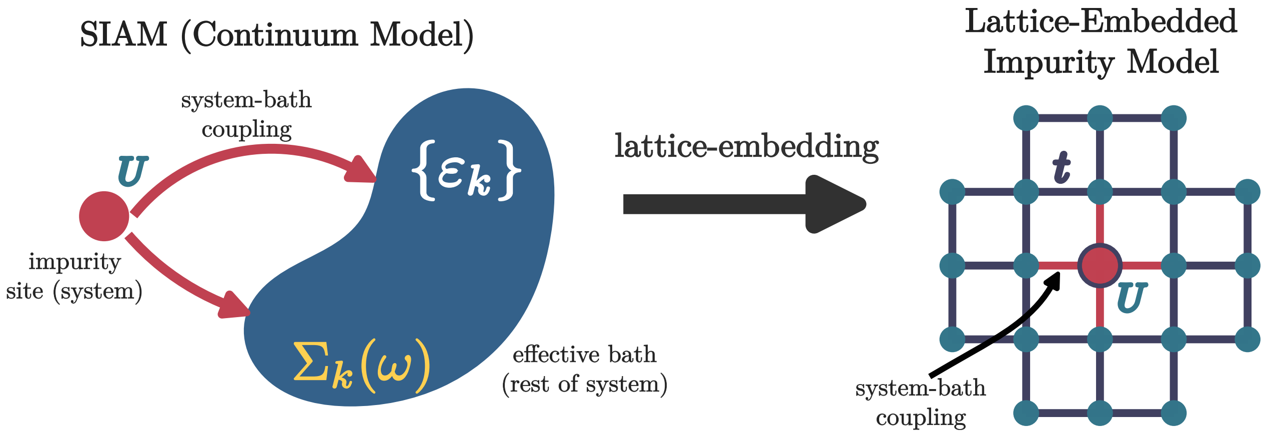

Figure 2.2

As a concrete example, we take the extended SIAM defined in eq. 2.2, and write down its lattice-embedded form on a 2D square lattice, with the impurity placed at the site r_d. This is done by allowing the impurity to hybridise with the four nearest-neighbours, through V and J_K. The same four sites also carry the local bath interaction. The complete auxiliary model is

$$H = -t \sum_{\langle i,j \rangle,\sigma} (c^\dagger_{i,\sigma} c_{j,\sigma} + \text{h.c.}) + \sum_{\langle i,r_d \rangle} \left[-\frac{V}{Z} \sum_\sigma (c^\dagger_{r_d,\sigma} c_{i,\sigma} + \text{h.c.}) + \frac{J_K}{Z} \boldsymbol{S}_{r_d} \cdot \boldsymbol{S}_i - \frac{W}{2Z} \left(n^2_{i,\uparrow} - n^2_{i,\downarrow} \right) \right] + \varepsilon_d \sum_\sigma n_{\boldsymbol{r}_d,\sigma} + U_d n_{\boldsymbol{r}_d,\uparrow} n_{\boldsymbol{r}_d,\downarrow}, \tag{2.3}$$

where Z is the coordination number of the lattice, and the second summation $\sum_{\langle i,r_d \rangle}$ is carried out over all sites i that are nearest-neighbour to the site r_d. Note that the one-particle hybridisation V and the spin-exchange coupling J_K have their *symmetry lowered* from the $s-$wave spherical symmetry of the continuum impurity models to a C_4-symmetry consistent with the underlying lattice. This feature is crucial in explaining the phenomenology of the various low-dimensional materials.

Various enhancements can be applied to such a model. The lattice can be modified to study different materials. More complicated geometries such as a two-layer model can be studied using a two-impurity auxiliary model, where each impurity couples with it's own conduction bath as well as to each other.

2.3 Mapping Auxiliary Model To Lattice Model

In this subsection, we provide an explicit example of constructing a lattice model from a lattice-embedded impurity model. This will involve the use of manybody lattice translation operators, so we begin with a brief discussion of them.

2.3.1 Manybody Translation Operators

In order to create the bulk model, we need to translate the local auxiliary model over the entire lattice. For this, we define *many-particle global* translation operators $T(\boldsymbol{a})$ that translate all positions by a vector $\boldsymbol{a}$. In terms of manybody states and operators, their action is defined as

$$\begin{aligned} T^\dagger(\boldsymbol{a})\left|\boldsymbol{r}_1, \boldsymbol{r}_2, \ldots\right\rangle &= \left|\boldsymbol{r}_1+\boldsymbol{a}\right\rangle \otimes \left|\boldsymbol{r}_2+\boldsymbol{a}\right\rangle \ldots \otimes \left|\boldsymbol{r}_n+\boldsymbol{a}\right\rangle \\ T^\dagger(\boldsymbol{a}) O\left(\boldsymbol{r}_1, \boldsymbol{r}_2, \ldots\right) T(\boldsymbol{a}) &= O\left(\boldsymbol{r}_1+\boldsymbol{a}, \boldsymbol{r}_2+\boldsymbol{a}, \ldots\right) , \end{aligned} \tag{3.1}$$

where $\left|\boldsymbol{r}_1, \boldsymbol{r}_2, \ldots\right\rangle$ is a state in the manyparticle Fock-space basis with the particles localised at the specified positions. For example, for a local fermionic creation operator $c^\dagger(\boldsymbol{r})$, we have

$$T^\dagger(\boldsymbol{a}) c^\dagger(\boldsymbol{r}) T(\boldsymbol{a}) = c^\dagger(\boldsymbol{r}+\boldsymbol{a}) . \tag{3.2}$$

It acts similarly on the auxiliary model Hamiltonian:

$$T^\dagger(\boldsymbol{a}) H_{\text{aux}}(\boldsymbol{r}_d) T(\boldsymbol{a}) = H_{\text{aux}}(\boldsymbol{r}_d+\boldsymbol{a}) , \tag{3.3}$$

translating all sites by the vector $\boldsymbol{a}$. By introducing the Fourier transform to momentum space,

$$\left|\boldsymbol{r}_1, \boldsymbol{r}_2, \ldots\right\rangle = \otimes_{j=1}^N \int d\boldsymbol{k}_j e^{i\boldsymbol{r}_j\cdot\boldsymbol{k}_j} \left|\boldsymbol{k}_j\right\rangle , \tag{3.4}$$

it is easy to see that the total momentum states are eigenstates of the global translation operators:

$$\begin{aligned} T^\dagger(\boldsymbol{a})\left|\boldsymbol{k}_1, \boldsymbol{k}_2, \ldots\right\rangle &= \otimes_{j=1}^N \int d\boldsymbol{r}_j e^{-i\boldsymbol{r}_j\cdot\boldsymbol{k}_j} T^\dagger(\boldsymbol{a})\left|\boldsymbol{r}_j\right\rangle , \\ &= \otimes_{j=1}^N \int d\boldsymbol{r}_j e^{-i\boldsymbol{r}_j\cdot\boldsymbol{k}_j} \left|\boldsymbol{r}_j+\boldsymbol{a}\right\rangle \\ &= \otimes_{j=1}^N e^{i\boldsymbol{a}\cdot\boldsymbol{k}_j} \int d\boldsymbol{r}_j e^{-i\boldsymbol{r}_j\cdot\boldsymbol{k}_j} \left|\boldsymbol{r}_j\right\rangle \\ &= e^{i\boldsymbol{a}\cdot\boldsymbol{k}_{\text{tot}}} \left|\boldsymbol{k}_1, \boldsymbol{k}_2, \ldots\right\rangle , \end{aligned} \tag{3.5}$$

where $\boldsymbol{k}_{\text{tot}} = \sum_j \boldsymbol{k}_j$ is the total momentum. This gives a compact form for the translation operator:

$$T^\dagger(\boldsymbol{a}) = e^{i\boldsymbol{a}\cdot\hat{\boldsymbol{k}}_{\text{tot}}} , \tag{3.6}$$

where $\hat{\boldsymbol{k}}_{\text{tot}}$ is the total momentum operator.

We end this section by calculating the effect of the translation operators on the $k-$space annihilation operators. We have

$$c(\boldsymbol{k}) = \frac{1}{\sqrt{N}} \sum_{\boldsymbol{r}} e^{-i\boldsymbol{k}\cdot\boldsymbol{r}} c(\boldsymbol{r}) . \tag{3.7}$$

Applying the translation operator to this decomposition, we get

$$\begin{aligned} T(\vec{a})\, c(\boldsymbol{k}) T^\dagger(\vec{a}) &= \frac{1}{\sqrt{N}} \sum_{\boldsymbol{r}} e^{-i\boldsymbol{k}\cdot\boldsymbol{r}} T(\vec{a})\, c(\boldsymbol{r}) T^\dagger(\vec{a}) \\ &= \frac{1}{\sqrt{N}} \sum_{\boldsymbol{r}} e^{-i\boldsymbol{k}\cdot\boldsymbol{r}} c(\boldsymbol{r}-\vec{a}) \\ &= e^{-i\boldsymbol{k}\cdot\vec{a}} c(\boldsymbol{k}) . \end{aligned} \tag{3.8}$$

2.3.2 Periodisation of Auxiliary Model Into Extended Hubbard Model

We recall the lattice-embedded impurity model defined in eq. 2.3. As discussed above, the route to restoring translation symmetry and reconstructing the lattice model is to translate the auxiliary model using manybody translation operators. Accordingly, we identify the "unit cell" of translation. Since we expect the impurity site to capture most of the physics arising from the correlated electrons, a good choice for the unit cell is the impurity site and it's surrounding correlated environment. In order to retain information of the lattice geometry, we convert the hybridisation and bath interaction terms from real space to $k-$space. The Hamiltonian for the unit cell is therefore

$$\begin{aligned} H_{\text{clust}}(\boldsymbol{r}_d) = &-\sum_{\boldsymbol{k},\sigma} e^{i\boldsymbol{k}\cdot\boldsymbol{r}_d} V(\boldsymbol{k})(c^\dagger_{\boldsymbol{r}_d,\sigma} c_{\boldsymbol{k},\sigma} + \text{h.c.}) + \sum_{\boldsymbol{k}_1,\boldsymbol{k}_2} e^{i(\boldsymbol{k}_1-\boldsymbol{k}_2)\cdot\boldsymbol{r}_d} J_K(\boldsymbol{k}_1,\boldsymbol{k}_2)\boldsymbol{S}_{\boldsymbol{r}_d}\cdot\boldsymbol{S}_{\boldsymbol{k}_1,\boldsymbol{k}_2} + \varepsilon_d n_{\boldsymbol{r}_d} \\ &+ U_d n_{\boldsymbol{r}_d,\uparrow} n_{\boldsymbol{r}_d,\downarrow} - \frac{1}{2}\sum_{\{\boldsymbol{k}_i\},\sigma,\sigma'} \sigma\,\sigma'\, e^{i(\boldsymbol{k}_1-\boldsymbol{k}_2+\boldsymbol{k}_3-\boldsymbol{k}_4)\cdot\boldsymbol{r}_d} W(\{\boldsymbol{k}_i\}) c^\dagger_{\boldsymbol{k}_1,\sigma} c_{\boldsymbol{k}_2,\sigma} c^\dagger_{\boldsymbol{k}_3,\sigma'} c_{\boldsymbol{k}_4,\sigma'} \ , \end{aligned} \tag{3.9}$$

where $\boldsymbol{S}_{\boldsymbol{k}_1,\boldsymbol{k}_2} = \sum_{\alpha,\beta}\boldsymbol{\sigma}_{\alpha,\beta} c^\dagger_{\boldsymbol{k}_1,\alpha} c_{\boldsymbol{k}_2,\beta}$, and the Fourier transforms are defined by the following relations:

$$\begin{aligned} \sum_{\boldsymbol{k}} e^{i\boldsymbol{k}\cdot\boldsymbol{r}_d} V(\boldsymbol{k}) c^\dagger_{\boldsymbol{r}_d,\sigma} c_{\boldsymbol{k},\sigma} &= \frac{V}{Z}\sum_{r\to\text{NN of }\boldsymbol{r}_d} c^\dagger_{\boldsymbol{r}_d,\sigma} c_{\boldsymbol{r},\sigma} \ , \\ \sum_{\boldsymbol{k}_1,\boldsymbol{k}_2} e^{i(\boldsymbol{k}_1-\boldsymbol{k}_2)\cdot\boldsymbol{r}_d} J_K(\boldsymbol{k}_1,\boldsymbol{k}_2)\vec{S}_{\boldsymbol{r}_d}\cdot\vec{S}_{\boldsymbol{k}_1,\boldsymbol{k}_2} &= \frac{J_K}{Z}\sum_{r\to\text{NN of }\boldsymbol{r}_d} \boldsymbol{S}_{\boldsymbol{r}_d}\cdot\boldsymbol{S}_{\boldsymbol{r}} \ , \\ \sum_{\boldsymbol{k}_1,\boldsymbol{k}_2,\boldsymbol{q}_1,\boldsymbol{q}_2} e^{i(\boldsymbol{k}_1-\boldsymbol{k}_2+\boldsymbol{k}_3-\boldsymbol{k}_4)\cdot\boldsymbol{r}_d} W(\{\boldsymbol{k}_i\}) c^\dagger_{\boldsymbol{k}_1,\sigma} c_{\boldsymbol{k}_2,\sigma} c^\dagger_{\boldsymbol{k}_3,\sigma'} c_{\boldsymbol{k}_4,\sigma'} &= \frac{W}{2Z}\sum_{r\to\text{NN of }\boldsymbol{r}_d} n_{\boldsymbol{r},\sigma} n_{\boldsymbol{r},\sigma'} \ , \end{aligned} \tag{3.10}$$

We now translate this unit cell to all sites of the lattice, and reconstruct a translationally-invariant correlated lattice model:

$$H_{\text{latt}} = \sum_{\boldsymbol{r}} T^\dagger(\boldsymbol{r}-\boldsymbol{r}_d) H_{\text{clust}}(\boldsymbol{r}_d) T(\boldsymbol{r}-\boldsymbol{r}_d) \ . \tag{3.11}$$

We consider the effect of the translation operations on each part of the Hamiltonian. The purely local terms translate directly into a chemical potential term and a Hubbard interaction term:

$$\sum_{\boldsymbol{r}} T^\dagger(\boldsymbol{r}-\boldsymbol{r}_d)(\varepsilon_d\sum_{\sigma} n_{\boldsymbol{r}_d,\sigma} + U_d n_{\boldsymbol{r}_d,\uparrow} n_{\boldsymbol{r}_d,\downarrow}) T(\boldsymbol{r}-\boldsymbol{r}_d) = \varepsilon_d \sum_{\boldsymbol{r},\sigma} n_{\boldsymbol{r},\sigma} + U_d\sum_{\boldsymbol{r}} n_{\boldsymbol{r},\uparrow} n_{\boldsymbol{r},\downarrow} \ . \tag{3.12}$$

The one-particle hybridisation term maps onto a tight-binding hopping term:

$$\begin{aligned} \sum_{\boldsymbol{r}} T^\dagger(\boldsymbol{r}-\boldsymbol{r}_d)\sum_{\boldsymbol{k}} e^{i\boldsymbol{k}\cdot\boldsymbol{r}_d} V(\boldsymbol{k}) c^\dagger_{\boldsymbol{r}_d,\sigma} c_{\boldsymbol{k},\sigma} T(\boldsymbol{r}-\boldsymbol{r}_d) &= \sum_{\boldsymbol{r}}\sum_{\boldsymbol{k}} e^{i\boldsymbol{k}\cdot\boldsymbol{r}} V(\boldsymbol{k}) c^\dagger_{\boldsymbol{r},\sigma} c_{\boldsymbol{k},\sigma} \\ &= \frac{V}{Z}\sum_{\boldsymbol{r}}\sum_{\boldsymbol{r}'\to\text{NN of }\boldsymbol{r}} c^\dagger_{\boldsymbol{r},\sigma} c_{\boldsymbol{r}',\sigma} \\ &= \frac{2V}{Z}\sum_{\langle\boldsymbol{r},\boldsymbol{r}'\rangle} c^\dagger_{\boldsymbol{r},\sigma} c_{\boldsymbol{r}',\sigma} \ , \end{aligned} \tag{3.13}$$

where we used eq. 3.8 and the first equation from 3.10, and the fact that each nearest-neighbour pair $\boldsymbol{r}, \boldsymbol{r}'$ appears twice in the summation.

The Kondo terms map onto a nearest-neighbour spin-exchange term:

$$\begin{aligned}\sum_{\boldsymbol{r},\boldsymbol{k}_1,\boldsymbol{k}_2} T^\dagger(\boldsymbol{r}-\boldsymbol{r}_d)e^{i(\boldsymbol{k}_1-\boldsymbol{k}_2)\cdot\boldsymbol{r}_d} J_K(\boldsymbol{k}_1,\boldsymbol{k}_2)\vec{S}_{\boldsymbol{r}_d}\cdot\vec{S}_{\boldsymbol{k}_1,\boldsymbol{k}_2}T(\boldsymbol{r}-\boldsymbol{r}_d) &= \sum_{\boldsymbol{r},\boldsymbol{k}_1,\boldsymbol{k}_2} e^{i(\boldsymbol{k}_1-\boldsymbol{k}_2)\cdot\boldsymbol{r}} J_K(\boldsymbol{k}_1,\boldsymbol{k}_2)\vec{S}_{\boldsymbol{r}}\cdot\vec{S}_{\boldsymbol{k}_1,\boldsymbol{k}_2} \\ &= \frac{J_K}{Z}\sum_{\boldsymbol{r}}\sum_{\boldsymbol{r}'\to\text{NN of }\boldsymbol{r}} \vec{S}_{\boldsymbol{r}}\cdot\vec{S}_{\boldsymbol{r}'}~, \\ &= \frac{2J_K}{Z}\sum_{\langle\boldsymbol{r},\boldsymbol{r}'\rangle}\vec{S}_{\boldsymbol{r}}\cdot\vec{S}_{\boldsymbol{r}'}~,\end{aligned} \tag{3.14}$$

where we used the second equation from 3.10. Finally, the bath interaction term W maps onto another Hubbard term

$$\begin{aligned}&\sum_{\boldsymbol{r},\boldsymbol{k}_1,\boldsymbol{k}_2,\boldsymbol{k}_3,\boldsymbol{k}_4} T^\dagger(\boldsymbol{r}-\boldsymbol{r}_d)e^{i(\boldsymbol{k}_1-\boldsymbol{k}_2+\boldsymbol{k}_3-\boldsymbol{k}_4)\cdot\boldsymbol{r}_d}W(\{\boldsymbol{k}_i\})c^\dagger_{\boldsymbol{k}_1,\sigma}c_{\boldsymbol{k}_2,\sigma}c^\dagger_{\boldsymbol{k}_3,\sigma'}c_{\boldsymbol{k}_4,\sigma'}T(\boldsymbol{r}-\boldsymbol{r}_d) \\ &= \sum_{\boldsymbol{r},\boldsymbol{k}_1,\boldsymbol{k}_2,\boldsymbol{k}_3,\boldsymbol{k}_4} e^{i(\boldsymbol{k}_1-\boldsymbol{k}_2+\boldsymbol{k}_3-\boldsymbol{k}_4)\cdot\boldsymbol{r}}W(\{\boldsymbol{k}_i\})c^\dagger_{\boldsymbol{k}_1,\sigma}c_{\boldsymbol{k}_2,\sigma}c^\dagger_{\boldsymbol{k}_3,\sigma'}c_{\boldsymbol{k}_4,\sigma'} \\ &= \frac{W}{2Z}\sum_{\boldsymbol{r}}\sum_{\boldsymbol{r}'\to\text{NN of }\boldsymbol{r}} n_{\boldsymbol{r}',\sigma}n_{\boldsymbol{r}',\sigma'} \\ &= \frac{W}{2}\sum_{\boldsymbol{r}} n_{\boldsymbol{r},\sigma}n_{\boldsymbol{r},\sigma'}~,\end{aligned} \tag{3.15}$$

where we used the fact that each point $\boldsymbol{r}'$ (which we relabel as r) appears Z times in the summation.

The complete lattice Hamiltonian, after applying the translation operators, is

$$H_\text{latt} = \varepsilon_d\sum_{\boldsymbol{r},\sigma}n_{\boldsymbol{r},\sigma} + U_d\sum_{\boldsymbol{r}}n_{\boldsymbol{r},\uparrow}n_{\boldsymbol{r},\downarrow} - \frac{2V}{Z}\sum_{\langle\boldsymbol{r},\boldsymbol{r}'\rangle}c^\dagger_{\boldsymbol{r},\sigma}c_{\boldsymbol{r}',\sigma} + \frac{2J_K}{Z}\sum_{\langle\boldsymbol{r},\boldsymbol{r}'\rangle}\vec{S}_{\boldsymbol{r}}\cdot\vec{S}_{\boldsymbol{r}'} - \frac{W}{2}\sum_{\boldsymbol{r}}\left(n_{\boldsymbol{r},\uparrow}-n_{\boldsymbol{r},\downarrow}\right)^2~. \tag{3.16}$$

This is clearly an extended Hubbard model, with both 1-particle and spin-exchange nearest-neighbour hybridisation. The general form of the model is

$$H_\text{ex-Hub} = -t\sum_{\langle i,j\rangle,\sigma}(c^\dagger_{i,\sigma}c_{j,\sigma}+\text{h.c.}) + J\sum_{\langle i,j\rangle}\boldsymbol{S}_i\cdot\boldsymbol{S}_j + U\sum_i n_{i,\uparrow}n_{i,\downarrow} - \mu\sum_{i,\sigma}n_{i,\sigma}~, \tag{3.17}$$

such that the effective couplings resulting from the tiling operation is

$$t = \frac{2V}{Z},~J = \frac{2J_K}{Z},~U = U_d - W,~\mu = -\varepsilon_d + \frac{W}{2}~. \tag{3.18}$$

2.3.3 Form Of The Eigenstates: Bloch's Theorem

The tiled Hamiltonian defined in eq. 3.11 is manifestly symmetric under discrete translation, given that it's expressed in terms of the translation operators. The fact that the Hamiltonian H_latt commutes with the manybody translation operator implies that the total crystal momentum is a conserved quantity. We will now use this to write down a "tiling" relation between the eigenstates of the auxiliary model and those of the lattice model.

In the tight-binding approach to lattice problems, the full Hamiltonian is obtained by translating the dynamics of the localised orbitals at each site over the entire lattice. The full eigenstate $|\Psi\rangle$ is obtained by constructing liner combinations of the eigenstates $|\psi(i)\rangle$ of the local Hamiltonians. According to Bloch's theorem, $|\Psi\rangle$ is of the form $|\Psi_{n,\boldsymbol{K}}\rangle = \sum_i e^{i\boldsymbol{K}\cdot\boldsymbol{r}_i} |\psi_n(i)\rangle$, where $\boldsymbol{K}$ is the single-particle crystal momentum, n is the band-index that labels various states within the local orbital, and $\boldsymbol{r}_i$ sums over the positions of the local Hamiltonians. Bloch's theorem ensures that eigenstates satisfy the following relation under a translation operation by an arbitrary number of lattice spacings $n\boldsymbol{a}$:

$$T^\dagger(l\boldsymbol{a})\,|\Psi_{n,\boldsymbol{K}}\rangle = \sum_i e^{i\boldsymbol{K}\cdot\boldsymbol{r}_i}\,|\psi_n(i+l)\rangle = e^{-il\boldsymbol{K}\cdot\boldsymbol{a}}\,|\Psi_{n,\boldsymbol{K}}\rangle\ . \tag{3.19}$$

As discussed earlier, the eigenstates $|\Psi_{n,\boldsymbol{K}}\rangle$ of the lattice Hamiltonian obtained using eq. 3.11 enjoy the discrete translation symmetry of the lattice, and therefore enjoy a *many-body* Bloch's theorem [205]. This means that the *local* eigenstates $|\psi_n\left(\boldsymbol{r}_d\right)\rangle$ (with the impurity located at an arbitrary position $\boldsymbol{r}_d$) of the unit cell auxiliary model Hamiltonian $H_{\text{clust}}(\boldsymbol{r}_d)$ defined in eq. 3.9 can be used to construct eigenstates of the lattice Hamiltonian. The index $n(=0,1,\ldots)$ in the subscript indicates that it is the n^{th} eigenstate of the auxiliary model. The following combination of the auxiliary model eigenstates satisfies a many-particle equivalent of Bloch's theorem [205]:

$$|\Psi_s\rangle \equiv |\Psi_{\boldsymbol{K},n}\rangle = \frac{1}{\sqrt{N}}\sum_{\boldsymbol{r}_d} e^{i\boldsymbol{K}\cdot\boldsymbol{r}_d}\,|\psi_n\left(\boldsymbol{r}_d\right)\rangle\ , \tag{3.20}$$

where N is the total number of lattice sites and $\boldsymbol{r}_d$ is summed over all lattice spacings. The set of $n=0$ states form the lowest band in the spectrum of the lattice, while higher values of n produce the more energetic bands.

2.4 Properties of the Bloch states

2.4.1 Translation invariance

It is easy to verify that $\Psi_{\vec{k}}\left(\{\boldsymbol{r}_k\}\right)$ transforms like Bloch functions under translation by a displacement $\boldsymbol{r}$:

$$\begin{aligned}\Psi_{\vec{k}}\left(\{\boldsymbol{r}_k+\boldsymbol{r}\}\right) = \frac{1}{\lambda_{|\vec{k}\rangle}\mathcal{Z}N}\sum_{\boldsymbol{r}_i,\boldsymbol{a}} e^{i\vec{k}\cdot\vec{R}_i}\psi_{\text{aux}}\left(\boldsymbol{r}_i-\boldsymbol{r},\boldsymbol{a}\right) &= \frac{1}{\lambda_{|\vec{k}\rangle}\mathcal{Z}N}\sum_{\boldsymbol{r}_j,\boldsymbol{a}} e^{i\vec{k}\cdot\left(\vec{R}_j+\boldsymbol{r}\right)}\psi_{\text{aux}}\left(\boldsymbol{r}_j,\boldsymbol{a}\right)\\ &= e^{i\vec{k}\cdot\vec{R}}\Psi_{\vec{k}}\left(\{\boldsymbol{r}_k\}\right)\end{aligned} \tag{4.1}$$

In the last equation, we transformed $\boldsymbol{r}_i \to \boldsymbol{r}_j = \boldsymbol{r}_i - \boldsymbol{r}$. Note that the argument $\boldsymbol{a}$ does not change under a translation of the system, because that vector always represents the difference between the impurity lattice position and its nearest neighbours, irrespective of the absolute position of the impurity.

The wavefunction can be even brought into the familiar Bloch function form:

$$\begin{aligned}\Psi^n_{\vec{k}}\left(\{\boldsymbol{r}_k\}\right) &= \sum_{\boldsymbol{r}_i,\boldsymbol{a}} \frac{e^{i\vec{k}\cdot\vec{R}_i}}{\lambda_{|\vec{k}\rangle}\mathcal{Z}N}\psi_{\text{aux}}\left(\{\boldsymbol{r}_d-\boldsymbol{r}_i, \boldsymbol{r}_0-\boldsymbol{r}_i-\boldsymbol{a}\}\right) \\ &= \frac{e^{i\vec{k}\cdot\frac{1}{N}\sum_k\vec{r}_k}}{\lambda_{|\vec{k}\rangle}\mathcal{Z}N}\sum_{\boldsymbol{r}_i,\boldsymbol{a}} e^{-i\vec{k}\cdot\left(\frac{1}{N}\sum_k\boldsymbol{r}_k-\vec{R}_i\right)}\psi_{\text{aux}}\left(\{\boldsymbol{r}_d-\boldsymbol{r}_i, \boldsymbol{r}_0-\boldsymbol{r}_i-\boldsymbol{a}\}\right) \\ &= e^{i\vec{k}\cdot\vec{r}_{\text{COM}}}\eta_{\vec{k}}\left(\{\boldsymbol{r}_k\}\right)\end{aligned} \tag{4.2}$$

where $\boldsymbol{r}_{\text{COM}} = \frac{1}{N}\sum_k \boldsymbol{r}_k$ is the center-of-mass coordinate and

$$\eta_{\vec{k}}\left(\{\boldsymbol{r}_k\}\right) = \frac{1}{\lambda_{|\vec{k}\rangle}\mathcal{Z}N}\sum_{\boldsymbol{r}_i} e^{-i\vec{k}\cdot\left(\frac{1}{N}\sum_k\boldsymbol{r}_k-\vec{R}_i\right)}\psi^n_{\text{aux}}\left(\{\boldsymbol{r}_k-\boldsymbol{r}_i\}\right)\ , \tag{4.3}$$

is the translation symmetric function. This form of the eigenstate allows the interpretation that tuning the Bloch momentum $\vec{k}$ corresponds to a translation of the center of mass of the system (or in this case, of the auxiliary models that comprise the system).

2.4.2 Orthonormality

It is straightforward to show that these states form an orthonormal basis. We start by writing down the inner product of two distinct such states:

$$\begin{aligned}\langle\Psi_{\vec{k}'}|\Psi_{\vec{k}}\rangle &= \frac{1}{\lambda^*_{|\vec{k}'\rangle}\lambda_{|\vec{k}\rangle}\mathcal{Z}^2N^2}\sum_{\boldsymbol{r}_i,\boldsymbol{r}_j,\boldsymbol{a},\boldsymbol{a}'} e^{i\left(\vec{k}\cdot\vec{R}_i-\vec{k}'\cdot\vec{R}_j\right)}\left\langle\psi_{\text{aux}}\left(\boldsymbol{r}_j,\boldsymbol{a}'\right)|\psi_{\text{aux}}\left(\boldsymbol{r}_i,\boldsymbol{a}\right)\right\rangle \\ &= \frac{1}{\lambda^*_{|\vec{k}'\rangle}\lambda_{|\vec{k}\rangle}\mathcal{Z}^2N^2}\sum_{\boldsymbol{r}_i,\boldsymbol{r}_j,\boldsymbol{a},\boldsymbol{a}'} e^{i\left(\vec{k}\cdot\vec{R}_i-\vec{k}'\cdot\vec{R}_j\right)}\left\langle\psi_{\text{aux}}\left(\boldsymbol{r}_0,\boldsymbol{a}'\right)T^\dagger\left(\boldsymbol{r}_0-\boldsymbol{r}_j\right)T\left(\boldsymbol{r}_0-\boldsymbol{r}_i\right)|\psi_{\text{aux}}\left(\boldsymbol{r}_0,\boldsymbol{a}\right)\right\rangle\end{aligned} \tag{4.4}$$

At this point, we insert a complete basis of momentum eigenkets $1 = \sum_{\vec{q}}|\vec{q}\rangle\langle\vec{q}|$ to resolve the translation operators:

$$\begin{aligned}\langle\Psi_{\vec{k}'}|\Psi_{\vec{k}}\rangle &= \frac{1}{\lambda^*_{|\vec{k}'\rangle}\lambda_{|\vec{k}\rangle}\mathcal{Z}^2N^2}\sum_{\boldsymbol{r}_i,\boldsymbol{r}_j,\boldsymbol{a},\boldsymbol{a}',\vec{q}} e^{i\left(\vec{k}\cdot\vec{R}_i-\vec{k}'\cdot\vec{R}_j\right)}\left\langle\psi_{\text{aux}}\left(\boldsymbol{r}_0,\boldsymbol{a}'\right)T\left(\boldsymbol{r}_j-\boldsymbol{r}_i\right)|\vec{q}\rangle\langle\vec{q}|\,\psi_{\text{aux}}\left(\boldsymbol{r}_0,\boldsymbol{a}\right)\right\rangle \\ &= \frac{1}{\lambda^*_{|\vec{k}'\rangle}\lambda_{|\vec{k}\rangle}\mathcal{Z}^2N^2}\sum_{\boldsymbol{r}_i,\boldsymbol{r}_j,\boldsymbol{a},\boldsymbol{a}',\vec{q}} e^{i\left(\vec{k}\cdot\vec{R}_i-\vec{k}'\cdot\vec{R}_j\right)}\left\langle\psi_{\text{aux}}\left(\boldsymbol{r}_0,\boldsymbol{a}'\right)e^{i\vec{q}\cdot\left(\boldsymbol{r}_j-\boldsymbol{r}_i\right)}|\vec{q}\rangle\langle\vec{q}|\,\psi_{\text{aux}}\left(\boldsymbol{r}_0,\boldsymbol{a}\right)\right\rangle \\ &= \frac{1}{|\lambda_{|\vec{k}\rangle}|^2\mathcal{Z}^2}\delta_{\vec{k},\vec{k}'}\sum_{\boldsymbol{a},\boldsymbol{a}'}\left\langle\psi_{\text{aux}}\left(\boldsymbol{r}_0,\boldsymbol{a}'\right)|\vec{k}\rangle\langle\vec{k}|\,\psi_{\text{aux}}\left(\boldsymbol{r}_0,\boldsymbol{a}\right)\right\rangle\end{aligned} \tag{4.5}$$

At the last step, we used the summation form of the Kronecker delta function: $\sum_{\boldsymbol{r}_i} e^{i\boldsymbol{r}_i\left(\vec{k}-\vec{q}\right)}\sum_{\boldsymbol{r}_j} e^{i\boldsymbol{r}_j\left(\vec{q}-\vec{k}'\right)} = N\delta_{\vec{k},\vec{q}}N\delta_{\vec{k}',\vec{q}}$. The final remaining step is to identify that the inner products inside the summation are actually independent of the direction $\boldsymbol{a}, \boldsymbol{a}'$ of the connecting vector, and both are equal to the normalisation factor $\lambda_{|\vec{k}\rangle}$:

$$\langle \Psi_{\vec{k}'} | \Psi_{\vec{k}} \rangle = \frac{1}{|\lambda_{|\vec{k}\rangle}|^2 \mathcal{Z}^2} \delta_{\vec{k},\vec{k}'} \sum_{\boldsymbol{a},\boldsymbol{a}'} |\lambda_{|\vec{k}\rangle}|^2 = \frac{1}{|\lambda_{|\vec{k}\rangle}|^2 \mathcal{Z}^2} \delta_{\vec{k},\vec{k}'} w^2 |\lambda_{|\vec{k}\rangle}|^2 = \delta_{\vec{k},\vec{k}'} \tag{4.6}$$

This concludes the proof of orthonormality.

2.4.3 Eigenstates

To demonstrate that these are indeed eigenstates of the bulk Hamiltonian, we will calculate the matrix elements of the full Hamiltonian between these states:

$$\begin{aligned}
\langle \Psi_{\vec{k}'} | H_{\text{H-H}} | \Psi_{\vec{k}} \rangle &= \frac{1}{\lambda^*_{|\vec{k}'\rangle} \lambda_{|\vec{k}\rangle} \mathcal{Z}^2 N^2} \sum_{\boldsymbol{r}_i,\boldsymbol{r}_j,\boldsymbol{r}_k,\boldsymbol{a},\boldsymbol{a}'} e^{i\left(\vec{k}\cdot\vec{R}_i - \vec{k}'\cdot\vec{R}_j\right)} \langle \psi_{\text{aux}} (\boldsymbol{r}_j, \boldsymbol{a}') | H_{\text{aux}} (\boldsymbol{r}_k) | \psi_{\text{aux}} (\boldsymbol{r}_i, \boldsymbol{a}) \rangle \\
&= \frac{1}{\lambda^*_{|\vec{k}'\rangle} \lambda_{|\vec{k}\rangle} \mathcal{Z}^2 N^2} \sum_{\boldsymbol{r}_i,\boldsymbol{r}_j,\boldsymbol{r}_k,\boldsymbol{a},\boldsymbol{a}'} e^{i\left(\vec{k}\cdot\vec{R}_i - \vec{k}'\cdot\vec{R}_j\right)} \langle \psi_{\text{aux}} (\boldsymbol{r}_0, \boldsymbol{a}') | T^\dagger_{\boldsymbol{r}_0 - \boldsymbol{r}_j} T_{\boldsymbol{r}_0 - \boldsymbol{r}_k} H_{\text{aux}} (\boldsymbol{r}_0) T^\dagger_{\boldsymbol{r}_0 - \boldsymbol{r}_k} T_{\boldsymbol{r}_0 - \boldsymbol{r}_i} | \psi_{\text{aux}} (\boldsymbol{r}_0, \boldsymbol{a}) \rangle \\
&= \frac{1}{\lambda^*_{|\vec{k}'\rangle} \lambda_{|\vec{k}\rangle} \mathcal{Z}^2 N^2} \sum_{\boldsymbol{r}_i,\boldsymbol{r}_j,\boldsymbol{r}_k,\boldsymbol{a},\boldsymbol{a}'} e^{i\left(\vec{k}\cdot\vec{R}_i - \vec{k}'\cdot\vec{R}_j\right)} \langle \psi_{\text{aux}} (\boldsymbol{r}_0, \boldsymbol{a}') | T^\dagger_{\boldsymbol{r}_k - \boldsymbol{r}_j} H_{\text{aux}} (\boldsymbol{r}_0) T_{\boldsymbol{r}_k - \boldsymbol{r}_i} | \psi_{\text{aux}} (\boldsymbol{r}_0, \boldsymbol{a}) \rangle
\end{aligned} \tag{4.7}$$

To resolve the translation operators, we will now insert the momentum eigenstates $|\vec{k}\rangle$ of the full model: $1 = \sum_{\vec{q}} |\vec{q}\rangle \langle \vec{q}|$ in between the operators.

$$\begin{aligned}
\langle \Psi_{\vec{k}'} | H_{\text{H-H}} | \Psi_{\vec{k}} \rangle &= \sum_{\substack{\boldsymbol{r}_i,\boldsymbol{r}_j,\boldsymbol{r}_k,\\ \boldsymbol{a},\boldsymbol{a}',\vec{q},\vec{q}'}} \frac{e^{i\left(\vec{k}\cdot\vec{R}_i - \vec{k}'\cdot\vec{R}_j\right)}}{\lambda^*_{|\vec{k}'\rangle} \lambda_{|\vec{k}\rangle} \mathcal{Z}^2 N^2} \langle \psi_{\text{aux}} (\boldsymbol{r}_0, \boldsymbol{a}') | \vec{q}' \rangle \langle \vec{q}' | T^\dagger_{\boldsymbol{r}_k - \boldsymbol{r}_j} H_{\text{aux}} (\boldsymbol{r}_0) T_{\boldsymbol{r}_k - \boldsymbol{r}_i} | \vec{q} \rangle \langle \vec{q} | \psi_{\text{aux}} (\boldsymbol{r}_0, \boldsymbol{a}) \rangle \\
&= \sum_{\substack{\boldsymbol{r}_i,\boldsymbol{r}_j,\boldsymbol{r}_k,\\ \boldsymbol{a},\boldsymbol{a}',\vec{q},\vec{q}'}} \frac{e^{i\left(\vec{k}\cdot\vec{R}_i - \vec{k}'\cdot\vec{R}_j\right)}}{\lambda^*_{|\vec{k}'\rangle} \lambda_{|\vec{k}\rangle} \mathcal{Z}^2 N^2} \langle \psi_{\text{aux}} (\boldsymbol{r}_0, \boldsymbol{a}') | \vec{q}' \rangle e^{-i\vec{q}'\cdot(\boldsymbol{r}_k - \boldsymbol{r}_j)} e^{i\vec{q}\cdot(\boldsymbol{r}_k - \boldsymbol{r}_i)} \langle \vec{q}' | H_{\text{aux}} (\boldsymbol{r}_0) | \vec{q} \rangle \langle \vec{q} | \psi_{\text{aux}} (\boldsymbol{r}_0, \boldsymbol{a}) \rangle \\
&= \frac{N}{\lambda^*_{|\vec{k}'\rangle} \lambda_{|\vec{k}\rangle} \mathcal{Z}^2} \sum_{\boldsymbol{a},\boldsymbol{a}',\vec{q},\vec{q}'} \delta_{\vec{k},\vec{q}} \delta_{\vec{q},\vec{q}'} \delta_{\vec{q}',\vec{k}} \langle \psi_{\text{aux}} (\boldsymbol{r}_0, \boldsymbol{a}') | \vec{q}' \rangle \langle \vec{q}' | H_{\text{aux}} (\boldsymbol{r}_0) | \vec{q} \rangle \langle \vec{q} | \psi_{\text{aux}} (\boldsymbol{r}_0, \boldsymbol{a}) \rangle \\
&= \frac{N}{|\lambda_{|\vec{k}\rangle}|^2 \mathcal{Z}^2} \delta_{kk'} \sum_{\boldsymbol{a},\boldsymbol{a}'} \langle \psi_{\text{aux}} (\boldsymbol{r}_0, \boldsymbol{a}') | \vec{k} \rangle \langle \vec{k} | H_{\text{aux}} (\boldsymbol{r}_0) | \vec{k} \rangle \langle \vec{k} | \psi_{\text{aux}} (\boldsymbol{r}_0, \boldsymbol{a}) \rangle \\
&= \frac{N}{|\lambda_{|\vec{k}\rangle}|^2 \mathcal{Z}^2} \delta_{kk'} \langle \vec{k} | H_{\text{aux}} (\boldsymbol{r}_0) | \vec{k} \rangle \underbrace{\sum_{\boldsymbol{a},\boldsymbol{a}'} \langle \psi_{\text{aux}} (\boldsymbol{r}_0, \boldsymbol{a}') | \vec{k} \rangle \langle \vec{k} | \psi_{\text{aux}} (\boldsymbol{r}_0, \boldsymbol{a}) \rangle}_{w^2 |\lambda_{|\vec{k}\rangle}|^2} \\
&= N \delta_{kk'} \langle \vec{k} | H_{\text{aux}} (\boldsymbol{r}_0) | \vec{k} \rangle
\end{aligned} \tag{4.8}$$

The momentum eigenket $|\vec{k}\rangle$ can be expressed as a sum over position eigenkets at varying distances from $\boldsymbol{r}_0$. This form allows a systematic method of improving the energy eigenvalue estimate, because the interacting in the impurity model is extremely localised, so the overlap between two auxiliary models decreases rapidly with distance. The majority of the contribution will come from the state at $\boldsymbol{r}_0$, and improvements are then made by considering auxiliary models at further distances.

2.5 Mapping Greens Functions from Auxiliary Model to Lattice Model

2.5.1 One-particle $k-$space Greens functions

In the previous part, we proposed a form for the eigenstates $|\Psi_s\rangle$ of the bulk Hamiltonian in terms of the ground-states $|\psi_n\rangle$ of the auxiliary models. In this section, we will relate one-particle Greens functions of the lattice model to those of the auxiliary model. We define the retarded time-domain lattice $k-$space Greens function at zero temperature as

$$G(\boldsymbol{k}\sigma;t) = -i\theta(t)\left\langle\Psi_{\text{gs}}|\{c_{\boldsymbol{k}\sigma}(t), c^\dagger_{\boldsymbol{k}\sigma}\}|\Psi_{\text{gs}}\right\rangle \ . \tag{5.1}$$

where $|\Psi_{\text{gs}}\rangle$ is the lattice groundstate. Time evolution of the operators is governed by the lattice Hamiltonian H_{latt}:

$$c_{\boldsymbol{k}\sigma}(t) = e^{itH_{\text{latt}}}c_{\boldsymbol{k}\sigma}e^{-itH_{\text{latt}}} \ . \tag{5.2}$$

By using eq. 3.8 and the fact that the translation operators commute with H_{latt}, we have the relation

$$T\left(\vec{a}\right)c(\boldsymbol{k})(t)T^\dagger\left(\vec{a}\right) = e^{-i\boldsymbol{k}\cdot\vec{a}}c(\boldsymbol{k})(t) \ , \tag{5.3}$$

which combined with eq. 3.8 gives

$$T\left(\boldsymbol{a}\right)c(\boldsymbol{k})(t)c^\dagger(\boldsymbol{k})T^\dagger\left(\boldsymbol{b}\right) = e^{i\boldsymbol{k}\cdot(\boldsymbol{b}-\boldsymbol{a})}c(\boldsymbol{k})(t)T^\dagger(\boldsymbol{b}-\boldsymbol{a})c^\dagger(\boldsymbol{k}) \ . \tag{5.4}$$

In order to make use of the mapping between auxiliary model and lattice model Hamiltonians, we would like to express this lattice Greens function in terms of auxiliary model Greens functions – objects that can be computed purely from within a single auxiliary model. This will be a major simplification in our computations, because it is easier to work with a quantum impurity model, as compared to an interacting lattice model.

To make progress towards this, we consider the equivalent object

$$G(\boldsymbol{k}\sigma;t) = -i\theta(t)\frac{1}{2}\left[\left\langle\Psi_{\text{gs}}|\{c_{\boldsymbol{k}\sigma}(t), c^\dagger_{\boldsymbol{k}\sigma}\}|\Psi_{\text{gs}}\right\rangle + \left\langle\Psi_{\text{gs}}|\{c_{\boldsymbol{k}\sigma}, c^\dagger_{\boldsymbol{k}\sigma}(-t)\}|\Psi_{\text{gs}}\right\rangle\right] \ . \tag{5.5}$$

We consider only the first term for now. We express the lattice groundstate in terms of the auxiliary model groundstates (using eq. 3.20):

$$\left\langle\Psi_{\text{gs}}|\{c_{\boldsymbol{k}\sigma}(t), c^\dagger_{\boldsymbol{k}\sigma}\}|\Psi_{\text{gs}}\right\rangle = \frac{1}{N}\sum_{\boldsymbol{r}_1,\boldsymbol{r}_2}\left\langle\psi_{\text{gs}}(\boldsymbol{r}_2)|\{c_{\boldsymbol{k}\sigma}(t), c^\dagger_{\boldsymbol{k}\sigma}\}|\psi_{\text{gs}}(\boldsymbol{r}_1)\right\rangle \ . \tag{5.6}$$

We take one of the terms in the anticommutator and proceed to simplify it:

$$\begin{aligned}\sum_{\boldsymbol{r}_1,\boldsymbol{r}_2} \langle\psi_{\text{gs}}(\boldsymbol{r}_2)|c_{\boldsymbol{k}\sigma}(t)c^\dagger_{\boldsymbol{k}\sigma}|\psi_{\text{gs}}(\boldsymbol{r}_1)\rangle &= \sum_{\boldsymbol{r}_1,\boldsymbol{r}_2} \langle\psi_{\text{gs}}(\boldsymbol{r}_d)|T(\boldsymbol{r}_2-\boldsymbol{r}_d)c_{\boldsymbol{k}\sigma}(t)c^\dagger_{\boldsymbol{k}\sigma}T^\dagger(\boldsymbol{r}_1-\boldsymbol{r}_d)|\psi_{\text{gs}}(\boldsymbol{r}_d)\rangle \\ &= \sum_{\boldsymbol{r}_1,\boldsymbol{r}_2} e^{i\boldsymbol{k}(\boldsymbol{r}_1-\boldsymbol{r}_2)}\langle\psi_{\text{gs}}(\boldsymbol{r}_d)|c_{\boldsymbol{k}\sigma}(t)T^\dagger(\boldsymbol{r}_1-\boldsymbol{r}_2)c^\dagger_{\boldsymbol{k}\sigma}|\psi_{\text{gs}}(\boldsymbol{r}_d)\rangle \\ &= N\sum_{\boldsymbol{r}} e^{i\boldsymbol{k}\cdot\boldsymbol{r}}\langle\psi_{\text{gs}}(\boldsymbol{r}_d)|c_{\boldsymbol{k}\sigma}(t)T^\dagger(\boldsymbol{r})c^\dagger_{\boldsymbol{k}\sigma}|\psi_{\text{gs}}(\boldsymbol{r}_d)\rangle \ .\end{aligned} \tag{5.7}$$

where $\boldsymbol{r}_d$ is a reference lattice site, and we used eq. 5.4 in the second step. The other term in the anticommutator gives

$$\begin{aligned}\sum_{\boldsymbol{r}_1,\boldsymbol{r}_2} \langle\psi_{\text{gs}}(\boldsymbol{r}_2)|c^\dagger_{\boldsymbol{k}\sigma}c_{\boldsymbol{k}\sigma}(t)|\psi_{\text{gs}}(\boldsymbol{r}_1)\rangle &= \sum_{\boldsymbol{r}_1,\boldsymbol{r}_2} \langle\psi_{\text{gs}}(\boldsymbol{r}_d)|T(\boldsymbol{r}_2-\boldsymbol{r}_d)c^\dagger_{\boldsymbol{k}\sigma}c_{\boldsymbol{k}\sigma}(t)T^\dagger(\boldsymbol{r}_1-\boldsymbol{r}_d)|\psi_{\text{gs}}(\boldsymbol{r}_d)\rangle \\ &= \sum_{\boldsymbol{r}_1,\boldsymbol{r}_2} e^{-i\boldsymbol{k}(\boldsymbol{r}_1-\boldsymbol{r}_2)}\langle\psi_{\text{gs}}(\boldsymbol{r}_d)|c^\dagger_{\boldsymbol{k}\sigma}T^\dagger(\boldsymbol{r}_1-\boldsymbol{r}_2)c_{\boldsymbol{k}\sigma}(t)|\psi_{\text{gs}}(\boldsymbol{r}_d)\rangle \\ &= N\sum_{\boldsymbol{r}} e^{-i\boldsymbol{k}\cdot\boldsymbol{r}}\langle\psi_{\text{gs}}(\boldsymbol{r}_d)|c^\dagger_{\boldsymbol{k}\sigma}T^\dagger(\boldsymbol{r})c_{\boldsymbol{k}\sigma}(t)|\psi_{\text{gs}}(\boldsymbol{r}_d)\rangle \ .\end{aligned} \tag{5.8}$$

To resolve the time evolution of the operators, we insert $1 = \sum_m |\psi_m(\boldsymbol{r}_d)\rangle\langle\psi_m(\boldsymbol{r}_d)|$ into the equation. We again work with just the first term of the anticommutator for convenience:

$$\begin{aligned}&\sum_{\boldsymbol{r}_1,\boldsymbol{r}_2} \langle\psi_{\text{gs}}(\boldsymbol{r}_2)|c_{\boldsymbol{k}\sigma}(t)c^\dagger_{\boldsymbol{k}\sigma}|\psi_{\text{gs}}(\boldsymbol{r}_1)\rangle \\ &= N\sum_{\boldsymbol{r},n_1,n_2} e^{i\boldsymbol{k}\cdot\boldsymbol{r}}\langle\psi_{\text{gs}}(\boldsymbol{r}_d)|c_{\boldsymbol{k}\sigma}(t)\,|\psi_{n_1}(\boldsymbol{r}_d)\rangle\langle\psi_{n_1}(\boldsymbol{r}_d)|\,T^\dagger(\boldsymbol{r})\,|\psi_{n_2}(\boldsymbol{r}_d)\rangle\langle\psi_{n_2}(\boldsymbol{r}_d)|\,c^\dagger_{\boldsymbol{k}\sigma}|\psi_{\text{gs}}(\boldsymbol{r}_d)\rangle \\ &= N\sum_{\boldsymbol{r},n_1,n_2} e^{i\boldsymbol{k}\cdot\boldsymbol{r}}\langle\psi_{\text{gs}}(\boldsymbol{r}_d)|c_{\boldsymbol{k}\sigma}(t)\,|\psi_{n_1}(\boldsymbol{r}_d)\rangle\langle\psi_{n_1}(\boldsymbol{r}_d)|\psi_{n_2}(\boldsymbol{r}_d+\boldsymbol{r})\rangle\langle\psi_{n_2}(\boldsymbol{r}_d)|\,c^\dagger_{\boldsymbol{k}\sigma}|\psi_{\text{gs}}(\boldsymbol{r}_d)\rangle \ .\end{aligned} \tag{5.9}$$

The overlap $\langle\psi_{n_1}(\boldsymbol{r}_d)|\psi_{n_2}(\boldsymbol{r}_d+\boldsymbol{r})\rangle$ between the eigenstates of the auxiliary models at $\boldsymbol{r}_d$ and $\boldsymbol{r}_d+\boldsymbol{r}$ are appreciable only for $n_1 = n_2$. This is because, for small $\boldsymbol{r}$, the auxiliary models are very similar and the states are therefore almost orthogonal. For large $\boldsymbol{r}$, the overlap is again suppressed because the auxiliary model eigenstates are localised around the impurity. We therefore make the simplifying assumption: $\langle\psi_{n_1}(\boldsymbol{r}_d)|\psi_{n_2}(\boldsymbol{r}_d+\boldsymbol{r})\rangle = \delta_{n_1,n_2}\langle\psi_{n_1}(\boldsymbol{r}_d)|\psi_{n_1}(\boldsymbol{r}_d+\boldsymbol{r})\rangle = \delta_{n_1,n_2}F_{n_1}(\boldsymbol{r})$. Substituting this, we get

$$\sum_{\boldsymbol{r}_1,\boldsymbol{r}_2} \langle\psi_{\text{gs}}(\boldsymbol{r}_2)|c_{\boldsymbol{k}\sigma}(t)c^\dagger_{\boldsymbol{k}\sigma}|\psi_{\text{gs}}(\boldsymbol{r}_1)\rangle = N\sum_n F_n(\boldsymbol{k})\langle\psi_{\text{gs}}(\boldsymbol{r}_d)|c_{\boldsymbol{k}\sigma}(t)\,|\psi_n(\boldsymbol{r}_d)\rangle\langle\psi_n(\boldsymbol{r}_d)|\,c^\dagger_{\boldsymbol{k}\sigma}|\psi_{\text{gs}}(\boldsymbol{r}_d)\rangle \ , \tag{5.10}$$

where we have defined the Fourier transform $F_n(\boldsymbol{k}) = \sum_{\boldsymbol{r}} e^{i\boldsymbol{k}\cdot\boldsymbol{r}}F_n(\boldsymbol{r})$ of the overlap function.

We now consider more carefully the transition operator $c_{\boldsymbol{k}\sigma}$ for the 1-particle excitation giving rise to the above Greens function. Within our auxiliary model approach, gapless excitations within the lattice model are represented by gapless excitations of the impurity site, specifically those that try to screen the impurity site. As a result, the uncoordinated scattering process $c_{\boldsymbol{k}\sigma}$ for the lattice model must be replaced by a coherent excitation $\mathcal{T}$ within the impurity model that captures those gapless excitations that occur in connection with the impurity, and projects out the uncorrelated excitations that take place even when the impurity site is decoupled from the bath.

In order to construct this auxiliary model $\mathcal{T}-$matrix, we allow for all excitations of the impurity site in conjunction with those of the $k-$state:

$$\mathcal{T}_{\boldsymbol{k},\sigma} = \left[c^\dagger_{\boldsymbol{r}_d\sigma} + S^\sigma_{\boldsymbol{r}_d} + C^+_{\boldsymbol{r}_d}\right] c_{\boldsymbol{k}\sigma} \ , \tag{5.11}$$

where $S^+_{\boldsymbol{r}_d}$ and $C^+_{\boldsymbol{r}_d}$ are spin and charge flip operators on the impurity site that cause transition from $|\downarrow\rangle \to |\uparrow\rangle$ and $|0\rangle \to |\uparrow\downarrow\rangle$ respectively. The first term encodes the local component of the $k-$space Greens function by creating an excitation on the impurity site (which is then translated across the lattice), and the rest of the terms encode the non-local components by tracking impurity-bath excitations.

Apart from modifying the excitation operator, we will now simplify the time-evolution as well. The first overlap on the right-hand side of eq. 5.10 expands to

$$\langle\psi_\text{gs}(\boldsymbol{r}_d)|\mathcal{T}_{\boldsymbol{k},\sigma}(t)|\psi_n(\boldsymbol{r}_d)\rangle = \langle\psi_\text{gs}(\boldsymbol{r}_d)|e^{i(H_\text{aux}(\boldsymbol{r}_d)+H_\text{cav}(\boldsymbol{r}_d))t}\mathcal{T}_{\boldsymbol{k},\sigma}e^{-i(H_\text{aux}(\boldsymbol{r}_d)+H_\text{cav}(\boldsymbol{r}_d))t}|\psi_n(\boldsymbol{r}_d)\rangle \ , \tag{5.12}$$

where $H_\text{aux}(\boldsymbol{r}_d)$ (eq. 2.3) is the auxiliary model Hamiltonian consisting of the cluster Hamiltonian $H_\text{clust}(\boldsymbol{r}_d)$ along with all one-particle terms H_1p in the Hamiltonian (in a typical lattice model, this will be he tight-binding hopping terms), and $H_\text{cav}(\boldsymbol{r}_d)$ is the "cavity" Hamiltonian consisting of all interaction (2-particle and beyond) terms in the Hamiltonian. Together, they create the full lattice Hamiltonian: $H_\text{latt} = H_\text{aux}(\boldsymbol{r}_d) + H_\text{cav}(\boldsymbol{r}_d)$. Using the Zassenhaus form of the Baker-Campbell-Hausdorff formula, we have

$$e^{i(H_\text{aux}(\boldsymbol{r}_d)+H_\text{cav}(\boldsymbol{r}_d))t} \approx e^{iH_\text{aux}(\boldsymbol{r}_d)t}e^{iH_\text{cav}(\boldsymbol{r}_d)t}e^{[H_\text{aux}(\boldsymbol{r}_d),\ H_\text{cav}(\boldsymbol{r}_d)]t^2/2} \ , \tag{5.13}$$

where we have dropped the higher-order terms because the commutators involve only the boundary degrees of freedom and become subsequently smaller as the order increases. One can think of the operator $e^{iH_\text{cav}(\boldsymbol{r}_d)t}e^{[H_\text{aux}(\boldsymbol{r}_d),\ H_\text{cav}(\boldsymbol{r}_d)]t^2/2}$ as renormalising the auxiliary model $\mathcal{T}-$matrix through hybridisation with the rest of the system and formally leading to a new $\mathcal{T}-$matrix:

$$\mathcal{T}^*_{\boldsymbol{k},\sigma} = e^{iH_\text{cav}(\boldsymbol{r}_d)t}e^{[H_\text{aux}(\boldsymbol{r}_d),\ H_\text{cav}(\boldsymbol{r}_d)]t^2/2}\mathcal{T}_{\boldsymbol{k},\sigma}e^{-[H_\text{aux}(\boldsymbol{r}_d),\ H_\text{cav}(\boldsymbol{r}_d)]t^2/2}e^{-iH_\text{cav}(\boldsymbol{r}_d)t} \ . \tag{5.14}$$

One can also absorb the modulating phase factor $F_n(\boldsymbol{k})$ (defined above and appearing in eq. 5.10) into this renormalisation:

$$\mathcal{T}^*_{\boldsymbol{k},\sigma} = e^{iH_\text{cav}(\boldsymbol{r}_d)t}e^{[H_\text{aux}(\boldsymbol{r}_d),\ H_\text{cav}(\boldsymbol{r}_d)]t^2/2}\mathcal{T}_{\boldsymbol{k},\sigma}F_n(\boldsymbol{k})\mathcal{P}_n(\boldsymbol{r}_d)\ e^{-[H_\text{aux}(\boldsymbol{r}_d),\ H_\text{cav}(\boldsymbol{r}_d)]t^2/2}e^{-iH_\text{cav}(\boldsymbol{r}_d)t} \ , \tag{5.15}$$

where $\mathcal{P}_n(\boldsymbol{r}_d)$ projects onto the n^th eigenstate $|\psi_n(\boldsymbol{r}_d)\rangle$.

Since the overlap is calculated in the eigenstates of the auxiliary model at $\boldsymbol{r}_d$, we adopt the zeroth order approximation of ignoring this renormalisation. Later, we will specify a prescription of systematically improving our computation. The overlap expression becomes

$$\langle\psi_\text{gs}(\boldsymbol{r}_d)|\mathcal{T}_{\boldsymbol{k},\sigma}(t)|\psi_n(\boldsymbol{r}_d)\rangle = \langle\psi_\text{gs}(\boldsymbol{r}_d)|e^{i(H_\text{clust}(\boldsymbol{r}_d)+H_\text{1p})t}\mathcal{T}_{\boldsymbol{k},\sigma}e^{-i(H_\text{clust}(\boldsymbol{r}_d)+H_\text{1p})t}|\psi_n(\boldsymbol{r}_d)\rangle \ . \tag{5.16}$$

To make the problem tractable, we treat the effect of H_{1p} at the level of an expectation value and note that the effect of the transition operator is to create a hole of momentum $\boldsymbol{k}$ (eq. 5.11). Let the Hamiltonian H_{1p} be diagonal in certain modes $c_{\nu,\sigma}$ with energies ϵ_ν, and the operator $c_{\boldsymbol{k},\sigma}$ can be expressed in terms of these modes: $c_{\boldsymbol{k},\sigma} = \sum_\nu C_{\nu,\boldsymbol{k}} c_{\nu,\sigma}$. Expanding the $\mathcal{T}-$matrix in this fashion gives

$$\langle\psi_{\text{gs}}(\boldsymbol{r}_d)|\mathcal{T}_{\boldsymbol{k},\sigma}(t)|\psi_n(\boldsymbol{r}_d)\rangle = \sum_\nu C_{\nu,\boldsymbol{k}} \langle\psi_{\text{gs}}(\boldsymbol{r}_d)|e^{i(H_{\text{clust}}(\boldsymbol{r}_d)+H_{1\text{p}})t}\mathcal{T}_{\nu,\sigma}e^{-i(H_{\text{clust}}(\boldsymbol{r}_d)+H_{1\text{p}})t}|\psi_n(\boldsymbol{r}_d)\rangle \ . \tag{5.17}$$

Each excitation $\mathcal{T}_{\nu,\sigma}$ therefore leads to an energy cost $-\epsilon_\nu$. Moreover, $|\psi_n(\boldsymbol{r}_d)\rangle$ and $|\psi_{\text{gs}}(\boldsymbol{r}_d)\rangle$ are eigenstates of $H_{\text{clust}}(\boldsymbol{r}_d)$ with energies E_n and E_0. This leads to the following simplified expression:

$$\langle\psi_{\text{gs}}(\boldsymbol{r}_d)|\mathcal{T}_{\boldsymbol{k},\sigma}(t)|\psi_n(\boldsymbol{r}_d)\rangle = \sum_\nu C_{\nu,\boldsymbol{k}} e^{-i(E_n-E_0+\epsilon_\nu)t} \langle\psi_{\text{gs}}(\boldsymbol{r}_d)|\mathcal{T}_{\nu,\sigma}|\psi_n(\boldsymbol{r}_d)\rangle \ . \tag{5.18}$$

Substituting the expression back into 5.10 and then into 5.6, we get an expression for one half of the first part of the retarded Greens function (the full form is in eq. 5.5):

$$\begin{aligned} & -i\theta(t)\langle\Psi_{\text{gs}}|c_{\boldsymbol{k}\sigma}(t)c^\dagger_{\boldsymbol{k}\sigma}|\Psi_{\text{gs}}\rangle \\ & = -i\theta(t)\sum_{\nu,n} C_{\nu,\boldsymbol{k}} e^{-i(E_n-E_0+\epsilon_\nu)t} \langle\psi_{\text{gs}}(\boldsymbol{r}_d)|\mathcal{T}_{\nu,\sigma}|\psi_n(\boldsymbol{r}_d)\rangle \langle\psi_n(\boldsymbol{r}_d)|\mathcal{T}^\dagger_{\boldsymbol{k},\sigma}|\psi_{\text{gs}}(\boldsymbol{r}_d)\rangle \ . \end{aligned} \tag{5.19}$$

Working through the other term of eq. 5.6 with the same strategy, we get the full expression:

$$\begin{aligned} & -i\theta(t)\langle\Psi_{\text{gs}}|\{c_{\boldsymbol{k}\sigma}(t), c^\dagger_{\boldsymbol{k}\sigma}\}|\Psi_{\text{gs}}\rangle \\ & = -i\theta(t)\sum_{\nu,n} C_{\nu,\boldsymbol{k}} e^{-i(E_n-E_0+\epsilon_\nu)t} \langle\psi_{\text{gs}}(\boldsymbol{r}_d)|\mathcal{T}_{\nu,\sigma}|\psi_n(\boldsymbol{r}_d)\rangle \langle\psi_n(\boldsymbol{r}_d)|\mathcal{T}^\dagger_{\boldsymbol{k},\sigma}|\psi_{\text{gs}}(\boldsymbol{r}_d)\rangle \\ & \quad -i\theta(t)\sum_{\nu,n} C_{\nu,\boldsymbol{k}} e^{i(E_n-E_0-\epsilon_\nu)t} \langle\psi_{\text{gs}}(\boldsymbol{r}_d)|\mathcal{T}^\dagger_{\boldsymbol{k},\sigma}|\psi_n(\boldsymbol{r}_d)\rangle \langle\psi_n(\boldsymbol{r}_d)|\mathcal{T}_{\nu,\sigma}|\psi_{\text{gs}}(\boldsymbol{r}_d)\rangle \ . \end{aligned} \tag{5.20}$$

Finally, we consider the other term in eq. 5.5, $-i\theta(t)\langle\Psi_{\text{gs}}|\{c_{\boldsymbol{k}\sigma}, c^\dagger_{\boldsymbol{k}\sigma}(-t)\}|\Psi_{\text{gs}}\rangle$. Proceeding in the same way, eq. 5.18 is replaced by

$$\begin{aligned} \langle\psi_n(\boldsymbol{r}_d)|\mathcal{T}^\dagger_{\boldsymbol{k},\sigma}(-t)|\psi_{\text{gs}}(\boldsymbol{r}_d)\rangle &= \langle\psi_{\text{gs}}(\boldsymbol{r}_d)|\mathcal{T}_{\boldsymbol{k},\sigma}(-t)|\psi_n(\boldsymbol{r}_d)\rangle^* \\ &= \sum_\nu C^*_{\nu,\boldsymbol{k}} e^{-i(E_n-E_0+\epsilon_\nu)t} \langle\psi_n(\boldsymbol{r}_d)|\mathcal{T}^\dagger_{\nu,\sigma}|\psi_{\text{gs}}(\boldsymbol{r}_d)\rangle \ . \end{aligned} \tag{5.21}$$

Evaluating this term exactly as before and changing from time to frequency domain using $g(\omega) = \int dt\ e^{i\omega T} f(t)$ gives

$$\begin{aligned} G(\boldsymbol{k},\sigma;\omega) = \sum_{\nu,n} C_{\nu,\boldsymbol{k}} \Bigg[& \frac{\langle\psi_{\text{gs}}(\boldsymbol{r}_d)|\mathcal{T}_{\nu,\sigma}|\psi_n(\boldsymbol{r}_d)\rangle \langle\psi_n(\boldsymbol{r}_d)|\mathcal{T}^\dagger_{\boldsymbol{k},\sigma}|\psi_{\text{gs}}(\boldsymbol{r}_d)\rangle}{\omega - (E_n - E_0 + \epsilon_\nu)} \\ & + \frac{\langle\psi_{\text{gs}}(\boldsymbol{r}_d)|\mathcal{T}^\dagger_{\boldsymbol{k},\sigma}|\psi_n(\boldsymbol{r}_d)\rangle \langle\psi_n(\boldsymbol{r}_d)|\mathcal{T}_{\nu,\sigma}|\psi_{\text{gs}}(\boldsymbol{r}_d)\rangle}{\omega + (E_n - E_0 - \epsilon_\nu)} \Bigg] + \text{h.c.} \end{aligned} \tag{5.22}$$

We recognise the two parts of the lattice Greens function as a retarded off-diagonal Greens functions calculated within the auxiliary model at $\boldsymbol{r}_d$, but at shifted frequencies $\omega - \epsilon_\nu$:

$$G(\boldsymbol{k},\sigma;\omega) = \sum_\nu \left[C_{\nu,\boldsymbol{k}}\mathcal{G}(\mathcal{T}_{\nu,\sigma}, \mathcal{T}^\dagger_{\boldsymbol{k},\sigma}; \omega - \epsilon_\nu) + C^*_{\nu,\boldsymbol{k}}\mathcal{G}(\mathcal{T}_{\boldsymbol{k},\sigma}, \mathcal{T}^\dagger_{\nu,\sigma}; \omega - \epsilon_\nu)\right] \tag{5.23}$$

We now discuss how to systematically improve this estimate. One way is to include the effect of other auxiliary models in the overlap function $F_n(\boldsymbol{r})$. This will of course require us to calculate an off-diagonal Greens function. As a concrete example, upon including the effects of the auxiliary models nearest to the reference point $\boldsymbol{r}_d$, the expression for the Greens function becomes

$$\begin{aligned} G(\boldsymbol{k},\sigma;\omega) = \sum_{\nu,n} C_{\nu,\boldsymbol{k}} \Bigg[&\frac{\langle\psi_{\text{gs}}(\boldsymbol{r}_d)|\mathcal{T}^*_{\nu,\sigma}|\psi_n(\boldsymbol{r}_d)\rangle \langle\psi_n(\boldsymbol{r}_d)|\mathcal{T}^\dagger_{\boldsymbol{k},\sigma}|\psi_{\text{gs}}(\boldsymbol{r}_d)\rangle}{\omega - \epsilon_\nu - (E_n - E_0)} \\ &+ \frac{\langle\psi_{\text{gs}}(\boldsymbol{r}_d)|\mathcal{T}^\dagger_{\boldsymbol{k},\sigma}|\psi_n(\boldsymbol{r}_d)\rangle \langle\psi_n(\boldsymbol{r}_d)|\mathcal{T}^*_{\nu,\sigma}|\psi_{\text{gs}}(\boldsymbol{r}_d)\rangle}{\omega - \epsilon_\nu + (E_n - E_0)}\Bigg] , \end{aligned} \tag{5.24}$$

with $\tau^*_{\nu,\sigma} = \mathcal{T}_{\nu,\sigma}(F_n(0) + \sum_{\boldsymbol{a}} F_n(\boldsymbol{a})e^{i\boldsymbol{k}\cdot\boldsymbol{a}})$ where $\boldsymbol{a}$ sums over the primitive lattice vectors. Another way to include more non-local contributions is by considering the effects of the cavity Hamiltonian in eq. 5.15:

$$\mathcal{T}^*_{\boldsymbol{k},\sigma}(t) = e^{iH_{\text{cav}}(\boldsymbol{r}_d)t}(\mathcal{T}_{\boldsymbol{k},\sigma} + t^2/2\left[[H_{\text{aux}}(\boldsymbol{r}_d),\ H_{\text{cav}}(\boldsymbol{r}_d)], \mathcal{T}_{\boldsymbol{k},\sigma}\right])e^{-iH_{\text{cav}}(\boldsymbol{r}_d)t} . \tag{5.25}$$

The time-domain Greens function associated with this operator will then need to be calculated, which can then be used to calculate frequency-domain functions.

2.5.2 One-particle real-space Greens functions

We now work with Greens functions for real-space excitations. The retarded Greens function in this case is

$$G(\boldsymbol{r}\sigma;t) = -i\theta(t)\langle\Psi_{\text{gs}}|\{c_{\boldsymbol{r}_d+\boldsymbol{r},\sigma}(t), c^\dagger_{\boldsymbol{r}_d,\sigma}\}|\Psi_{\text{gs}}\rangle . \tag{5.26}$$

Since our lattice model has discrete translation symmetry, we instead calculate the equivalent quantity

$$G(\boldsymbol{r}\sigma;t) = -i\theta(t)\frac{1}{2}\sum_{\boldsymbol{r}_d}\left[\langle\Psi_{\text{gs}}|\{c_{\boldsymbol{r}_d+\boldsymbol{r},\sigma}(t), c^\dagger_{\boldsymbol{r}_d,\sigma}\}|\Psi_{\text{gs}}\rangle + \langle\Psi_{\text{gs}}|\{c_{\boldsymbol{r}_d+\boldsymbol{r},\sigma}, c^\dagger_{\boldsymbol{r}_d,\sigma}(-t)\}|\Psi_{\text{gs}}\rangle\right] . \tag{5.27}$$

Similar to the $k-$space Greens functions, we consider only the first term for now, and express the groundstates as a superposition of auxiliary model Greens functions:

$$\begin{aligned} \sum_{\boldsymbol{r}_d}\langle\Psi_{\text{gs}}|c_{\boldsymbol{r}_d+\boldsymbol{r},\sigma}(t)c^\dagger_{\boldsymbol{r}_d,\sigma}|\Psi_{\text{gs}}\rangle &= \frac{1}{N}\sum_{\boldsymbol{R},\boldsymbol{\Delta},\boldsymbol{r}_d}\langle\psi_{\text{gs}}(\boldsymbol{R}+\boldsymbol{\Delta})|c_{\boldsymbol{r}_d+\boldsymbol{r},\sigma}(t)c^\dagger_{\boldsymbol{r}_d,\sigma}|\psi_{\text{gs}}(\boldsymbol{R})\rangle \\ &= \frac{1}{N}\sum_{\boldsymbol{R},\boldsymbol{\Delta}}\langle\psi_{\text{gs}}(-\boldsymbol{r}_d)|c_{\boldsymbol{r}-\boldsymbol{R}-\boldsymbol{\Delta},\sigma}(t)T(\boldsymbol{\Delta})c^\dagger_{-\boldsymbol{R},\sigma}|\psi_{\text{gs}}(-\boldsymbol{r}_d)\rangle . \end{aligned} \tag{5.28}$$

Local Greens function: $r = 0$

We first consider the case of local Greens function, by setting $r = 0$. We also change the dummy variable r_d, R and Δ to $-r_d$, R and $-\Delta$ to make the expressions tidier:

$$\sum_{r_d} \langle \Psi_{gs}|c_{r_d,\sigma}(t)c^{\dagger}_{r_d,\sigma}|\Psi_{gs}\rangle = \frac{1}{N}\sum_{R,\Delta,r_d} \langle \psi_{gs}(r_d)|c_{R+\Delta,\sigma}(t)T^{\dagger}(\Delta)c^{\dagger}_{R,\sigma}|\psi_{gs}(r_d)\rangle \ . \tag{5.29}$$

We split the full summation into four parts, based on whether $R = r_d$ and whether $\Delta + R = r_d$:

$$\begin{aligned}\sum_{r_d} \langle \Psi_{gs}|c_{r_d,\sigma}(t)c^{\dagger}_{r_d,\sigma}|\Psi_{gs}\rangle = &\langle \psi_{gs}(r_d)|c_{r_d,\sigma}(t)c^{\dagger}_{r_d,\sigma}|\psi_{gs}(r_d)\rangle \\ &+ \sum_{\Delta\neq 0} \langle \psi_{gs}(r_d)|c_{r_d+\Delta,\sigma}(t)T^{\dagger}(\Delta)c^{\dagger}_{r_d,\sigma}|\psi_{gs}(r_d)\rangle \\ &+ \sum_{R\neq r_d} \langle \psi_{gs}(r_d)|c_{r_d,\sigma}(t)T^{\dagger}(r_d - R)c^{\dagger}_{R,\sigma}|\psi_{gs}(r_d)\rangle \\ &+ \sum_{R\neq r_d,\Delta\neq r_d} \langle \psi_{gs}(r_d)|c_{\Delta,\sigma}(t)T^{\dagger}(\Delta - R)c^{\dagger}_{R,\sigma}|\psi_{gs}(r_d)\rangle \ ,\end{aligned} \tag{5.30}$$

where we have used the fact that the location of the auxiliary model does not affect the correlation calculated within the auxiliary model in order to cancel out the summation over r_d with the factor of $1/N$. The reason for splitting the expression into these parts is to explicitly write out which operators act on the impurity site r_d and which act on other real-space sites away from the impurity site. We will now replace the real-space excitation operators away from the impurity site with the associated $T-$matrices, following the discussion above eq. 5.11:

$$\mathcal{T}_{r,\sigma} = \left[\left(\sum_{\sigma'} c^{\dagger}_{r_d\sigma} + \text{h.c.}\right) + \left(S^{+}_{r_d} + \text{h.c.}\right) + \left(C^{+}_{r_d} + \text{h.c.}\right)\right] c_{r\sigma} \ . \tag{5.31}$$

We also insert a complete basis of states of the auxiliary model at r_d, $1 = \sum_n |\psi_n(r_d)\rangle\langle\psi_n(r_d)|$, generating an overlap factor F_n as before:

$$\begin{aligned}\sum_{r_d} \langle \Psi_{gs}|c_{r_d,\sigma}(t)c^{\dagger}_{r_d,\sigma}|\Psi_{gs}\rangle = &\langle \psi_{gs}(r_d)|c_{r_d,\sigma}(t)c^{\dagger}_{r_d,\sigma}|\psi_{gs}(r_d)\rangle \\ &+ \sum_{r_1,r_2\neq r_d,n} F_n(r_1 - r_2)^* \langle \psi_{gs}(r_d)|\mathcal{T}_{r_1,\sigma}(t)|\psi_n(r_d)\rangle \langle \psi_n(r_d)|\mathcal{T}^{\dagger}_{r_2,\sigma}|\psi_{gs}(r_d)\rangle \ .\end{aligned} \tag{5.32}$$

Several terms have been dropped because those contributions vanish for systems where the eigenstates conserve total spin S_z.

We next treat the time-evolution of the operators exactly as in the $k-$space Greens function. We replace the one-particle Hamiltonian H_{1p} with its local component H_{loc} (such as any on-site term of inter-layer hopping, etc). As done previously, we will make use of the eigenstates $\nu(r)$ of H_{loc}, with energy ϵ^{loc}_{ν}. Without repeating the details, we state the final result here, for the frequency-domain real-space local Greens function:

$$\begin{aligned} G_{\mathrm{loc}}(\omega) =& \frac{1}{2}\sum_{\nu}\left[C_{\nu,\boldsymbol{r}_d}\mathcal{G}(\nu(r_d)\sigma, c^\dagger_{r_d,\sigma};\tilde{\omega}) + C^*_{\nu,\boldsymbol{r}_d}\mathcal{G}(c_{r_d,\sigma},\nu^\dagger(r_d)\sigma;\tilde{\omega})\right] \\ &+ \frac{1}{2}\sum_{\boldsymbol{r}_1,\boldsymbol{r}_2\neq r_d,\nu} F_n(\boldsymbol{r}_1-\boldsymbol{r}_2)^*\left[C_{\boldsymbol{r}_1,\nu}\mathcal{G}(\mathcal{T}_{\nu(\boldsymbol{r}_1)\sigma},\mathcal{T}^\dagger_{\boldsymbol{r}_2\sigma};\tilde{\omega}) + C^*_{\boldsymbol{r}_1,\nu}\mathcal{G}(\mathcal{T}_{\boldsymbol{r}_1\sigma},\mathcal{T}^\dagger_{\nu(\boldsymbol{r}_2)\sigma};\tilde{\omega})\right] , \end{aligned} \tag{5.33}$$

where the hermitian conjugate in each part comes from considering the second term in eq. 5.27. The shifted frequency $\tilde{\omega} = \omega - \epsilon^{\mathrm{loc}}_\nu$ captures the effect of local terms in $H_{1\mathrm{p}}$; the shift is defined as $e^{i\epsilon^{\mathrm{loc}}_\nu t}c^\dagger_\nu = e^{iH_{\mathrm{loc}}t}c^\dagger_\nu e^{-iH_{\mathrm{loc}}t}$. The operator $\nu^\dagger(\boldsymbol{r})$ represents the eigenstates of H_{loc} at any site $\boldsymbol{r}$. The coefficients $C_{\nu,\boldsymbol{r}}$ express the eigenstates in terms of local states: $c_{\boldsymbol{r}} = \sum_\nu C_{\nu,\boldsymbol{r}}c_{\nu(\boldsymbol{r})}$.

Of course, the states ν will differ from the local states $c_{\boldsymbol{r}}$ only if there are multiple local orbitals (say α) that can hybridise among each other. In such a case, one can also think of an inter-orbital Greens function, defined as

$$G_{\mathrm{loc}}(\alpha,\alpha';t) = -i\theta(t)\left\langle\Psi_{\mathrm{gs}}\right|\{c_{\boldsymbol{r}_d,\sigma,\alpha}(t), c^\dagger_{\boldsymbol{r}_d,\sigma,\alpha'}\}\left|\Psi_{\mathrm{gs}}\right\rangle . \tag{5.34}$$

Such a Greens function can also be calculated from our approach:

$$\begin{aligned} G_{\mathrm{loc}}(\alpha,\alpha';\omega) =& \frac{1}{2}\sum_{\nu}\left[C_{\nu,\alpha}\mathcal{G}(\nu(r_d,\alpha)\sigma, c^\dagger_{r_d,\sigma,\alpha'};\tilde{\omega}) + C^*_{\nu,\alpha'}\mathcal{G}(c_{r_d,\sigma,\alpha},\nu^\dagger(r_d,\alpha')\sigma;\tilde{\omega})\right] \\ &+ \frac{1}{2}\sum_{\boldsymbol{r}_1,\boldsymbol{r}_2\neq r_d,\nu} F_n(\boldsymbol{r}_1-\boldsymbol{r}_2)^*\left[C_{\alpha,\nu}\mathcal{G}(\mathcal{T}_{\nu(\boldsymbol{r}_1),\sigma,\alpha},\mathcal{T}^\dagger_{\boldsymbol{r}_2,\sigma,\alpha'};\tilde{\omega}) + C^*_{\alpha',\nu}\mathcal{G}(\mathcal{T}_{\boldsymbol{r}_1,\sigma,\alpha},\mathcal{T}^\dagger_{\nu(\boldsymbol{r}_2),\sigma,\alpha'};\tilde{\omega})\right] , \end{aligned} \tag{5.35}$$

Non-local Greens function: $\boldsymbol{r}\neq 0$

We next consider the case of non-local Greens function, where the excitations propagate a non-zero distance $\boldsymbol{r}$:

$$\sum_{\boldsymbol{r}_d}\left\langle\Psi_{\mathrm{gs}}\right|c_{\boldsymbol{r}_d+\boldsymbol{r},\sigma}(t)c^\dagger_{\boldsymbol{r}_d,\sigma}\left|\Psi_{\mathrm{gs}}\right\rangle = \frac{1}{N}\sum_{\boldsymbol{R},\boldsymbol{\Delta},\boldsymbol{r}_d}\left\langle\psi_{\mathrm{gs}}(\boldsymbol{r}_d)\right|c_{\boldsymbol{R}+\boldsymbol{\Delta}+\boldsymbol{r},\sigma}(t)T^\dagger(\boldsymbol{\Delta})c^\dagger_{\boldsymbol{R},\sigma}\left|\psi_{\mathrm{gs}}(\boldsymbol{r}_d)\right\rangle . \tag{5.36}$$

To bring this expression closer to the one for the local Greens function, we redefine $\Delta+\boldsymbol{r}\to\Delta$:

$$\sum_{\boldsymbol{r}_d}\left\langle\Psi_{\mathrm{gs}}\right|c_{\boldsymbol{r}_d+\boldsymbol{r},\sigma}(t)c^\dagger_{\boldsymbol{r}_d,\sigma}\left|\Psi_{\mathrm{gs}}\right\rangle = \frac{1}{N}\sum_{\boldsymbol{R},\boldsymbol{\Delta},\boldsymbol{r}_d}\left\langle\psi_{\mathrm{gs}}(\boldsymbol{r}_d)\right|c_{\boldsymbol{R}+\boldsymbol{\Delta},\sigma}(t)T^\dagger(\boldsymbol{\Delta})T(\boldsymbol{r})c^\dagger_{\boldsymbol{R},\sigma}\left|\psi_{\mathrm{gs}}(\boldsymbol{r}_d)\right\rangle . \tag{5.37}$$

This allows to again split the full summation based on whether $\boldsymbol{R} = \boldsymbol{r}_d$ and whether $\Delta + \boldsymbol{R} = \boldsymbol{r}_d$:

$$\begin{aligned}\sum_{\boldsymbol{r}_d} \langle \Psi_{\text{gs}} | c_{\boldsymbol{r}_d+\boldsymbol{r},\sigma}(t) c^\dagger_{\boldsymbol{r}_d,\sigma} | \Psi_{\text{gs}} \rangle = &\langle \psi_{\text{gs}}(\boldsymbol{r}_d) | c_{\boldsymbol{r}_d,\sigma}(t) T(\boldsymbol{r}) c^\dagger_{\boldsymbol{r}_d,\sigma} | \psi_{\text{gs}}(\boldsymbol{r}_d) \rangle \\ &+ \sum_{\boldsymbol{\Delta}\neq 0} \langle \psi_{\text{gs}}(\boldsymbol{r}_d) | c_{\boldsymbol{r}_d+\boldsymbol{\Delta},\sigma}(t) T^\dagger(\boldsymbol{\Delta} - \boldsymbol{r}) c^\dagger_{\boldsymbol{r}_d,\sigma} | \psi_{\text{gs}}(\boldsymbol{r}_d) \rangle \\ &+ \sum_{\boldsymbol{R}\neq \boldsymbol{r}_d} \langle \psi_{\text{gs}}(\boldsymbol{r}_d) | c_{\boldsymbol{r}_d,\sigma}(t) T^\dagger(\boldsymbol{r}_d - \boldsymbol{R} - \boldsymbol{r}) c^\dagger_{\boldsymbol{R},\sigma} | \psi_{\text{gs}}(\boldsymbol{r}_d) \rangle \\ &+ \sum_{\boldsymbol{R}\neq \boldsymbol{r}_d, \boldsymbol{\Delta}\neq \boldsymbol{r}_d} \langle \psi_{\text{gs}}(\boldsymbol{r}_d) | c_{\boldsymbol{\Delta},\sigma}(t) T^\dagger(\boldsymbol{\Delta} - \boldsymbol{R} - \boldsymbol{r}) c^\dagger_{\boldsymbol{R},\sigma} | \psi_{\text{gs}}(\boldsymbol{r}_d) \rangle \ .\end{aligned} \tag{5.38}$$

Introducing Fourier transforms of the operators and inserting eigenstates of the auxiliary models leads to

$$\begin{aligned}&\sum_{\boldsymbol{r}_d} \langle \Psi_{\text{gs}} | c_{\boldsymbol{r}_d+\boldsymbol{r},\sigma}(t) c^\dagger_{\boldsymbol{r}_d,\sigma} | \Psi_{\text{gs}} \rangle \\ &= \sum_n F^*_n(\boldsymbol{r}) \langle \psi_{\text{gs}}(\boldsymbol{r}_d) | c_{\boldsymbol{r}_d,\sigma}(t) | \psi_n(\boldsymbol{r}_d) \rangle \langle \psi_n(\boldsymbol{r}_d) | c^\dagger_{\boldsymbol{r}_d,\sigma} | \psi_{\text{gs}}(\boldsymbol{r}_d) \rangle \\ &+ \sum_{\boldsymbol{k},n} F'_n(\boldsymbol{k}) F^*_n(\boldsymbol{r}) \langle \psi_{\text{gs}}(\boldsymbol{r}_d) | \mathcal{T}_{\boldsymbol{k},\sigma}(t) | \psi_n(\boldsymbol{r}_d) \rangle \langle \psi_n(\boldsymbol{r}_d) | c^\dagger_{\boldsymbol{r}_d,\sigma} | \psi_{\text{gs}}(\boldsymbol{r}_d) \rangle \\ &+ \sum_{\boldsymbol{k},n} F'_n(\boldsymbol{k})^* F^*_n(\boldsymbol{r}) \langle \psi_{\text{gs}}(\boldsymbol{r}_d) | c_{\boldsymbol{r}_d,\sigma}(t) | \psi_n(\boldsymbol{r}_d) \rangle \langle \psi_n(\boldsymbol{r}_d) | \mathcal{T}^\dagger_{\boldsymbol{k},\sigma} | \psi_{\text{gs}}(\boldsymbol{r}_d) \rangle \\ &+ \sum_{\boldsymbol{k}_1,\boldsymbol{k}_2} F'_n(\boldsymbol{k}_1) F'_n(\boldsymbol{k}_2)^* F^*_n(\boldsymbol{r}) \langle \psi_{\text{gs}}(\boldsymbol{r}_d) | \mathcal{T}_{\boldsymbol{k}_1,\sigma}(t) | \psi_n(\boldsymbol{r}_d) \rangle \langle \psi_n(\boldsymbol{r}_d) | \mathcal{T}^\dagger_{\boldsymbol{k}_2,\sigma} | \psi_{\text{gs}}(\boldsymbol{r}_d) \rangle \ ,\end{aligned} \tag{5.39}$$

The net difference from the diagonal Greens function is the presence of the overlap factor $F^*_n(\boldsymbol{r})$.

2.5.3 Equal-Time Ground State Correlators

Apart from dynamical correlations such as Greens functions, we would also like to use our approach to calculate static correlation functions in terms of auxiliary model quantities. We first consider a real-space correlation that tracks the propagation of a general excitation described by the operator $\mathcal{O}(\boldsymbol{r})$ over a distance $\boldsymbol{\Delta}$:

$$C_{\mathcal{O}}(\boldsymbol{r}) = \langle \Psi_{\text{gs}} | \mathcal{O}(\boldsymbol{r}_d + \boldsymbol{r}) \mathcal{O}^\dagger(\boldsymbol{r}_d) | \Psi_{\text{gs}} \rangle \ . \tag{5.40}$$

We first consider a local correlation: $\boldsymbol{r} = 0$. To obtain a tractable expression for this, we note that the quantity being computed here is almost identical to the one computed in eq. 5.28, the only difference being the time-evolution of the operator. For the local correlation, we therefore assume the result of eq. 5.31, dropping the time evolution because the correlators in this section are calculated at equal time. We also transform from $k-$space back to real-space because there are no time-evolution operators which need extended states in order to be resolved. Putting all this together, we get

$$\begin{aligned} C^{\text{loc}}_O &= \langle\psi_{\text{gs}}(\boldsymbol{r}_d)|O(\boldsymbol{r}_d)O^\dagger(\boldsymbol{r}_d)|\psi_{\text{gs}}(\boldsymbol{r}_d)\rangle \\ &\quad + \sum_{\boldsymbol{r}\neq 0,n} F_n(\boldsymbol{r})\,\langle\psi_{\text{gs}}(\boldsymbol{r}_d)|\mathcal{T}_{O(\boldsymbol{r}+\boldsymbol{r}_d)}|\psi_n(\boldsymbol{r}_d)\rangle\,\langle\psi_n(\boldsymbol{r}_d)|O^\dagger(\boldsymbol{r}_d)|\psi_{\text{gs}}(\boldsymbol{r}_d)\rangle \\ &\quad + \sum_{\boldsymbol{r}\neq 0,n} F_n(\boldsymbol{r})\,\langle\psi_{\text{gs}}(\boldsymbol{r}_d)|O(\boldsymbol{r}+\boldsymbol{r}_d)|\psi_n(\boldsymbol{r}_d)\rangle\,\langle\psi_n(\boldsymbol{r}_d)|\mathcal{T}^\dagger_{O(\boldsymbol{r}_d)}|\psi_{\text{gs}}(\boldsymbol{r}_d)\rangle \\ &\quad + \sum_{\boldsymbol{r}_1\neq 0,\boldsymbol{r}_2\neq 0,n} F_n(\boldsymbol{r})\,\langle\psi_{\text{gs}}(\boldsymbol{r}_d)|\mathcal{T}_{O(\boldsymbol{r}_1+\boldsymbol{r}_d)}|\psi_n(\boldsymbol{r}_d)\rangle\,\langle\psi_n(\boldsymbol{r}_d)|\mathcal{T}^\dagger_{O(\boldsymbol{r}_2+\boldsymbol{r}_d)}|\psi_{\text{gs}}(\boldsymbol{r}_d)\rangle\ , \end{aligned} \tag{5.41}$$

where the transition operator $\mathcal{T}_{O(\boldsymbol{r}+\boldsymbol{r}_d)}$ represents impurity-bath correlation:

$$\mathcal{T}_{O(\boldsymbol{r}+\boldsymbol{r}_d),\sigma} = \left[c^\dagger_{\boldsymbol{r}_d\sigma} + S^\sigma_{\boldsymbol{r}_d} + C^+_{\boldsymbol{r}_d}\right] O(\boldsymbol{r}+\boldsymbol{r}_d)\ , \tag{5.42}$$

We next consider non-local correlations over a distance $\boldsymbol{\Delta}$. Adapting eq. 5.39 for the non-local real-space Greens function in the same manner, we get

$$\begin{aligned} C_O(\boldsymbol{\Delta}) &= \sum_n F^*_n(\boldsymbol{\Delta})\,\langle\psi_{\text{gs}}(\boldsymbol{r}_d)|c_{\boldsymbol{r}_d,\sigma}(t)|\psi_n(\boldsymbol{r}_d)\rangle\,\langle\psi_n(\boldsymbol{r}_d)|c^\dagger_{\boldsymbol{r}_d,\sigma}|\psi_{\text{gs}}(\boldsymbol{r}_d)\rangle \\ &\quad + \sum_{\boldsymbol{r}\neq 0,n} F_n(\boldsymbol{r}+\boldsymbol{\Delta})\,\langle\psi_{\text{gs}}(\boldsymbol{r}_d)|\mathcal{T}_{O(\boldsymbol{r}+\boldsymbol{\Delta}+\boldsymbol{r}_d)}|\psi_n(\boldsymbol{r}_d)\rangle\,\langle\psi_n(\boldsymbol{r}_d)|O^\dagger(\boldsymbol{r}_d)|\psi_{\text{gs}}(\boldsymbol{r}_d)\rangle \\ &\quad + \sum_{\boldsymbol{r}\neq 0,n} F_n(\boldsymbol{r}+\boldsymbol{\Delta})\,\langle\psi_{\text{gs}}(\boldsymbol{r}_d)|O(\boldsymbol{r}+\boldsymbol{\Delta}+\boldsymbol{r}_d)|\psi_n(\boldsymbol{r}_d)\rangle\,\langle\psi_n(\boldsymbol{r}_d)|\mathcal{T}^\dagger_{O(\boldsymbol{r}_d)}|\psi_{\text{gs}}(\boldsymbol{r}_d)\rangle \\ &\quad + \sum_{\boldsymbol{r}_1\neq 0,\boldsymbol{r}_2\neq 0,n} F_n(\boldsymbol{r}+\boldsymbol{\Delta})\,\langle\psi_{\text{gs}}(\boldsymbol{r}_d)|\mathcal{T}_{O(\boldsymbol{r}_1+\boldsymbol{\Delta}+\boldsymbol{r}_d)}|\psi_n(\boldsymbol{r}_d)\rangle\,\langle\psi_n(\boldsymbol{r}_d)|\mathcal{T}^\dagger_{O(\boldsymbol{r}_2+\boldsymbol{r}_d)}|\psi_{\text{gs}}(\boldsymbol{r}_d)\rangle\ , \end{aligned} \tag{5.43}$$

2.6 Entanglement Measures

We will now describe the prescription of calculating entanglement measures of the lattice model from within our auxiliary model treatment. In this section, we are interested mainly in two such measures, the entanglement entropy and the mutual information. Given a pure state $|\Psi\rangle$ describing the complete system, the entanglement entropy $S_{\text{EE}}(\nu)$ of a subsystem ν quantifies the entanglement of ν with the rest of the subsystem, and is defined as

$$S_{\text{EE}}(\nu) = -\text{Tr}\left[\rho(\nu)\log\rho(\nu)\right],\quad \rho(\nu) = \text{Tr}_\nu\left[|\Psi\rangle\langle\Psi|\right] \tag{6.1}$$

where Tr $[\cdot]$ is the trace operator, and $\rho(\nu)$ is the reduced density matrix (RDM) for the subsystem ν obtained by taking the partial trace Tr_ν (over the states of ν) of the full density matrix $\rho = |\Psi\rangle\langle\Psi|$. If the subsystem ν describes local regions in real space (or states in $k-$space), we might be interested in the entanglement between two such subsystems ν_1 and ν_2. The correct measure to quantify such entanglement is the mutual information:

$$I_2(\nu_1,\nu_2) = S_{\text{EE}}(\nu_1) + S_{\text{EE}}(\nu_2) - S_{\text{EE}}(\nu_1\cup\nu_2)\ , \tag{6.2}$$

where $\nu_1 \cup \nu_2$ is a larger subsystem formed by combining ν_1 and ν_2.

We start by recalling the decomposition of entanglement entropy in terms of multipartite information as laid out in Subsection **??**. Given a partitioning of the system into mutually exclusive subsystems $\{\nu_i\}$, the average entanglement entropy $S^{(1)}_{\mathrm{EE}}$ over all subsystems can be expressed as

$$\begin{aligned} S^{(1)}_{\mathrm{EE}} &= \frac{1}{N}\sum_{n=2}^{N}(-1)^n\binom{N}{n}\bar{I}_n, \\ \bar{I}_n &= \frac{(N-n)!n!}{N!}\sum_{S_n\in C_n(\mathcal{S}_N)} I_n(S_n), \end{aligned} \tag{6.3}$$

where $\bar{I}_n$ is the average multipartite information I_n over all distinct combinations of n subsystems.

We first take a look at real-space entanglement. Real-space entanglement measures can be used to probe delocalisation-localisation transitions. The simplest such measure is the entanglement of a lattice site in real-space. Since the local entanglement entropy will be uniform at each lattice site for a system with translation invariance, it suffices to calculate the real-space averaged entanglement entropy $S^{(1)}_{\mathrm{EE}}$. The corresponding multipartite measures $\bar{I}_n$ involve collections of lattice sites. We write such a measure as

$$\bar{I}_n = \frac{(N-n)!n!}{N!}\frac{1}{N}\sum_{I(\boldsymbol{r}_d)}\sum_{B^{n-1}(\boldsymbol{r}_d)} I_n(I(\boldsymbol{r}_d);B^{n-1}(\boldsymbol{r}_d)), \tag{6.4}$$

where $I(\boldsymbol{r}_d)$ sums over all lattice sites in the form of an impurity site and $B^{n-1}(\boldsymbol{r}_d)$ considers all possible subsets of size $n-1$ of the remaining bath sites. Together, this choice corresponds to an auxiliary model with the impurity placed at $\boldsymbol{r}_d$. The additional factor of $1/N$ takes care of the overcounting arising from each choice appearing in N auxiliary models. We can therefore write the multipartite information as an average over auxiliary models, each auxiliary model contributing an average I_n over its bath sites:

$$\bar{I}_n = \frac{n}{N^2}\sum_{\boldsymbol{r}_d} I^{(\mathrm{tot})}_n(\boldsymbol{r}_d)\ . \tag{6.5}$$

The quantity $I^{(\mathrm{tot})}_n(\boldsymbol{r}_d)$ is defined as $I^{(\mathrm{tot})}_n(\boldsymbol{r}_d) = \frac{(N-n)!(n-1)!}{(N-1)!}\sum_{B^{n-1}(\boldsymbol{r}_d)} I_n(I(\boldsymbol{r}_d);B^{n-1}(\boldsymbol{r}_d))$. By translation invariance, such a measure will be uniform across auxiliary models and it suffices to calculate the measure for a single auxiliary model located at a reference point $\boldsymbol{r}_d$; the whole expression therefore simplifies to

$$S^{(1)}_{\mathrm{EE}} = \frac{1}{N}\sum_{n=2}^{N}(-1)^n\binom{N-1}{n-1}I^{(\mathrm{tot})}_n(\boldsymbol{r}_d) = \frac{1}{N}\sum_{n=2}^{N}(-1)^n\sum_{B^{n-1}(\boldsymbol{r}_d)} I_n(I(\boldsymbol{r}_d);B^{n-1}(\boldsymbol{r}_d))\ . \tag{6.6}$$

For example, truncating the summation to $n=3$, we have the explicit expression

$$S^{(1)}_{\mathrm{EE}} = \frac{N-1}{N}I^{(\mathrm{tot})}_2(\boldsymbol{r}_d) - \frac{(N-1)(N-2)}{2N}I^{(\mathrm{tot})}_3(\boldsymbol{r}_d) \tag{6.7}$$

Chapter 3

Kondo Breakdown in a Heavy-Fermion Model

3.1 Bilayer extended Hubbard model: Impurity model

[206] We approach the heavy-fermion problem by starting from a bilayer extended Hubbard model, consisting of two layers (f and c). Towards studying this lattice model, we adopt a two-layer impurity problem that hosts a correlated impurity site in each layer (S_f and S_d):

$$H_{\text{aux}} = H_{1\text{p}} + H_f + H_d + H_{fd} \ , \tag{1.1}$$

where $H_{1\text{p}}$ is the Hamiltonian for the non-interacting itinerant electrons of either layer,

$$H_{1\text{p}} = -t \sum_{\sigma,\alpha} \sum_{\langle i,j \rangle} \left(c^\dagger_{i,\sigma,\alpha} c_{j,\sigma,\alpha} + \text{h.c.} \right) - \sum_{i,\sigma,\alpha} \mu_\alpha n_{i,\sigma,\alpha} \ , \tag{1.2}$$

such that α sums over the two layers f and d. H_f and H_d describe the dynamics of the correlated impurity sites (and their local neighbourhood) in each layer:

$$\begin{aligned} H_f &= -\frac{U_f}{2}(n_{f,\uparrow} - n_{f,\downarrow})^2 + \sum_{Z \in \text{NN}} \left[-V \sum_{\sigma} \left(c^\dagger_{f,\sigma} c_{Z,\sigma,f} + \text{h.c.} \right) + \frac{J}{2} \sum_{\alpha,\beta} \mathbf{S}_f \cdot \boldsymbol{\sigma}_{\alpha\beta} c^\dagger_{Z,\alpha,f} c_{Z,\beta,f} \right. \\ &\qquad \left. - \frac{W_f}{2} \left(n_{Z,\uparrow,f} - n_{Z,\downarrow,f} \right)^2 \right] \ , \\ H_d &= -\eta_d n_d - \frac{U_d}{2}(n_{d,\uparrow} - n_{d,\downarrow})^2 + \sum_{Z \in \text{NN}} \left[-V \sum_{\sigma} \left(c^\dagger_{d,\sigma} c_{Z,\sigma,d} + \text{h.c.} \right) + \frac{J}{2} \sum_{\alpha,\beta} \mathbf{S}_d \cdot \boldsymbol{\sigma}_{\alpha\beta} c^\dagger_{Z,\alpha,d} c_{Z,\beta,d} \right] \ , \end{aligned} \tag{1.3}$$

where $c^\dagger_{\nu,\sigma,\alpha}$ can refer to creation operator for either the f−layer. Finally, H_{fd} represents the inter-layer hybridisation:

$$H_{fc} = -V_\perp \sum_{\sigma} \left(c^\dagger_{d,\sigma} c_{f,\sigma} + \text{h.c.} \right) \ . \tag{1.4}$$

3.2 Tiling Into Bilayer Extended Hubbard Model

Tiling the impurity model leads to a bilayer extended Hubbard model on the 2D square lattice. In order to tile our auxiliary model to restore translation invariance, we identify the cluster which acts as the "unit" cell of translation. As discussed in the text surrounding eq. 3.9, a good choice for the unit cell is the impurity site and it's surrounding correlated environment. Placing the impurity sites at $\boldsymbol{r}_d$ in each layer, we get

$$H_{\text{clust}}(\boldsymbol{r}_d) = H_f(\boldsymbol{r}_d) + H_d(\boldsymbol{r}_d) + H_{fc}(\boldsymbol{r}_d) \ . \tag{2.1}$$

The details of tiling a cluster Hamiltonian were already shown in Sec. 2.3.2; the general idea is to use manybody translation operators (details in Sec. 2.3.1) to center the cluster Hamiltonian at various sites of the lattice, such that combining all such terms leads to a periodised Hamiltonian. we write down the final result here:

$$\begin{aligned} H_{\text{latt}} &= \sum_r T^\dagger(\boldsymbol{r}) H_{\text{clust}}(\boldsymbol{r}_d) T(\boldsymbol{r}) \\ &= -\frac{2V}{Z}\sum_{\langle i,j\rangle,\sigma}\left(c^\dagger_{i,\sigma}c_{j,\sigma} + \text{h.c.}\right) - \sum_i\left[\eta_d n_{i,d} + \frac{U_d}{2}(n_{i,\uparrow,d} - n_{i,\downarrow,d})^2\right] + \frac{2J}{Z}\sum_{\langle i,j\rangle}\boldsymbol{S}_{i,d}\cdot\boldsymbol{S}_{j,d} \\ &\quad -\frac{2V}{Z}\sum_{\langle i,j\rangle,\sigma}\left(f^\dagger_{i,\sigma}f_{j,\sigma} + \text{h.c.}\right) - \frac{U_f - W_f}{2}\sum_i\left(n_{i,\uparrow,f} - n_{i,\downarrow,f}\right)^2 + \frac{2J}{Z}\sum_{\langle i,j\rangle}\boldsymbol{S}_{i,f}\cdot\boldsymbol{S}_{j,f} \\ &\quad - V_\perp\sum_{i,\sigma}(f^\dagger_{i,\sigma}c_{i,\sigma} + \text{h.c.}) \end{aligned} \tag{2.2}$$

Tiling the impurity model therefore leads to a bilayer extended Hubbard model, with effective couplings that depend on the auxiliary model couplings. The value of η_d defines the effective chemical potential of the $d-$layer; a non-zero value moves the layer away from half-filling. The $f-$layer is pinned at half-filling.

Following Sec. 2.5.1, the retarded $k-$space Greens function in either layer can be expressed in terms of auxiliary model Greens functions:

$$G(\boldsymbol{k},\sigma,\alpha;\omega) = \sum_\nu C_{\nu,\boldsymbol{k}\alpha}\mathcal{G}(\mathcal{T}_{\nu,\sigma,\alpha};\omega - \mathcal{E}_\nu(\boldsymbol{k})) \tag{2.3}$$

where $\alpha = f, d$ denotes the layer in which the Greens function is being calculated, the modes ν indicate the operators which diagonalise the Hamiltonian $H_{1\text{p}}$, with energies $\mathcal{E}_\nu(\boldsymbol{k})$:

$$H_{1\text{p}} = -\frac{2V}{Z}\sum_{\langle i,j\rangle,\sigma}\left(c^\dagger_{i,\sigma}c_{j,\sigma} + \text{h.c.} + f^\dagger_{i,\sigma}f_{j,\sigma} + \text{h.c.}\right) - V_\perp\sum_{i,\sigma}(f^\dagger_{i,\sigma}c_{i,\sigma} + \text{h.c.}) - \sum_i \eta_d n_{i,d} \ , \tag{2.4}$$

and the $\mathcal{T}-$matrices are defined in eq. 5.11:

$$\mathcal{T}_{\nu,\sigma,\alpha} = \left[\left(\sum_{\sigma'} c^\dagger_{\boldsymbol{\alpha}\sigma} + \text{h.c.}\right) + \left(S^+_{\boldsymbol{\alpha}} + \text{h.c.}\right) + \left(C^+_{\boldsymbol{\alpha}} + \text{h.c.}\right)\right] c_{\nu,\sigma} \ , \tag{2.5}$$

In order to calculate $\mathcal{E}_\nu(\boldsymbol{k})$, we introduce the $k-$space modes via a Fourier transform:

$$c^\dagger_{i,\sigma} = \frac{1}{\sqrt{N}}\sum_{\boldsymbol{k}} e^{-i\boldsymbol{k}\cdot\boldsymbol{r}_i} c^\dagger_{\boldsymbol{k},\sigma} , \quad f^\dagger_{i,\sigma} = \frac{1}{\sqrt{N}}\sum_{\boldsymbol{k}} e^{-i\boldsymbol{k}\cdot\boldsymbol{r}_i} f^\dagger_{\boldsymbol{k},\sigma} . \tag{2.6}$$

In terms of these modes, the 1−particle Hamiltonian decouples in $k-$space:

$$H_{1\mathrm{p}} = \sum_{\boldsymbol{k},\sigma} \left[(\epsilon_{\boldsymbol{k}} - \eta_d)(n_{\boldsymbol{k},\sigma,d} + n_{\boldsymbol{k},\sigma,f}) - V_\perp(c^\dagger_{\boldsymbol{k},\sigma} f_{\boldsymbol{k},\sigma} + \text{h.c.})\right] . \tag{2.7}$$

The diagonal modes are given by

$$\nu^\dagger_{\boldsymbol{k},\sigma,\pm} = C_{\pm,k} c^\dagger_{\boldsymbol{k},\sigma} + \sqrt{1 - C^2_{\pm,k}}\, f^\dagger_{\boldsymbol{k},\sigma} , \tag{2.8}$$

the coefficients obtained from diagonalising the matrix

$$H_{1\mathrm{p}}(\boldsymbol{k}) = \begin{pmatrix} \epsilon_{\boldsymbol{k}} & -V_\perp \\ -V_\perp & \epsilon_{\boldsymbol{k}} - \eta_d \end{pmatrix} . \tag{2.9}$$

The energies are given by $\mathcal{E}_\pm(\boldsymbol{k}) = \epsilon_k - \eta_d/2 \pm \Delta$, with ϵ_k being $-\frac{4V}{Z}(\cos k_x + \cos k_y)$ for a 2D square lattice and $\Delta = \sqrt{V_\perp^2 + \eta_d^2/4}$. Substituting into the expression for the lattice Greens function, we get

$$G(\boldsymbol{k}, d; \omega) = C_{+,\boldsymbol{k}}\mathcal{G}(\mathcal{T}_{\boldsymbol{k},+,d}; \omega - (\epsilon_k - \eta_d/2 + \Delta)) + C_{-,\boldsymbol{k}}\mathcal{G}(\mathcal{T}_{\boldsymbol{k},-,d}; \omega - (\epsilon_k - \eta_d/2 - \Delta)) . \tag{2.10}$$

Following eq. 5.33, we can similarly write down an expression the local lattice Greens function for the $f-$ and $d-$electrons. For that, we define the local diagonal modes $\nu_\pm$ obtained from the local Hamiltonian

$$H_{\mathrm{loc}} = \begin{pmatrix} 0 & -V_\perp \\ -V_\perp & 0 - \eta_d \end{pmatrix} , \tag{2.11}$$

and their associated energies $\varepsilon_\pm$. Within the impurity model, these local bonding/antibonding modes can be constructed on the impurity sites ($\nu_\pm$) or on the neighbouring bath sites, ($\nu^Z_\pm$). The local Greens function can be expressed in terms of these modes:

$$\begin{aligned} G_{dd}(\omega) =& \frac{1}{2}\sum_{\nu}\left[C_{\nu,d}\mathcal{G}(\nu\sigma, d\sigma; \omega - \varepsilon_\nu) + C^*_{\nu,d}\mathcal{G}(d\sigma, \nu\sigma; \omega - \varepsilon_\nu)\right] \\ &+ \frac{1}{2}\sum_{\nu^Z}\left[C^*_{\nu^Z,Z_d}\mathcal{G}(\mathcal{T}_{Z_d,\sigma}, \mathcal{T}_{\nu^Z,\sigma}; \omega - \varepsilon_{\nu^Z}) + C_{\nu^Z,Z_d}\mathcal{G}(\mathcal{T}_{\nu^Z,\sigma}, \mathcal{T}_{Z_d,\sigma}; \omega - \varepsilon_{\nu^Z})\right] . \end{aligned} \tag{2.12}$$

The presence of local hybridisation between the $d-$ and $f-$ orbitals makes the inter-orbital Greens function important as well. Following eq. 5.35, we have

$$\begin{aligned} G_{fd}(\omega) =& \frac{1}{2}\sum_{\nu}\left[C_{\nu,f}\mathcal{G}(\nu\sigma, d\sigma; \omega - \varepsilon_\nu) + C^*_{\nu,d}\mathcal{G}(f\sigma, \nu\sigma; \omega - \varepsilon_\nu)\right] \\ &+ \frac{1}{2}\sum_{\nu^Z}\left[C_{\nu^Z,Z_f}\mathcal{G}(\mathcal{T}_{\nu^Z,\sigma}, \mathcal{T}_{Z_d,\sigma}; \omega - \varepsilon_{\nu^Z}) + C^*_{\nu^Z,Z_d}\mathcal{G}(\mathcal{T}_{Z_f,\sigma}, \mathcal{T}_{\nu^Z,\sigma}; \omega - \varepsilon_{\nu^Z})\right] . \end{aligned} \tag{2.13}$$

3.3 Unitary RG Analysis of The Bilayer Lattice-Embedded ES-IAM

In the limit of large U_α, we carry out a Schrieffer-Wolff transformation and work with the following low-energy Hamiltonian:

$$H_{\rm aux} = \sum_{\mathbf{k},\sigma,\alpha} \epsilon_{\mathbf{k},\alpha} n_{\mathbf{k},\sigma,\alpha} + \sum_\alpha \sum_{Z\in \rm NN}\left[\frac{1}{2}J_\alpha \sum_{\sigma,\sigma'} \mathbf{S}_\alpha\cdot \boldsymbol{\sigma}_{\alpha\beta} c^\dagger_{Z,\sigma,\alpha}c_{Z,\sigma',\alpha} - \frac{W_\alpha}{2}\left(n_{Z,\uparrow,\alpha} - n_{Z,\downarrow,\alpha}\right)^2\right] + J\mathbf{S}_f\cdot\mathbf{S}_d\ . \tag{3.1}$$

The Kondo coupling and bath interaction acquire $k-$dependence:

$$\begin{aligned} J(\mathbf{k},\mathbf{q}) &= \frac{J}{2}\left[\cos(\mathbf{k}_x - \mathbf{q}_x) + \cos(\mathbf{k}_y - \mathbf{q}_y)\right]\ , \\ W(\mathbf{k},\mathbf{q},\mathbf{k}',\mathbf{q}') &= \frac{W}{2}\left[\cos(\mathbf{k}_x - \mathbf{q}_x + \mathbf{k}'_x - \mathbf{q}'_x) + \cos(\mathbf{k}_y - \mathbf{q}_y + \mathbf{k}'_y - \mathbf{q}'_y)\right]\ . \end{aligned} \tag{3.2}$$

The particle-hole transformed state $\bar{\mathbf{q}}$ associated with $\mathbf{q}$ is defined as $\bar{\mathbf{q}} = \mathbf{q} + (\pi,\pi)$ if $\mathbf{q}$ is in the first or third quadrant and $\mathbf{q} + (\pi,-\pi)$ is in the second or fourth quadrant.

In order to study the low-energy physics of the impurity model, we iteratively integrate out high-energy degrees of freedom using the unitary RG method. Let the momentum states being decoupled from the two layers be $|\mathbf{q},\sigma,f\rangle$ and $|\mathbf{q},\sigma,d\rangle$ if they are above the Fermi energy (unoccupied in the low-energy configuration) and $|\bar{\mathbf{q}},\sigma,f\rangle$ and $|\bar{\mathbf{q}},\sigma,d\rangle$ if they are below the Fermi energy (occupied in the low-energy configuration) For every shell of conduction bath states of width ΔD integrated out, the Hamiltonian couplings renormalise by folding in the effects of virtual excitations to these states. We already have the renormalisation group equations for the couplings J_α in the case of $J=0$:

$$\frac{\Delta J_f(\mathbf{k}_1,\mathbf{k}_2)}{\Delta D} = -\int_{UV} d\mathbf{q}\,\rho(\mathbf{q})\left[J_f(\mathbf{k}_1,\mathbf{q})J_f(\mathbf{q},\mathbf{k}_2) + 4J_f(\mathbf{q},\bar{\mathbf{q}})W_f(\mathbf{k}_1,\mathbf{q},\bar{\mathbf{q}},\mathbf{k}_2)\right]G^{(0)}_f(\mathbf{q})\ , \tag{3.3}$$

where $\mathbf{q}$ sums over the UV states being integrated out, $\rho(\mathbf{q})$ is the density of states for the momentum states being integrated out and $G^{(0)}_f$ is the propagator for the excited intermediate state (for $J=0$):

$$G^{(0)}_f(\mathbf{q}) = \frac{1}{\omega - |\epsilon_f(\mathbf{q})|/2 + J_f(\mathbf{q})/4 + W_f(\mathbf{q})/2}\ . \tag{3.4}$$

An identical equations holds for ΔJ_d, with al $f-$quantities replaced with $d-$quantities. Switching on J modifies the excitation energy in the propagator:

$$G_f(\mathbf{q}) = \frac{1}{1/G^{(0)}_f - J\vec{S}_f\cdot\vec{S}_d}\ . \tag{3.5}$$

This expression can be simplified by noting that the operator in the denominator has only two eigenvalues, $-3/4$ in the singlet sector and $1/4$ in the triplet sector. Defining the projectors $\mathcal{P}_0$ and $\mathcal{P}_1$ for the two sectors, we have

$$\vec{S}_f\cdot\vec{S}_d = \frac{-3}{4}\mathcal{P}_0 + \frac{1}{4}\mathcal{P}_1, \quad \mathcal{P}_0 + \mathcal{P}_1 = \mathbb{I}\ . \tag{3.6}$$

Using these relations, we can write

$$\begin{aligned} G_f(\mathbf{q}) &= \frac{1}{1/G_f^{(0)} + 3J/4}\mathcal{P}_0 + \frac{1}{1/G_f^{(0)} - J/4}\mathcal{P}_1 \\ &= \left(\frac{1/4}{1/G_f^{(0)} + 3J/4} + \frac{3/4}{1/G_f^{(0)} - J/4}\right)\mathbb{I} + \left(\frac{1}{1/G_f^{(0)} + 3J/4} + \frac{1}{1/G_f^{(0)} - J/4}\right)\vec{S}_f \cdot \vec{S}_d \; . \end{aligned} \tag{3.7}$$

Noting that only the first operator renormalises the Kondo interaction, we get the following modified RG equation for the Kondo coupling J_α:

$$\begin{aligned} \frac{\Delta J_\alpha}{\Delta D} = \int_{UV} d\mathbf{q}\, \rho(\mathbf{q}) \left[J_\alpha(\mathbf{k}_1, \mathbf{q}) J_\alpha(\mathbf{q}, \mathbf{k}_2) + J_\alpha(\mathbf{q}, \bar{\mathbf{q}}) W_\alpha(\mathbf{k}_1, \mathbf{q}, \bar{\mathbf{q}}, \mathbf{k}_2)\right] \left(\frac{1}{4}\frac{1}{1/G_\alpha^{(0)}(\mathbf{q}) + 3J/4}\right. \\ \left. + \frac{3}{4}\frac{1}{1/G_\alpha^{(0)}(\mathbf{q}) - J/4}\right) \; , \end{aligned} \tag{3.8}$$

We now turn to the renormalisation of J, arising from hybridisation of $f-$ and $d-$layers. In order to capture coherent scattering processes between the two layers, we project onto the following rotated basis of excited states:

$$\begin{aligned} |H\rangle_\pm &= \frac{1}{\sqrt{2}}\left(|H(\mathbf{q})\rangle_f \pm |H(\mathbf{q})\rangle_d\right) \; , \\ \mathcal{P}(\mathbf{q})_H &= |H\rangle_+ \langle H|_+ + |H\rangle_- \langle H|_- \; , \end{aligned} \tag{3.9}$$

where $|H(\mathbf{q})\rangle_\alpha$ is the state obtained upon exciting both states in the layer α: $|H(q)\rangle_\alpha = c^\dagger_{\mathbf{q},\alpha} c_{\bar{\mathbf{q}},\alpha} |L\rangle_f |L\rangle_d$, and $\mathcal{P}(\mathbf{q})_H$ projects onto this rotated excited basis. We focus on the spin-flip component $JS_f^+ S_d^-$ of the Hamiltonian. There are two kinds of processes that renormalise this coupling. We first consider one that involves a spin-flip of the $d-$layer followed by a spin-flip of the $f-$layer:

$$\Delta H = \frac{1}{N^2}\sum_{\mathbf{q}\in \text{UV}} \langle L|\frac{1}{2}J_f(\bar{\mathbf{q}}, \mathbf{q}) S_f^+ c^\dagger_{\mathbf{q}-\downarrow, f} c_{\mathbf{q}+\uparrow, f} \mathcal{P}(\mathbf{q})_H G \mathcal{P}(\mathbf{q})_H \frac{1}{2} J_d(\mathbf{q}, \bar{\mathbf{q}}) S_d^- c^\dagger_{\mathbf{q}\uparrow, d} c_{\bar{\mathbf{q}}\downarrow, d} |L\rangle \; , \tag{3.10}$$

where $|L\rangle = |L\rangle_f |L\rangle_d$ is the initial configuration for both layers. G is the propagator for the excited state:

$$G = \frac{1}{\omega - H_D} \; , \tag{3.11}$$

where H_D is the diagonal part of the Hamiltonian corresponding to the excited(intermediate) state. The diagonal part consists of the kinetic energy and the Ising part of the Kondo interaction:

$$H_D = \sum_{\mathbf{q}, \sigma \in \mathbf{q}_\pm, \alpha} \epsilon_\alpha(\mathbf{q}) \tau_{\mathbf{q},\sigma} + \frac{1}{2}\sum_\alpha J_\alpha S_\alpha^z \sum_{\mathbf{q},\sigma} \sigma n_{\mathbf{q},\sigma,\alpha} + J\vec{S}_f \cdot \vec{S}_d \; . \tag{3.12}$$

Calculating the matrix elements of the propagator in the rotated basis gives

$$\mathcal{P}(\mathbf{q})_H G \mathcal{P}(\mathbf{q})_H = \frac{1}{2}\left(G_f + G_d\right)\left(|H\rangle_+ \langle H|_+ + |H\rangle_- \langle H|_-\right) + \frac{1}{2}\left(G_f - G_d\right)\left(|H\rangle_+ \langle H|_- + |H\rangle_- \langle H|_+\right) \; , \tag{3.13}$$

where

$$G_\alpha(\mathbf{q}) = \frac{1}{\omega - \frac{1}{2}\left[\epsilon_\alpha(\mathbf{q}) - \epsilon_\alpha(\bar{\mathbf{q}})\right] + \frac{1}{2}J_\alpha(\mathbf{q}) + W_\alpha(\mathbf{q}) - \frac{1}{4}J} \ . \tag{3.14}$$

In eq. 3.10, since the first process is an excitation into a $d-$state and the second process is a de-excitation from an $f-$state, the expression can be simplified into

$$\begin{aligned}\langle L|S_f^+ c^\dagger_{\mathbf{q}-\downarrow,f} c_{\mathbf{q}+\uparrow,f}\mathcal{P}(\mathbf{q})_H G\mathcal{P}(\mathbf{q})_H S_d^- c^\dagger_{\mathbf{q}+\uparrow,d} c_{\mathbf{q}-\downarrow,d}|L\rangle &= S_f^+ S_d^- \langle H_f|\mathcal{P}(\mathbf{q})_H G\mathcal{P}(\mathbf{q})_H|H_d\rangle \\ &= \frac{1}{2}\left(G_f + G_d\right) S_f^+ S_d^- \ .\end{aligned} \tag{3.15}$$

The net Hamiltonian renormalisation is finally a sum over the contribution from each mode:

$$\Delta H = \frac{1}{8N^2}\sum_{\mathbf{q}\in \mathrm{UV}} J_f(\bar{\mathbf{q}},\mathbf{q}) J_d(\mathbf{q},\bar{\mathbf{q}})\left(G_f(\mathbf{q}) + G_d(\mathbf{q})\right) S_f^+ S_d^- \ . \tag{3.16}$$

Converting the sums to integrals (assuming a density of states $\rho(\mathbf{q})$) gives the final expression

$$\frac{\Delta H}{\Delta D} = \frac{1}{8N} S_f^+ S_d^- \int_{UV} d\mathbf{q}\, \rho(\mathbf{q}) J_f(\bar{\mathbf{q}},\mathbf{q}) J_d(\mathbf{q},\bar{\mathbf{q}})\left(G_f(\mathbf{q}) + G_d(\mathbf{q})\right) \ . \tag{3.17}$$

The time-reversed of this process can be obtained simply by exchanging the $d-$interaction with the $f-$interaction. Since the renormalisation itself is symmetric, the contribution is equal to what we obtained here. The total renormalisation in the coupling for the term $S_f^+ S_d^-$ is therefore

$$\begin{aligned}\frac{\Delta J}{\Delta D} &= \frac{1}{2}\int_{UV} d\mathbf{q}\, \rho(\mathbf{q}) J_f(\bar{\mathbf{q}},\mathbf{q}) J_d(\mathbf{q},\bar{\mathbf{q}})\left(G_f(\mathbf{q}) + G_d(\mathbf{q})\right) \\ &= \frac{1}{2}\int_{UV} d\mathbf{q}\, \rho(\mathbf{q})\left[\frac{J_f(\bar{\mathbf{q}},\mathbf{q}) J_d(\mathbf{q},\bar{\mathbf{q}})}{\omega - \frac{1}{2}\bar{\epsilon}_f(\mathbf{q}) + \frac{1}{2}J_f(\mathbf{q}) + W_f(\mathbf{q}) - \frac{1}{4}J} + \frac{J_f(\bar{\mathbf{q}},\mathbf{q}) J_d(\mathbf{q},\bar{\mathbf{q}})}{\omega - \frac{1}{2}\bar{\epsilon}_d(\mathbf{q}) + \frac{1}{2}J_d(\mathbf{q}) + W_d(\mathbf{q}) - \frac{1}{4}J}\right] ,\end{aligned} \tag{3.18}$$

where $\bar{\epsilon}_\alpha(\mathbf{q}) = \epsilon_f(\mathbf{q}) - \epsilon_f(\bar{\mathbf{q}})$ is the total excitation energy for the pair of states $\mathbf{q}, \bar{\mathbf{q}}$.

Since the bath interaction doesn't renormalise, it's functional form remains unchanged and can be substituted into the RG equations. In fact, we can write the expressions in a compact matrix form for efficient computation:

$$\frac{\Delta J_\alpha}{\Delta D} = -J_\alpha \mathcal{G}_\alpha J_\alpha^\dagger - 4\mathcal{W}_\alpha \mathrm{Tr}\left[\mathcal{G}'_\alpha\right] , \tag{3.19}$$

where all objects are matrices. J_α is the Kondo coupling matrix. $\mathcal{G}_\alpha$ is a diagonal matrix with matrix elements

$$\mathcal{G}_\alpha[q,q] = \begin{cases}\rho(q)\left(\frac{1}{8}\frac{1}{1/G_\alpha^{(0)}(\mathbf{q})+\mu/2+3J/4} + \frac{3}{8}\frac{1}{1/G_\alpha^{(0)}(\mathbf{q})+\mu/2-J/4}\right), & \mathbf{q}\in \mathrm{PS} , \\ \rho(q)\left(\frac{1}{8}\frac{1}{1/G_\alpha^{(0)}(\mathbf{q})-\mu/2+3J/4} + \frac{3}{8}\frac{1}{1/G_\alpha^{(0)}(\mathbf{q})-\mu/2-J/4}\right), & \mathbf{q}\in \mathrm{HS} ,\end{cases} \tag{3.20}$$

if q is in the UV subspace and zero otherwise. $\mathcal{W}_\alpha$ is the bath interaction matrix whose matrix elements are given by $\mathcal{W}_\alpha[k,k'] = \frac{W}{2}\left[\cos(k_x - k'_x) + \cos(k_y - k'_y)\right]$. Finally, $\mathcal{G}'_\alpha$ is another diagonal matrix whole matrix elements are defined as $\mathcal{G}'_\alpha[q,q] = \mathcal{G}[q,q] J_\alpha[q,\bar{q}]$.

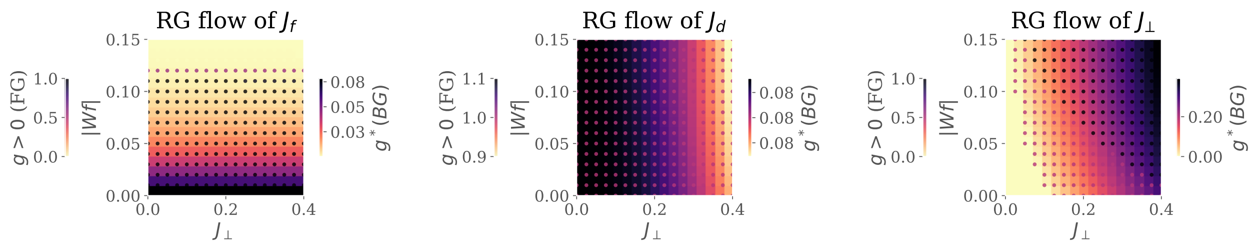

Figure 3.1: RG phase diagrams for various Kondo couplings: $f-$layer (left), $d-$layer (middle) and inter-layer (right). For each plot, the background color (quantified by the right colorbar) tracks the fixed point value of the coupling as a function of the inter-layer couplings $J_\perp$ and the $f-$layer bath interaction $|W_f|$. The marker color complements this by tracking whether the coupling is relevant, intermediate-coupling or irrelevant in the RG sense, by taking the values 1, $1/2$ or 0 respectively.

The RG equation for the inter-layer coupling J can be written as

$$\frac{\Delta J}{\Delta D} = \frac{1}{2}\mathbf{J} \cdot \mathbf{G} , \tag{3.21}$$

where $\mathbf{J}$ and $\mathbf{G}$ are vectors defined by the elements $\mathbf{J}[q] = J_f(\bar{\mathbf{q}}, \mathbf{q})J_d(\mathbf{q}, \bar{\mathbf{q}})$ and

$$\mathbf{G}[q] = \rho(\mathbf{q}) \left[\frac{1}{\omega - \frac{1}{2}\bar{\epsilon}_f(\mathbf{q}) + \frac{1}{2}J_f(\mathbf{q}) + W_f(\mathbf{q}) - \frac{1}{4}J} + \frac{1}{\omega - \frac{1}{2}\bar{\epsilon}_d(\mathbf{q}) + \frac{1}{2}J_d(\mathbf{q}) + W_d(\mathbf{q}) - \frac{1}{4}J}\right] . \tag{3.22}$$

3.4 Renormalisation Group Flows of Kondo Couplings: RG Phase Diagram

We first analyse the model by obtaining unitary RG flows of three kinds of Kondo couplings. Fig. 3.1 shows the fixed point values of the couplings under renormalisation to a low-energy theory. We view these values as a function of the $f-$layer bath interaction W_f and inter-layer Kondo coupling $J_\perp$. In each panel, the right (left) colorbar quantifies the background (marker) color of the plot. The background displays the fixed point value of each coupling whereas the marker color tracks whether the couplings is relevant (1), intermediate coupling (0.5) or irrelevant (0).

The Kondo coupling J_d (middle panel) of the $d-$layer remains relevant for the entirety of the chosen parameter regime. The Kondo coupling J_f (left panel) of the $f-$layer becomes RG-irrelevant for larger values of $|W_f|$; this leads to an in-plane Mott transition of the kind observed in our earlier work, where the local Fermi liquid excitations of the $f-$impurity give way to a pseudogapped non-Fermi liquid Mott metal phase which ultimately destabilises into a local moment phase. The RG flow of J_f does not appear to be significantly affected by the variation of $J_\perp$. Concomitant with the irrelevance of J_f under increasing $|W_f|$, the inter-plane coupling $J_\perp$ (right panel) becomes RG-relevant, as seen from the darkening markers as we go to higher $|W_f|$.

These simulations motivate an RG-based phase diagram with the following regimes: (i) weak $J_\perp$, weak $|W_f|$: decoupled metallic layers, each layer participating in transport independently, (ii) strong

$J_\perp$, strong $|W_f|$: Kondo insulator regime, where the local moments of the $f-$layer hybridise with the itinerant electrons of the $d-$layer to form a band insulator with the chemical potential situated in the hybridisation gap, and (iii) weak $J_\perp$, strong $|W_f|$: the so-called RKKY regime, where the $f-$layer forms local moments which decouple from the $d-$layer and are then free to order magnetically, and (iv) strong $J_\perp$, weak $|W_f|$: this is similar to the Kondo insulator regime, except here the $f-$electrons remain itinerant and it is they who hybridise with the $d-$electrons to again form two bands separated by a hybridisation gap.

3.5 Results at half-filling

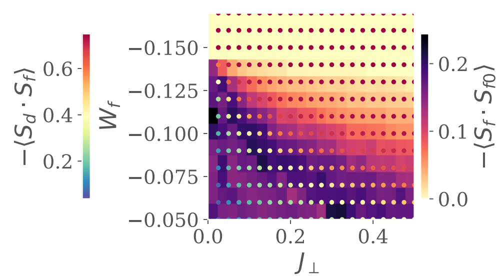

Figure 3.2: *Left: Real-space spin correlations for the auxiliary model. Background colour (right colourbar) represents that between the $f-$impurity and its nearest-neighbour sites, while the marker colour (left colourbar) represents that between the $f-$impurity and the $d-$impurity. While the former shrinks as $J_\perp$ or $|W_f|$ is increased, the latter grows. This marks a transfer of entanglement from intra-plane channel into inter-plane channel. Right:*

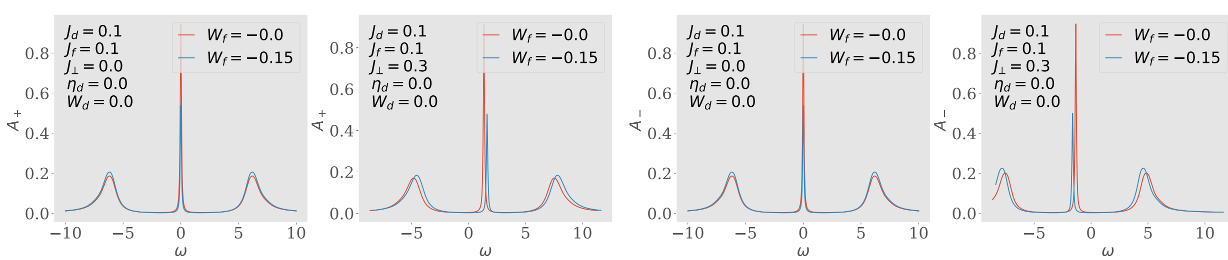

Figure 3.3: *Spectral functions of the tiled model, for excitations in the bonding (two on the left) and antibonding (two on the right) basis, defined by the Greens function $G_{k,\pm} = G(k_d \pm k_f; \omega)$. There is no hybridisation for $J_\perp = 0$ (first and third plots), and the only effect of W_f is to shrink the height of the central resonance. Introducing a non-zero $J_\perp$ splits the central resonance into an upper band (second from left) and a lower band (first from right). This leads to a Kondo insulator, where $\omega = 0$ resides inside the hybridisation gap.*

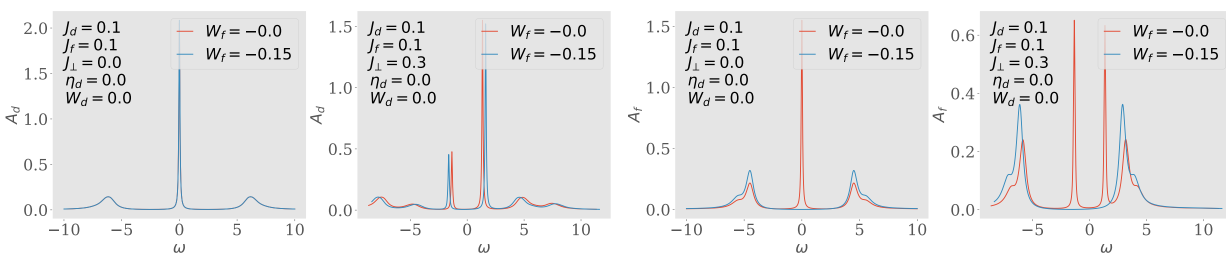

Figure 3.4: *Spectral functions A_d and A_f of the auxiliary model, for excitations in the $d-$ and $f-$layers respectively. Both show a central Kondo resonance for $J_\perp = 0$ which splits into two bands upon adding a non-zero $J_\perp$. Increasing $|W_f|$ to large values gaps out the local Fermi liquid excitations of A_f.*

Bibliography

[1] John Hubbard. Electron correlations in narrow energy bands. In *Proceedings of the royal society of london a: mathematical, physical and engineering sciences*, volume 276, pages 238–257. The Royal Society, 1963.

[2] N. F. Mott. Metal-insulator transition. *Rev. Mod. Phys.*, 40:677–683, Oct 1968.

[3] R F Milligan and G A Thomas. The metal-insulator transition. *Annual Review of Physical Chemistry*, 36(1):139–158, 1985.

[4] Masatoshi Imada, Atsushi Fujimori, and Yoshinori Tokura. Metal-insulator transitions. *Reviews of modern physics*, 70(4):1039, 1998.

[5] Patrick A. Lee, Naoto Nagaosa, and Xiao-Gang Wen. Doping a mott insulator: Physics of high-temperature superconductivity. *Rev. Mod. Phys.*, 78:17–85, Jan 2006.

[6] A. Pustogow, M. Bories, A. Löhle, R. Rösslhuber, E. Zhukova, B. Gorshunov, S. Tomić, J. A. Schlueter, R. Hübner, T. Hiramatsu, Y. Yoshida, G. Saito, R. Kato, T.-H. Lee, V. Dobrosavljević, S. Fratini, and M. Dressel. Quantum spin liquids unveil the genuine mott state. *Nature Materials*, 17(9):773–777, Sep 2018.

[7] G. R. Stewart. Non-fermi-liquid behavior in d- and f-electron metals. *Rev. Mod. Phys.*, 73:797–855, Oct 2001.

[8] Yuan Cao, Valla Fatemi, Ahmet Demir, Shiang Fang, Spencer L. Tomarken, Jason Y. Luo, Javier D. Sanchez-Yamagishi, Kenji Watanabe, Takashi Taniguchi, Efthimios Kaxiras, Ray C. Ashoori, and Pablo Jarillo-Herrero. Correlated insulator behaviour at half-filling in magic-angle graphene superlattices. *Nature*, 556(7699):80–84, Apr 2018.

[9] Yuan Cao, Valla Fatemi, Shiang Fang, Kenji Watanabe, Takashi Taniguchi, Efthimios Kaxiras, and Pablo Jarillo-Herrero. Unconventional superconductivity in magic-angle graphene superlattices. *Nature*, 556(7699):43–50, Apr 2018.

[10] J H de Boer and E J W Verwey. Semi-conductors with partially and with completely filled 3d-lattice bands. *Proceedings of the Physical Society*, 49(4S):59, aug 1937.

[11] N F Mott. The basis of the electron theory of metals, with special reference to the transition metals. *Proceedings of the Physical Society. Section A*, 62(7):416–422, jul 1949.

[12] N. F. Mott. On the transition to metallic conduction in semiconductors. *Canadian Journal of Physics*, 34(12A):1356–1368, 1956.

[13] N. F. Mott. The transition to the metallic state. *The Philosophical Magazine: A Journal of Theoretical Experimental and Applied Physics*, 6(62):287–309, 1961.

[14] Takayuki Isono, Taichi Terashima, Kazuya Miyagawa, Kazushi Kanoda, and Shinya Uji. Quantum criticality in an organic spin-liquid insulator κ-(bedt-ttf)2cu2(cn)3. *Nature Communications*, 7(1):13494, Nov 2016.

[15] P. W. Anderson. New approach to the theory of superexchange interactions. *Phys. Rev.*, 115:2–13, Jul 1959.

[16] Junjiro Kanamori. Electron Correlation and Ferromagnetism of Transition Metals. *Progress of Theoretical Physics*, 30(3):275–289, 09 1963.

[17] Martin C. Gutzwiller. Effect of correlation on the ferromagnetism of transition metals. *Phys. Rev. Lett.*, 10:159–162, Mar 1963.

[18] Antoine Georges, Gabriel Kotliar, Werner Krauth, and Marcelo J Rozenberg. Dynamical mean-field theory of strongly correlated fermion systems and the limit of infinite dimensions. *Reviews of Modern Physics*, 68(1):13, 1996.

[19] W. F. Brinkman and T. M. Rice. Application of gutzwiller's variational method to the metal-insulator transition. *Phys. Rev. B*, 2:4302–4304, Nov 1970.

[20] Martin C. Gutzwiller. Correlation of electrons in a narrow s band. *Phys. Rev.*, 137:A1726–A1735, Mar 1965.

[21] Peter Prelovšek, Jure Kokalj, Zala Lenarčič, and Ross H. McKenzie. Holon-doublon binding as the mechanism for the mott transition. *Phys. Rev. B*, 92:235155, Dec 2015.

[22] Sen Zhou, Yupeng Wang, and Ziqiang Wang. Doublon-holon binding, mott transition, and fractionalized antiferromagnet in the hubbard model. *Phys. Rev. B*, 89:195119, May 2014.

[23] Tsutomu Watanabe, Hisatoshi Yokoyama, Yukio Tanaka, and Jun-ichiro Inoue. Superconductivity and a mott transition in a hubbard model on an anisotropic triangular lattice. *Journal of the Physical Society of Japan*, 75(7):074707, 2006.

[24] Philip Phillips. *Advanced solid state physics*. Cambridge University Press, 2012. Discussion regarding f-sum rule and dynamical spectral weight transfer in pages 394-395, section 12.4 on Cooper pairs and computation of effective two particle energy.

[25] Richard M. Martin, Lucia Reining, and David M. Ceperley. *Interacting Electrons: Theory and Computational Approaches*. Cambridge University Press, 2016.

[26] Walter Metzner and Dieter Vollhardt. Correlated lattice fermions in $d = \infty$ dimensions. *Phys. Rev. Lett.*, 62:324–327, Jan 1989.

[27] J. Zaanen, G. A. Sawatzky, and J. W. Allen. Band gaps and electronic structure of transition-metal compounds. *Phys. Rev. Lett.*, 55:418, 1985.

[28] M. Imada, A. Fujimori, and Y. Tokura. Metal-insulator transitions, 1998.

[29] D. B. McWhan, T. M. Rice, and J. P. Remeika. Mott transition in cr-doped $\mathrm{v_2o_3}$. *Phys. Rev. Lett.*, 23:1384–1387, Dec 1969.

[30] P. Limelette, A. Georges, D. Jérome, P. Wzietek, P. Metcalf, and J. M. Honig. Universality and critical behavior at the mott transition. *Science*, 302(5642):89–92, 2003.

[31] M. J. Rozenberg, G. Kotliar, H. Kajueter, G. A. Thomas, D. H. Rapkine, J. M. Honig, and P. Metcalf. Optical conductivity in mott-hubbard systems. *Phys. Rev. Lett.*, 75:105–108, Jul 1995.

[32] M. M. Qazilbash, M. Brehm, Byung-Gyu Chae, P.-C. Ho, G. O. Andreev, Bong-Jun Kim, Sun Jin Yun, A. V. Balatsky, M. B. Maple, F. Keilmann, Hyun-Tak Kim, and D. N. Basov. Mott transition in vo2 revealed by infrared spectroscopy and nano-imaging. *Science*, 318(5857):1750–1753, 2007.

[33] H. Takagi, B. Batlogg, H. L. Kao, J. Kwo, R. J. Cava, J. J. Krajewski, and W. F. Peck. Systematic evolution of temperature-dependent resistivity in $_{2-x}sr_{x4}$. *Phys. Rev. Lett.*, 69:2975–2978, Nov 1992.

[34] Yoichi Ando, Seiki Komiya, Kouji Segawa, S. Ono, and Y. Kurita. Electronic phase diagram of high-T_c cuprate superconductors from a mapping of the in-plane resistivity curvature. *Phys. Rev. Lett.*, 93:267001, Dec 2004.

[35] T. R. Lemberger, I. Hetel, A. Tsukada, M. Naito, and M. Randeria. Superconductor-to-metal quantum phase transition in overdoped $\mathrm{la}_{2-x}\mathrm{sr}_x\mathrm{cuo}_4$. *Phys. Rev. B*, 83:140507, Apr 2011.

[36] Andrea Damascelli, Zahid Hussain, and Zhi-Xun Shen. Angle-resolved photoemission studies of the cuprate superconductors. *Rev. Mod. Phys.*, 75:473–541, Apr 2003.

[37] Jonathan A. Sobota, Yu He, and Zhi-Xun Shen. Angle-resolved photoemission studies of quantum materials. *Rev. Mod. Phys.*, 93:025006, May 2021.

[38] I M Vishik. Photoemission perspective on pseudogap, superconducting fluctuations, and charge order in cuprates: a review of recent progress. *Reports on Progress in Physics*, 81(6):062501, apr 2018.

[39] María Luisa Medarde. Structural, magnetic and electronic properties of perovskites (r = rare earth). *Journal of Physics: Condensed Matter*, 9(8):1679, feb 1997.

[40] Yuxin Wang, Kun Jiang, Jianjun Ying, Tao Wu, Jinguang Cheng, Jiangping Hu, and Xianhui Chen. Recent progress in nickelate superconductors. *National Science Review*, 12(10):nwaf373, 09 2025.

[41] Pascal Puphal, Thomas Schäfer, Bernhard Keimer, and Matthias Hepting. Superconductivity in infinite-layer and ruddlesden–popper nickelates. *Nature Reviews Physics*, 8(2):70–85, Feb 2026.

[42] Sajedeh Manzeli, Dmitry Ovchinnikov, Diego Pasquier, Oleg V. Yazyev, and Andras Kis. 2d transition metal dichalcogenides. *Nature Reviews Materials*, 2(8):17033, Jun 2017.

[43] Saju Joseph, Jainy Mohan, Seetha Lakshmy, Simil Thomas, Brahmananda Chakraborty, Sabu Thomas, and Nandakumar Kalarikkal. A review of the synthesis, properties, and applications of 2d transition metal dichalcogenides and their heterostructures. *Materials Chemistry and Physics*, 297:127332, 2023.

[44] J. A. Wilson, F. J. Di Salvo, and S. Mahajan. Charge-density waves in metallic, layered, transition-metal dichalcogenides. *Phys. Rev. Lett.*, 32:882–885, Apr 1974.

[45] J. A. Wilson, F. J. Di Salvo, and S. Mahajan. Charge-density waves and superlattices in the metallic layered transition metal dichalcogenides. *Advances in Physics*, 50(8):1171–1248, 2001.

[46] P. Fazekas and E. Tosatti. Electrical, structural and magnetic properties of pure and doped 1t-tas$_2$. *Philosophical Magazine B*, 39:229–244, 1979.

[47] L. Perfetti, T. A. Gloor, F. Mila, H. Berger, and M. Grioni. Unexpected periodicity in the quasi-two-dimensional mott insulator $1t-\mathrm{tas}_2$ revealed by angle-resolved photoemission. *Phys. Rev. B*, 71:153101, Apr 2005.

[48] B. Sipos, A. F. Kusmartseva, A. Akrap, H. Berger, L. Forró, and E. Tutiš. From mott state to superconductivity in 1t-tas2. *Nature Materials*, 7(12):960–965, Dec 2008.

[49] Doohee Cho, Sangmo Cheon, Ki-Seok Kim, Sung-Hoon Lee, Yong-Heum Cho, Sang-Wook Cheong, and Han Woong Yeom. Nanoscale manipulation of the mott insulating state coupled to charge order in 1t-tas2. *Nature Communications*, 7(1):10453, Jan 2016.

[50] S. V. Borisenko, A. A. Kordyuk, A. N. Yaresko, V. B. Zabolotnyy, D. S. Inosov, R. Schuster, B. Büchner, R. Weber, R. Follath, L. Patthey, and H. Berger. Pseudogap and charge density waves in two dimensions. *Phys. Rev. Lett.*, 100:196402, May 2008.

[51] S. V. Borisenko, A. A. Kordyuk, V. B. Zabolotnyy, D. S. Inosov, D. Evtushinsky, B. Büchner, A. N. Yaresko, A. Varykhalov, R. Follath, W. Eberhardt, L. Patthey, and H. Berger. Two energy gaps and fermi-surface “arcs” in nbse$_2$. *Phys. Rev. Lett.*, 102:166402, Apr 2009.

[52] V. Vescoli et al. Charge-density-wave correlations in the optical properties of layered transition-metal dichalcogenides. *Physical Review Letters*, 81:453–456, 1998.

[53] Lijun Li, Xiaoyu Deng, Zhen Wang, Yu Liu, Milinda Abeykoon, Eric Dooryhee, Aleksandra Tomic, Yanan Huang, John B. Warren, Emil S. Bozin, Simon J. L. Billinge, Yuping Sun, Yimei Zhu, Gabriel Kotliar, and Cedomir Petrovic. Superconducting order from disorder in 2h-tase2−xsx. *npj Quantum Materials*, 2(1):11, Feb 2017.

[54] Yoichi Kamihara, Takumi Watanabe, Masahiro Hirano, and Hideo Hosono. Iron-based layered superconductor la[o1-xfx]feas (x = 0.05−0.12) with tc = 26 k. *Journal of the American Chemical Society*, 130(11):3296–3297, 2008. PMID: 18293989.

[55] J. Paglione and R. L. Greene. High-temperature superconductivity in iron-based materials, 2010.

[56] P. J. Hirschfeld, M. M. Korshunov, and I. I. Mazin. Gap symmetry and structure of fe-based superconductors, 2011.

[57] G. R. Stewart. Superconductivity in iron compounds. *Rev. Mod. Phys.*, 83:1589–1652, Dec 2011.

[58] David C. Johnston. The puzzle of high temperature superconductivity in layered iron pnictides and chalcogenides. *Advances in Physics*, 59(6):803–1061, 2010.

[59] S. Kasahara, H. J. Shi, K. Hashimoto, S. Tonegawa, Y. Mizukami, T. Shibauchi, K. Sugimoto, T. Fukuda, T. Terashima, Andriy H. Nevidomskyy, and Y. Matsuda. Electronic nematicity above the structural and superconducting transition in bafe2(as1−xp x)2. *Nature*, 486(7403):382–385, Jun 2012.

[60] R. M. Fernandes, A. V. Chubukov, and J. Schmalian. What drives nematic order in iron-based superconductors? *Nature Physics*, 10(2):97–104, Feb 2014.

[61] James G. Analytis, H.-H. Kuo, Ross D. McDonald, Mark Wartenbe, P. M. C. Rourke, N. E. Hussey, and I. R. Fisher. Transport near a quantum critical point in bafe2(as1−xpx)2. *Nature Physics*, 10(3):194–197, Mar 2014.

[62] M. M. Qazilbash, J. J. Hamlin, R. E. Baumbach, Lijun Zhang, D. J. Singh, M. B. Maple, and D. N. Basov. Electronic correlations in the iron pnictides. *Nature Physics*, 5(9):647–650, Sep 2009.

[63] H. Shishido, A. F. Bangura, A. I. Coldea, S. Tonegawa, K. Hashimoto, S. Kasahara, P. M. C. Rourke, H. Ikeda, T. Terashima, R. Settai, Y. Ōnuki, D. Vignolles, C. Proust, B. Vignolle, A. McCollam, Y. Matsuda, T. Shibauchi, and A. Carrington. Evolution of the fermi surface of $\mathrm{bafe}_2(\mathrm{as}_{1-x}\mathbf{p}_x)_2$ on entering the superconducting dome. *Phys. Rev. Lett.*, 104:057008, Feb 2010.

[64] K. Terashima, Y. Sekiba, J. H. Bowen, K. Nakayama, T. Kawahara, T. Sato, P. Richard, Y.-M. Xu, L. J. Li, G. H. Cao, Z.-A. Xu, H. Ding, and T. Takahashi. Fermi surface nesting induced strong pairing in iron-based superconductors. *Proceedings of the National Academy of Sciences*, 106(18):7330–7333, 2009.

[65] J.-H. Chu, H.-H. Kuo, J. G. Analytis, and I. R. Fisher. Divergent nematic susceptibility in an iron arsenide superconductor. *Science*, 337:710–712, 2012.

[66] Hsueh-Hui Kuo, Jiun-Haw Chu, Johanna C. Palmstrom, Steven A. Kivelson, and Ian R. Fisher. Ubiquitous signatures of nematic quantum criticality in optimally doped fe-based superconductors. *Science*, 352(6288):958–962, 2016.

[67] D. S. Inosov, J. T. Park, P. Bourges, D. L. Sun, Y. Sidis, A. Schneidewind, K. Hradil, D. Haug, C. T. Lin, B. Keimer, and V. Hinkov. Normal-state spin dynamics and temperature-dependent spin-resonance energy in optimally doped $BaFe_{1.85}Co_{0.15}As_2$. *Nature Physics*, 6:178–181, 2010.

[68] Andrey Chubukov. Pairing mechanism in fe-based superconductors. *Annual Review of Condensed Matter Physics*, 3(Volume 3, 2012):57–92, 2012.

[69] G. R. Stewart. Heavy-fermion systems. *Reviews of Modern Physics*, 56:755–787, 1984.

[70] Steffen Wirth and Frank Steglich. Exploring heavy fermions from macroscopic to microscopic length scales. *Nature Reviews Materials*, 1(10):16051, Aug 2016.

[71] A. C. Hewson. *The Kondo Problem to Heavy Fermions*. Cambridge University Press, 1993.

[72] Hilbert v. Löhneysen, Achim Rosch, Matthias Vojta, and Peter Wölfle. Fermi-liquid instabilities at magnetic quantum phase transitions. *Rev. Mod. Phys.*, 79:1015–1075, Aug 2007.

[73] Philipp Gegenwart, Qimiao Si, and Frank Steglich. Quantum criticality in heavy-fermion metals. *Nature Physics*, 4(3):186–197, Mar 2008.

[74] J. Custers, P. Gegenwart, H. Wilhelm, K. Neumaier, Y. Tokiwa, O. Trovarelli, C. Geibel, F. Steglich, C. Pépin, and P. Coleman. The break-up of heavy electrons at a quantum critical point. *Nature*, 424(6948):524–527, Jul 2003.

[75] S. Paschen, T. Lühmann, S. Wirth, P. Gegenwart, O. Trovarelli, C. Geibel, F. Steglich, P. Coleman, and Q. Si. Hall-effect evolution across a heavy-fermion quantum critical point. *Nature*, 432(7019):881–885, Dec 2004.

[76] A. Schröder, G. Aeppli, R. Coldea, M. Adams, O. Stockert, H.v. Löhneysen, E. Bucher, R. Ramazashvili, and P. Coleman. Onset of antiferromagnetism in heavy-fermion metals. *Nature*, 407(6802):351–355, Sep 2000.

[77] O. Stockert, J. Arndt, E. Faulhaber, C. Geibel, H. S. Jeevan, S. Kirchner, M. Loewenhaupt, K. Schmalzl, W. Schmidt, Q. Si, and F. Steglich. Magnetically driven superconductivity in cecu2si2. *Nature Physics*, 7(2):119–124, Feb 2011.

[78] Hiroaki Shishido, Rikio Settai, Hisatomo Harima, and Yoshichika Ōnuki. A drastic change of the fermi surface at a critical pressure in cerhin5: dhva study under pressure. *Journal of the Physical Society of Japan*, 74(4):1103–1106, 2005.

[79] Qimiao Si and Frank Steglich. Heavy fermions and quantum phase transitions. *Science*, 329(5996):1161–1166, 2010.

[80] Y Kuramoto and T Watanabe. Theory of momentum-dependent magnetic response in heavy-fermion systems. In *Proceedings of the Yamada Conference XVIII on Superconductivity in Highly Correlated Fermion Systems*, pages 80–83. Elsevier, 1987.

[81] D. L. Cox and N. Grewe. Transport properties of the anderson lattice. *Zeitschrift für Physik B Condensed Matter*, 71(3):321–340, Sep 1988.

[82] X. Y. Zhang, M. J. Rozenberg, and G. Kotliar. Mott transition in the d=∞ hubbard model at zero temperature. *Phys. Rev. Lett.*, 70:1666–1669, Mar 1993.

[83] O. Parcollet, G. Biroli, and G. Kotliar. Cluster dynamical mean field analysis of the mott transition. *Phys. Rev. Lett.*, 92:226402, Jun 2004.

[84] Thomas Maier, Mark Jarrell, Thomas Pruschke, and Matthias H. Hettler. Quantum cluster theories. *Rev. Mod. Phys.*, 77:1027–1080, Oct 2005.

[85] G. Kotliar, S. Y. Savrasov, K. Haule, V. S. Oudovenko, O. Parcollet, and C. A. Marianetti. Electronic structure calculations with dynamical mean-field theory. *Rev. Mod. Phys.*, 78:865–951, Aug 2006.

[86] Takuma Ohashi, Tsutomu Momoi, Hirokazu Tsunetsugu, and Norio Kawakami. Finite temperature mott transition in hubbard model on anisotropic triangular lattice. *Phys. Rev. Lett.*, 100:076402, Feb 2008.

[87] Barbara M. Terhal, Michael M. Wolf, and Andrew C. Doherty. Quantum entanglement: A modern perspective. *Physics Today*, 56(4):46–52, 2003.

[88] Ryszard Horodecki, Paweł Horodecki, Michał Horodecki, and Karol Horodecki. Quantum entanglement. *Reviews of modern physics*, 81(2):865, 2009.

[89] J. Eisert, M. Cramer, and M. B. Plenio. Colloquium: Area laws for the entanglement entropy. *Rev. Mod. Phys.*, 82:277–306, Feb 2010.

[90] Nicolas Laflorencie. Quantum entanglement in condensed matter systems. *Physics Reports*, 646:1–59, 2016.

[91] Bei Zeng, Xie Chen, Duan-Lu Zhou, and Xiao-Gang Wen. *Quantum information meets quantum matter*. Springer, 2019.

[92] Manuel Erhard, Mario Krenn, and Anton Zeilinger. Advances in high-dimensional quantum entanglement. *Nature Reviews Physics*, 2(7):365–381, Jul 2020.

[93] X. G. Wen. Topological orders in rigid states. *International Journal of Modern Physics B*, 04(02):239–271, 1990.

[94] X. G. Wen and Q. Niu. Ground-state degeneracy of the fractional quantum hall states in the presence of a random potential and on high-genus riemann surfaces. *Phys. Rev. B*, 41:9377–9396, May 1990.

[95] Xiao-Gang Wen. Topological orders and edge excitations in fractional quantum hall states. *Advances in Physics*, 44(5):405–473, 1995.

[96] Qian Niu, Ds J Thouless, and Yong-Shi Wu. Quantized hall conductance as a topological invariant. *Physical Review B*, 31(6):3372, 1985.

[97] Saurya Das and S. Shankaranarayanan. How robust is the entanglement entropy-area relation? *Phys. Rev. D*, 73:121701, Jun 2006.

[98] M. Cramer, J. Eisert, M. B. Plenio, and J. Dreißig. Entanglement-area law for general bosonic harmonic lattice systems. *Phys. Rev. A*, 73:012309, Jan 2006.

[99] Oskar Vafek and Ashvin Vishwanath. Dirac fermions in solids: From high-tc cuprates and graphene to topological insulators and weyl semimetals. *Annual Review of Condensed Matter Physics*, 5(1):83–112, 2014.

[100] B. Andrei Bernevig and Taylor L. Hughes. *Topological Insulators and Topological Superconductors*. Princeton University Press, 2013.

[101] Jérôme Cayssol. Introduction to dirac materials and topological insulators. *Comptes Rendus Physique*, 14(9):760–778, 2013. Topological insulators / Isolants topologiques.

[102] Roderich Moessner and Joel E. Moore. *Topological Phases of Matter*. Cambridge University Press, 2021.

[103] Stephan Rachel. Interacting topological insulators: a review. *Reports on Progress in Physics*, 81(11):116501, oct 2018.

[104] Jinying Wang, Shibin Deng, Zhongfan Liu, and Zhirong Liu. ”The rare two-dimensional materials with Dirac cones”. *National Science Review*, 2(1):22–39, 01 2015.

[105] Menghao Wu, Zhijun Wang, Junwei Liu, Wenbin Li, Huahua Fu, Lei Sun, Xin Liu, Minghu Pan, Hongming Weng, Mircea Dincă, Liang Fu, and Ju Li. Conetronics in 2d metal-organic frameworks: double/half dirac cones and quantum anomalous hall effect. *2D Materials*, 4(1):015015, nov 2016.

[106] Xiaojuan Ni, Huaqing Huang, and Jean-Luc Brédas. Emergence of a two-dimensional topological dirac semimetal phase in a phthalocyanine-based covalent organic framework. *Chemistry of Materials*, 34(7):3178–3184, 2022.

[107] Shi-Liang Zhu, Baigeng Wang, and L.-M. Duan. Simulation and detection of dirac fermions with cold atoms in an optical lattice. *Phys. Rev. Lett.*, 98:260402, Jun 2007.

[108] L. Lepori, G. Mussardo, and A. Trombettoni. (3+1) massive dirac fermions with ultracold atoms in frustrated cubic optical lattices. *EPL (Europhysics Letters)*, 92(5):50003, dec 2010.

[109] O Boada, A Celi, J I Latorre, and M Lewenstein. Dirac equation for cold atoms in artificial curved spacetimes. *New Journal of Physics*, 13(3):035002, mar 2011.

[110] Yoshihito Kuno, Ikuo Ichinose, and Yoshiro Takahashi. Generalized lattice wilson–dirac fermions in $(1+1)$ dimensions for atomic quantum simulation and topological phases. *Scientific Reports*, 8(1):10699, Jul 2018.

[111] Kenjiro K. Gomes, Warren Mar, Wonhee Ko, Francisco Guinea, and Hari C. Manoharan. Designer dirac fermions and topological phases in molecular graphene. *Nature*, 483(7389):306–310, Mar 2012.

[112] Xiang Yuan, Cheng Zhang, Yanwen Liu, Awadhesh Narayan, Chaoyu Song, Shoudong Shen, Xing Sui, Jie Xu, Haochi Yu, Zhenghua An, Jun Zhao, Stefano Sanvito, Hugen Yan, and Faxian Xiu. Observation of quasi-two-dimensional dirac fermions in zrte5. *NPG Asia Materials*, 8(11):e325–e325, Nov 2016.

[113] Matthieu Bellec, Ulrich Kuhl, Gilles Montambaux, and Fabrice Mortessagne. Topological transition of dirac points in a microwave experiment. *Phys. Rev. Lett.*, 110:033902, Jan 2013.

[114] Guifre Vidal, José Ignacio Latorre, Enrique Rico, and Alexei Kitaev. Entanglement in quantum critical phenomena. *Physical review letters*, 90(22):227902, 2003.

[115] J. I. Latorre, E. Rico, and G. Vidal. Ground state entanglement in quantum spin chains. 2003.

[116] J. I. Latorre, C. A. Lütken, E. Rico, and G. Vidal. Fine-grained entanglement loss along renormalization-group flows. *Phys. Rev. A*, 71:034301, Mar 2005.

[117] Pasquale Calabrese and John Cardy. Entanglement entropy and quantum field theory. *Journal of Statistical Mechanics: Theory and Experiment*, 2004(06):P06002, 2004.

[118] Michael M Wolf, Frank Verstraete, Matthew B Hastings, and J Ignacio Cirac. Area laws in quantum systems: mutual information and correlations. *Physical review letters*, 100(7):070502, 2008.

[119] H Casini, C D Fosco, and M Huerta. Entanglement and alpha entropies for a massive dirac field in two dimensions. *Journal of Statistical Mechanics: Theory and Experiment*, 2005(07):P07007–P07007, jul 2005.

[120] J. L. Cardy, O. A. Castro-Alvaredo, and B. Doyon. Form factors of branch-point twist fields in quantum integrable models and entanglement entropy. *Journal of Statistical Physics*, 130(1):129–168, Jan 2008.

[121] H Casini and M Huerta. Reduced density matrix and internal dynamics for multicomponent regions. *Classical and Quantum Gravity*, 26(18):185005, sep 2009.

[122] Benjamin Doyon. Bipartite entanglement entropy in massive two-dimensional quantum field theory. *Phys. Rev. Lett.*, 102:031602, Jan 2009.

[123] Xiao Chen, William Witczak-Krempa, Thomas Faulkner, and Eduardo Fradkin. Two-cylinder entanglement entropy under a twist. *Journal of Statistical Mechanics: Theory and Experiment*, 2017(4):043104, apr 2017.

[124] Pablo Bueno and William Witczak-Krempa. Holographic torus entanglement and its renormalization group flow. *Phys. Rev. D*, 95:066007, Mar 2017.

[125] Max A. Metlitski, Carlos A. Fuertes, and Subir Sachdev. Entanglement entropy in the $o(n)$ model. *Phys. Rev. B*, 80:115122, Sep 2009.

[126] Raúl E Arias, David D Blanco, and Horacio Casini. Entanglement entropy as a witness of the aharonov–bohm effect in QFT. *Journal of Physics A: Mathematical and Theoretical*, 48(14):145401, mar 2015.

[127] Seth Whitsitt, William Witczak-Krempa, and Subir Sachdev. Entanglement entropy of large-n wilson-fisher conformal field theory. *Phys. Rev. B*, 95:045148, Jan 2017.

[128] Wei Zhu, Xiao Chen, Yin-Chen He, and William Witczak-Krempa. Entanglement signatures of emergent dirac fermions: Kagome spin liquid and quantum criticality. *Science Advances*, 4(11):eaat5535, 2018.

[129] Shijie Hu, W. Zhu, Sebastian Eggert, and Yin-Chen He. Dirac spin liquid on the spin-$1/2$ triangular heisenberg antiferromagnet. *Phys. Rev. Lett.*, 123:207203, Nov 2019.

[130] Valentin Crépel, Anna Hackenbroich, Nicolas Regnault, and Benoit Estienne. Universal signatures of dirac fermions in entanglement and charge fluctuations. *Phys. Rev. B*, 103:235108, Jun 2021.

[131] G.'t Hooft. A planar diagram theory for strong interactions. *Nuclear Physics B*, 72(3):461–473, 1974.

[132] A.M. Polyakov. Thermal properties of gauge fields and quark liberation. *Physics Letters B*, 72(4):477–480, 1978.

[133] A.M. Polyakov. *Gauge Fields and Strings (1st ed.)*. Routledge, 1987.

[134] David J. Gross and Washington Taylor. Two-dimensional qcd is a string theory. *Nuclear Physics B*, 400(1):181–208, 1993.

[135] Raphael Bousso. The holographic principle. *Rev. Mod. Phys.*, 74:825–874, Aug 2002.

[136] Jan Zaanen, Yan Liu, Ya-Wen Sun, and Koenraad Schalm. *Holographic Duality in Condensed Matter Physics*. Cambridge University Press, 2015.

[137] S. A. Hartnoll, A. Lucas, and S. Sachdev. *Holographic Quantum Matter*. The MIT Press, 2018.

[138] E.T. Akhmedov. A remark on the ads/cft correspondence and the renormalization group flow. *Physics Letters B*, 442(1):152–158, 1998.

[139] Enrique Álvarez and César Gómez. Geometric holography, the renormalization group and the c-theorem. *Nuclear Physics B*, 541(1):441–460, 1999.

[140] D. Z. Freedman, S. S. Gubser, K. Pilch, and N. P. Warner. Renormalization group flows from holography–supersymmetry and a c-theorem. 1999.

[141] M. Porrati and A. Starinets. Rg fixed points in supergravity duals of 4-d field theory and asymptotically ads spaces. *Physics Letters B*, 454(1):77–83, 1999.

[142] Shamit Kachru, Michael Schulz, and Eva Silverstein. Self-tuning flat domain walls in 5d gravity and string theory. *Phys. Rev. D*, 62:045021, Jul 2000.

[143] L. Girardello, M. Petrini, M. Porrati, and A. Zaffaroni. The supergravity dual of n=1 super yang–mills theory. *Nuclear Physics B*, 569(1):451–469, 2000.

[144] Jan De Boer, Erik Verlinde, and Herman Verlinde. On the holographic renormalization group. *Journal of High Energy Physics*, 2000(08):003, 2000.

[145] Jan de Boer. The holographic renormalization group. *Fortschritte der Physik*, 49(4-6):339–358, 2001.

[146] Clifford V Johnson, Kenneth J Lovis, and David C Page. Probing some N = 1 AdS/CFT RG flows. *Journal of High Energy Physics*, 2001(05):036–036, may 2001.

[147] Satoshi Yamaguchi. AdS branes corresponding to superconformal defects. *Journal of High Energy Physics*, 2003(06):002–002, jun 2003.

[148] Jacob D. Bekenstein. Black holes and entropy. *Phys. Rev. D*, 7:2333–2346, Apr 1973.

[149] S. W. Hawking. Black hole explosions? *Nature*, 248(5443):30–31, Mar 1974.

[150] S. W. Hawking. Particle creation by black holes. *Communications in Mathematical Physics*, 43(3):199–220, Aug 1975.

[151] Juan Maldacena. The large-n limit of superconformal field theories and supergravity. *International journal of theoretical physics*, 38(4):1113–1133, 1999.

[152] S.S. Gubser, I.R. Klebanov, and A.M. Polyakov. Gauge theory correlators from non-critical string theory. *Physics Letters B*, 428(1):105–114, 1998.

[153] Edward Witten. Anti de sitter space and holography. *arXiv preprint hep-th/9802150*, 2:253–291, 1998.

[154] Shinsei Ryu and Tadashi Takayanagi. Holographic derivation of entanglement entropy from the anti–de sitter space/conformal field theory correspondence. *Physical review letters*, 96(18):181602, 2006.

[155] Shinsei Ryu and Tadashi Takayanagi. Aspects of holographic entanglement entropy. *Journal of High Energy Physics*, 2006(08):045, 2006.

[156] P. K. Kovtun, D. T. Son, and A. O. Starinets. Viscosity in strongly interacting quantum field theories from black hole physics. *Phys. Rev. Lett.*, 94:111601, Mar 2005.

[157] Kedar Damle and Subir Sachdev. Nonzero-temperature transport near quantum critical points. *Phys. Rev. B*, 56:8714–8733, Oct 1997.

[158] Sean A. Hartnoll, Pavel K. Kovtun, Markus Müller, and Subir Sachdev. Theory of the nernst effect near quantum phase transitions in condensed matter and in dyonic black holes. *Phys. Rev. B*, 76:144502, Oct 2007.

[159] Sung-Sik Lee. Non-fermi liquid from a charged black hole: A critical fermi ball. *Phys. Rev. D*, 79:086006, Apr 2009.

[160] Hong Liu, John McGreevy, and David Vegh. Non-fermi liquids from holography. *Phys. Rev. D*, 83:065029, Mar 2011.

[161] Thomas Faulkner, Hong Liu, John McGreevy, and David Vegh. Emergent quantum criticality, fermi surfaces, and ads$_2$. *Phys. Rev. D*, 83:125002, Jun 2011.

[162] Mihailo Čubrović, Jan Zaanen, and Koenraad Schalm. String theory, quantum phase transitions, and the emergent fermi liquid. *Science*, 325(5939):439–444, 2009.

[163] Steven S. Gubser, Silviu S. Pufu, and Fabio D. Rocha. Quantum critical superconductors in string theory and m-theory. *Physics Letters B*, 683(2):201–204, 2010.

[164] Jerome P. Gauntlett, Julian Sonner, and Toby Wiseman. Holographic superconductivity in m theory. *Phys. Rev. Lett.*, 103:151601, Oct 2009.

[165] Sean A. Hartnoll, Christopher P. Herzog, and Gary T. Horowitz. Building a holographic superconductor. *Phys. Rev. Lett.*, 101:031601, Jul 2008.

[166] Sean A Hartnoll, Christopher P Herzog, and Gary T Horowitz. Holographic superconductors. *Journal of High Energy Physics*, 2008(12):015–015, dec 2008.

[167] RongGen Cai, Li Li, LiFang Li, and RunQiu Yang. Introduction to holographic superconductor models. *Science China Physics, Mechanics & Astronomy*, 58(6):1–46, Jun 2015.

[168] Chunyan Wang, Dan Zhang, Guoyang Fu, and Jian-Pin Wu. Analytical study of the holographic superconductor from higher derivative theory. *Advances in High Energy Physics*, 2020:5902473, Jan 2020.

[169] Aristomenis Donos and Jerome P. Gauntlett. Holographic striped phases. *Journal of High Energy Physics*, 2011(8):140, Aug 2011.

[170] Aristomenis Donos. Striped phases from holography. *Journal of High Energy Physics*, 2013(5):59, May 2013.

[171] Sera Cremonini, Li Li, and Jie Ren. Holographic fermions in striped phases. *Journal of High Energy Physics*, 2018(12):80, Dec 2018.

[172] Shamit Kachru, Xiao Liu, and Michael Mulligan. Gravity duals of lifshitz-like fixed points. *Phys. Rev. D*, 78:106005, Nov 2008.

[173] Mihailo Čubrović, Yan Liu, Koenraad Schalm, Ya-Wen Sun, and Jan Zaanen. Spectral probes of the holographic fermi ground state: Dialing between the electron star and ads dirac hair. *Phys. Rev. D*, 84:086002, Oct 2011.

[174] T. Senthil, Subir Sachdev, and Matthias Vojta. Fractionalized fermi liquids. *Phys. Rev. Lett.*, 90:216403, May 2003.

[175] Subir Sachdev. Model of a fermi liquid using gauge-gravity duality. *Phys. Rev. D*, 84:066009, Sep 2011.

[176] Sean A. "Hartnoll, Diego M. Hofman, and David" Vegh. "Stellar spectroscopy: Fermions and holographic Lifshitz criticality". *"JHEP"*, "08":"096", "2011".

[177] Hiroaki Matsueda. Multiscale entanglement renormalization ansatz for kondo problem. *arXiv:1208.2872*, 2012.

[178] Bartłomiej Czech. Einstein equations from varying complexity. *Phys. Rev. Lett.*, 120:031601, Jan 2018.

[179] Sung-Sik Lee. Background independent holographic description: from matrix field theory to quantum gravity. *Journal of High Energy Physics*, 2012(10):160, Oct 2012.

[180] Sung-Sik Lee. Quantum renormalization group and holography. *Journal of High Energy Physics*, 2014(1):76, 2014.

[181] Guifre Vidal. Entanglement renormalization. *Physical review letters*, 99(22):220405, 2007.

[182] Guifré Vidal. Class of quantum many-body states that can be efficiently simulated. *Physical review letters*, 101(11):110501, 2008.

[183] Brian Swingle. Entanglement renormalization and holography. *Physical Review D*, 86(6):065007, 2012.

[184] G. Evenbly and G. Vidal. Scaling of entanglement entropy in the (branching) multiscale entanglement renormalization ansatz. *Phys. Rev. B*, 89:235113, Jun 2014.

[185] Anirban Mukherjee and Siddhartha Lal. Holographic unitary renormalisation group for correlated electrons-i: a tensor network approach. *Nuclear Physics B*, 960:115170, 2020.

[186] Anirban Mukherjee and Siddhartha Lal. Holographic unitary renormalisation group for correlated electrons-ii: insights on fermionic criticality. *Nuclear Physics B*, 960:115163, 2020.

[187] Sung-Sik Lee. Holographic description of quantum field theory. *Nuclear Physics B*, 832(3):567–585, 2010.

[188] Sung-Sik Lee. Tasi lectures on emergence of supersymmetry, gauge theory and string in condensed matter systems. *arXiv:1009.5127*, 2010.

[189] Ashley Milsted and Guifre Vidal. Geometric interpretation of the multi-scale entanglement renormalization ansatz. 2018.

[190] Xiao-Liang Qi. Exact holographic mapping and emergent space-time geometry. *arXiv preprint arXiv:1309.6282*, 2013.

[191] Ching Hua Lee and Xiao-Liang Qi. Exact holographic mapping in free fermion systems. *Physical Review B*, 93(3):035112, 2016.

[192] Ki-Seok Kim, Suk Bum Chung, Chanyong Park, and Jae-Ho Han. *Phys. Rev. D*, 99:105012, 2019.

[193] Ki-Seok Kim, Miok Park, Jaeyoon Cho, and Chanyong Park. Emergent geometric description for a topological phase transition in the kitaev superconductor model. *Phys. Rev. D*, 96:086015, Oct 2017.

[194] Ki-Seok Kim and Chanyong Park. Emergent geometry from field theory: Wilson's renormalization group revisited. *Phys. Rev. D*, 93:121702, Jun 2016.

[195] Anirban Mukherjee and Siddhartha Lal. Superconductivity from repulsion in the doped 2d electronic hubbard model: an entanglement perspective. *Journal of Physics: Condensed Matter*, 34(27):275601, apr 2022.

[196] Brian Swingle. Entanglement entropy and the fermi surface. *Physical review letters*, 105(5):050502, 2010.

[197] Masaki Oshikawa. Topological approach to luttinger's theorem and the fermi surface of a kondo lattice. *Physical Review Letters*, 84(15):3370, 2000.

[198] Kazuhiro Seki and Seiji Yunoki. Topological interpretation of the luttinger theorem. *Physical Review B*, 96(8):085124, 2017.

[199] Anirban Mukherjee and Siddhartha Lal. Scaling theory for mott–hubbard transitions: I. t = 0 phase diagram of the 1/2-filled hubbard model. *New Journal of Physics*, 22(6):063007, jun 2020.

[200] Anirban Mukherjee and Siddhartha Lal. Scaling theory for mott–hubbard transitions-II: quantum criticality of the doped mott insulator. *New Journal of Physics*, 22(6):063008, jun 2020.

[201] Anirban Mukherjee and Siddhartha Lal. Superconductivity from repulsion in the doped 2d electronic hubbard model: an entanglement perspective. *Journal of Physics: Condensed Matter*, 34(27):275601, apr 2022.

[202] Siddhartha Patra and Siddhartha Lal. Origin of topological order in a cooper-pair insulator. *Phys. Rev. B*, 104:144514, Oct 2021.

[203] Santanu Pal, Anirban Mukherjee, and Siddhartha Lal. Correlated spin liquids in the quantum kagome antiferromagnet at finite field: a renormalization group analysis. *New Journal of Physics*, 21(2):023019, feb 2019.

[204] Daria Gazizova and J. P. F. LeBlanc. Emergent nearest-neighbor attraction in the fully renormalized interactions of the single-band repulsive hubbard model at weak coupling, arxiv:2307.02360, 2023.

[205] Alexandrina Stoyanova. *Delocalized and correlated wave functions for excited states in extended systems*. PhD thesis, University of Groningen, 2006.

[206] Abhirup Mukherjee, N S Vidhyadhiraja, A Taraphder, and Siddhartha Lal. Kondo frustration via charge fluctuations: a route to mott localisation. *New Journal of Physics*, 25(11):113011, nov 2023.